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THE

JULY 6 & 13, 2026

NEW YORKER



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By [Rachel Syme](#), [Emily Nussbaum](#), [Jane Bua](#), [Sheldon Pearce](#), [Inkoo Kang](#), [Vince Aletti](#), and [Richard Brody](#).

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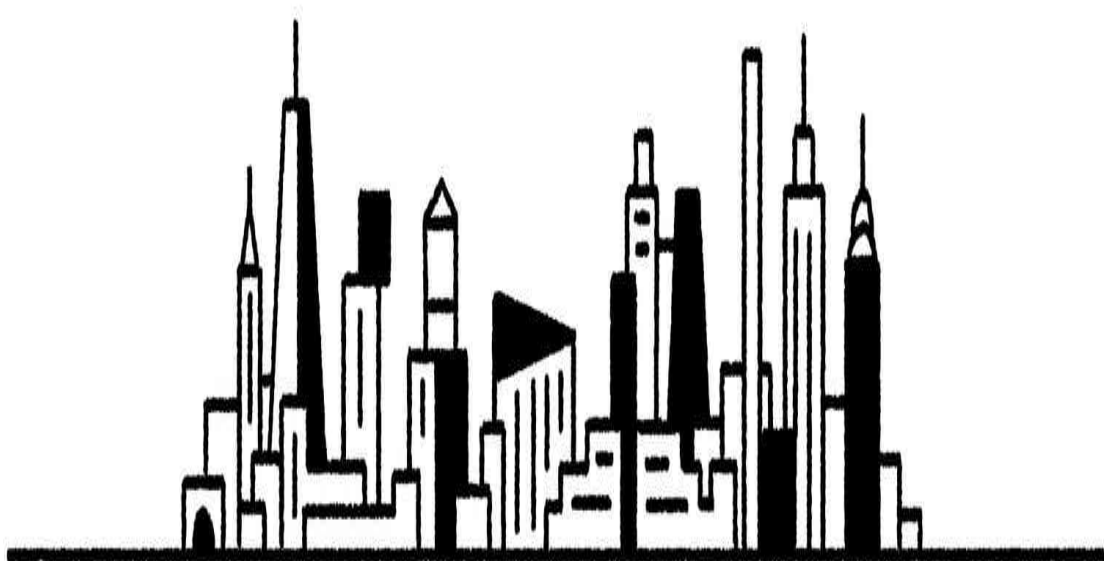
The first tarot cards, from fifteenth-century Italy, were, essentially, playing cards. Known as *carte de trionfi*, the cards formed the basis of a popular game played in Milanese courts: in addition to the fifty-two pip cards one might find in a standard set of playing cards, the tarot deck added twenty-one cards to the mix, the *trionfi*, or “triumphs,” which featured mystical imagery combined with scenes of daily Renaissance life. Few of these original tarot cards remain—the scant number that are left are part of what are known as the Visconti-Sforza decks, none of which are complete—and they had more to do with passing the time than telling one’s fortune. It was not until the eighteenth century that people began to use the cards for occult readings.



(Left to right) Queen of Swords, Death, and Time from the Visconti-Sforza tarot cards by Bonifacio Bembo. Milan or Cremona, Italy, ca. 1456-58. Art works by Bonifacio Bembo / Courtesy Morgan Library & Museum

In the years since, tarot has evolved into not only a New Age trend but a bustling form of artistic expression; beyond the classic Rider-Waite-Smith deck of 1909, you can now find a tarot deck customized for nearly every interest—there’s cat tarot, “Peanuts” tarot, pasta tarot, and even “Tarot del Toro,” inspired by Guillermo del Toro films. The evolution of the cards over generations forms the basis of [“Tarot! Renaissance Symbols, Modern](#)

[Visions](#),” a new, two-part exhibition running at the Morgan Library & Museum, through Oct. 4. The first section of the show displays rare early tarot decks, while the second highlights artists, such as the Surrealists Leonora Carrington and Niki de Saint Phalle, who have reinterpreted the Major Arcana—twenty-two cards representing significant aspects of life—in wild and wacky ways.—*Rachel Syme*



About Town

Off Broadway

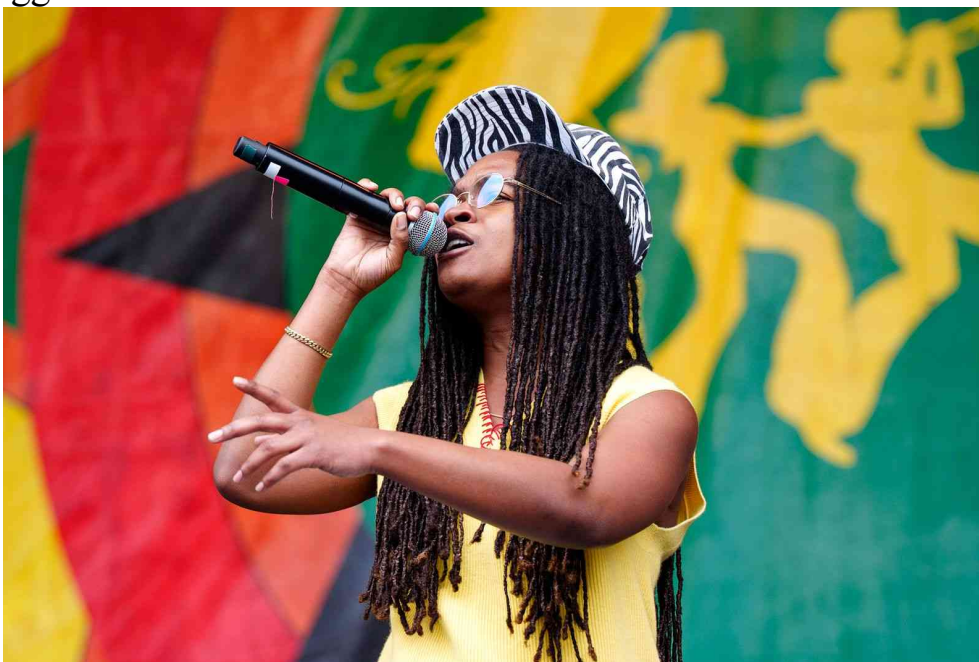
In Victoria Lynne Barclay’s startlingly intimate play “**Camping**,” two trailer-park girls share a beat-up tent, every few years, over three decades, as their paths fork along class lines. They are best friends, secret-sharers; their battles feel brutal, world-shattering. “Camping” is part of a modern wave of smart art about the humiliations and passions of girlhood, from “John Proctor Is the Villain” to the TV series “PEN15”; it shares some DNA with “Heated Rivalry,” as well, in its exploration of buried desire. But the defiant two-hander, directed by Adrienne Campbell-Holt for Colt Coeur, has a distinctive, gritty force all its own, deepened by the committed performances

of Alice Kremelberg and Colby Minifie, whose chemistry builds toward an inferno.—*Emily Nussbaum* ([HERE](#); through July 11.)

Classical

In 1896, a formerly enslaved hotel waiter named William Dorsey Swann was convicted in Washington, D.C., for “keeping a disorderly house.” The disorder in question? Hosting drag balls. Swann is widely considered to be the first self-identifying drag queen in America, putting on revelrous evenings complete with gowns, booze, and queer joy—a bright act of expression amid the mud of nineteenth-century society. Swann bravely sought a pardon for his charge, and, though denied, his efforts solidified him as a pioneer of L.G.B.T.Q.I.A.+ resistance. National Sawdust spotlights Swann’s legacy with a sing-through of Tamar-kali’s vocal chamber work “**The Swann**,” a meditation on identity, external judgment, and self-celebration.—*Jane Bua* ([National Sawdust](#); July 10.)

Reggae



Original Koffee performs at the 2026 New Orleans Jazz & Heritage Festival. Photograph by Erika Goldring / Getty

Few artists have been more of a load-bearer for a scene—in this case, the reggae revival of the twenty-tens—than **Original Koffee**, previously just Koffee. She seems to perfectly synthesize the music of Jamrock with sounds

of the broader Black diaspora. Support from the legendary Jamaican sprinter Usain Bolt launched her into a transformational run for the genre. For her fluid, rule-challenging 2019 EP, “Rapture,” she was the first woman to win the Grammy for best reggae album, and her début LP, “Gifted,” from 2022, placed her at the forefront of a sonic evolution that exploded distinctions of “old” and “new” in reggae music. She is a carrier of the movement’s legacy, but her voice refuses to be confined by its history.—*Sheldon Pearce* ([*Paradise Coney Island*](#); July 11.)

Television

The Netflix dramedy “**The Four Seasons**,” co-created by Tina Fey, is a remake of a 1981 film by Alan Alda. Like the film, the series follows a trio of well-heeled couples, including the high-strung Kate (Fey) and her soft-hearted husband, Jack (Will Forte), and the more sexually adventurous Danny (Colman Domingo) and Claude (Marco Calvani), who are all close enough to travel together every few months—and enmeshed enough that when the third couple, Nick (Steve Carell) and Anne (Kerri Kenney-Silver), split up, the remaining pairs spin out. “The Four Seasons,” now in its second season—which benefits tremendously from having shaken off the source material—is in many respects an overly tidy, milquetoast entertainment, but Fey’s treatment of relationships in midlife feels refreshingly rooted in reality.—*Inkoo Kang* (Now streaming.)

Art



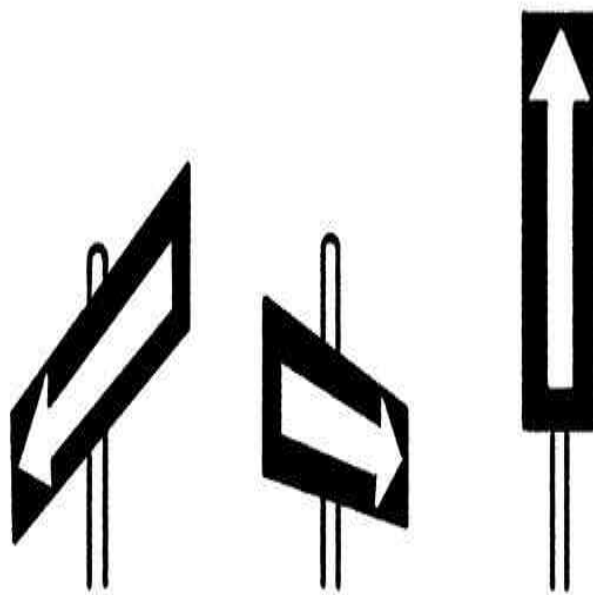
"Composite of Shirley Berman and Madison Grammar School, Topanga Canyon," from 1956. Photograph by Edmund Teske / Courtesy © Edmund Teske Archives / Laurence Bump and Nils Vidstrand

Deborah Bell runs a small, eccentric photo gallery in Chelsea. Her speciality is vintage material, but that never limits her range. Bell's current show, "[Abstract Visions & Uncommon Portraits](#)," is a miscellany that includes portraiture (August Sander, Clarence John Laughlin) but focusses on discoveries in avant-garde still-life and abstraction. Edmund Teske slips between genres with two small, dreamy images that layer a woman's face over the façades of old buildings, like memories that refuse to fade. László Moholy-Nagy hovers low over a gaggle of pigeons, suspending the viewer uneasily between protector and predator. In a room full of little mysteries, Rose Mandel's stand out. Her study of tangled foliage is classic modernism, but her two seascapes, seen through luminescent bubbles, are visionary in the vein of Hilma af Klint.—*Vince Aletti (Deborah Bell; through July 2.)*

Movies

In "**The Death of Robin Hood**," starring Hugh Jackman, the writer and director Michael Sarnoski presents the legendary hero as a thirteenth-century serial killer who, with his crude companion Little John (Bill Skarsgård), leaves a bloody trail of poor people in the English countryside. Badly wounded in a battle of revenge, Robin Hood is brought to a priory and tended by Sister Brigid (Jodie Comer), whose generosity awakens his finer

feelings and spurs him to virtuous, self-sacrificing action. But neither part of the drama comes alive; the early scenes, in which Robin Hood bathes in gore, are a hollow churn, and the redemptive warmth of the protagonist's monastic retreat comes off as a superficial, sentimental slog.—*Richard Brody (In wide release.)*



Pick Three

In Steven Spielberg's "Disclosure Day," he presents aliens as saviors. Richard Brody shares some of the wider and wilder alien movies available to stream.



Photograph from Everett

1. The Cold War gave rise to paranoid visions of space invasions, as in the low-budget, high-anxiety thriller **“Earth vs. the Flying Saucers,”** from 1956. The creatures’ arrival is marked by violence; their weaponry, brought to life with stark special effects, heralds an intergalactic war of conquest and sparks an arms race; and a destructive showdown in Washington, D.C., is heavily symbolic and inescapably political.

2. John Sayles’s poignant and rowdy 1984 satire, **“The Brother from Another Planet,”** stars Joe Morton as an extraterrestrial who has taken the form of a Black man and then makes his way to Harlem. The visitor—who understands English but can’t speak—gradually learns to fit in while deploying his few but startling paranormal abilities. He also reveals that he was enslaved in space, leading to bitterly ironic results.

3. Bruno Dumont’s three-hour-plus comedy **“Coincoin and the Extra-Humans,”** from 2018, is a sequel to his 2014 drama, “Li'l Quinquin,” but doesn’t depend on it. The action, set on France’s rugged northern coast, is anchored by teen romance, involves far-right anti-migrant activists, and is complicated by blue goo from space that creates clones—including of the two loopy police officers who investigate. The spectacularly antic tale of originals and doubles, locals and outsiders, visually and philosophically lampoons the very concept of identity.

P.S. Good stuff on the internet:

- [*Good gardening looks*](#)
- [*It's cherry season*](#)
- [*How to dance*](#)

By [Sheldon Pearce](#) and [Rachel Syme](#)

*You're reading the Goings On newsletter, a guide to what we're watching, listening to, and doing this week. **Sign up to receive it in your inbox.***

In a letter to his wife, Abigail, from July 3, 1776, John Adams wrote of his idealized vision for future celebrations of American independence: “It ought to be solemnized with Pomp and Parade, with Shews, Games, Sports, Guns, Bells, Bonfires and Illuminations from one End of this continent to the other from this Time forward forever more.” Now two hundred and fifty years deep into the American experiment, his wish has become a reality, as all the hallmarks of his proposed ritual ring forth for a monumental birthday bash in New York City.



Photograph by Kent Andreasen / Kintzing

Games and sports lead the festivities. If baseball is the American pastime, there may be no better place to spend the Fourth of July than at [Yankee Stadium](#), where the Bronx Bombers play the Minnesota Twins, led by the U.S. national-team captain Aaron Judge. Outside Nathan's Famous on Coney Island, one of the more beloved Independence Day traditions returns: the [Hot Dog Eating Contest](#). As competitors down dogs for ten straight minutes with a ravenous efficiency that feels worlds apart from snacking on a frank at the ballpark, eating escalates into an elite physical spectacle nearing performance art. The undefeated women's champ and world-record holder Miki Sudo looks to raise her eleventh mustard belt, while the face of

competitive eating, Joey Chestnut, seeks his eighteenth victory. But Chestnut seems to be running down his own shadow, chasing the record seventy-six dogs he ate in 2021, after a seventy-dog performance last summer.

There is also plenty of pomp, along with bells, bonfires, and illuminations. The **Times Square ball** will drop on a day that's not New Year's Eve for the first time in history, marking midnight in each American time zone to kick off birthday celebrations. In keeping with Operation Sail, a series of events commemorating special occasions over the years, [Sail4th 250](#) ushers a fleet of tall ships from more than twenty countries into New York Harbor and toward the Statue of Liberty. The fiftieth [Macy's Fourth of July fireworks show](#) leads what are sure to be explosive demonstrations across town, featuring the musicians Post Malone, Noah Kahan, Shaboozey, and more — "[Red, White & Views](#)" presents a waterfront view of the show from the South Street Seaport Museum, and **Edge NYC's** [sky-high fireworks](#) lets partygoers enjoy the display from its observation deck with music all night.



A photograph from the series "Migrantes," part of the exhibition "Declaring America: 1776 and Beyond." Photograph by Joseph Rodriguez / Courtesy NYPL

Any appreciation of this milestone anniversary would be well-balanced with a consideration of history, and the New York Public Library's ongoing exhibition reckoning with the complicated American legacy, "[Declaring America: 1776 and Beyond](#)," honors the signing of the Declaration of Independence with a viewing of the document itself (on display July 1-7).

For a less traditional journey through the last two hundred and fifty years, there's "[Growing America](#)," at the New York Botanical Garden, an exhibition exploring American agricultural history, featuring botanical illustrations, planting ledgers, and objects from Presidential libraries alongside living-plant displays tracing the evolution of our food traditions (July 4-Oct. 18). The Met offers an ode to the melting pot with performances of "[the national anthems](#)," the 2014 choral composition by the Pulitzer Prize-winning composer David Lang (July 4 at 2 *P.M.*, 4 *P.M.*, and 6 *P.M.*). Devised by taking phrases from national anthems across the globe and translating them into English, and performed by the Clarion Choir with Catalyst Quartet, the piece imagines national identity as not a xenophobic position but an act of coalition building.—*Sheldon Pearce*

On and Off the Avenue

Rachel Syme investigates the latest in Day-Glo fashion.

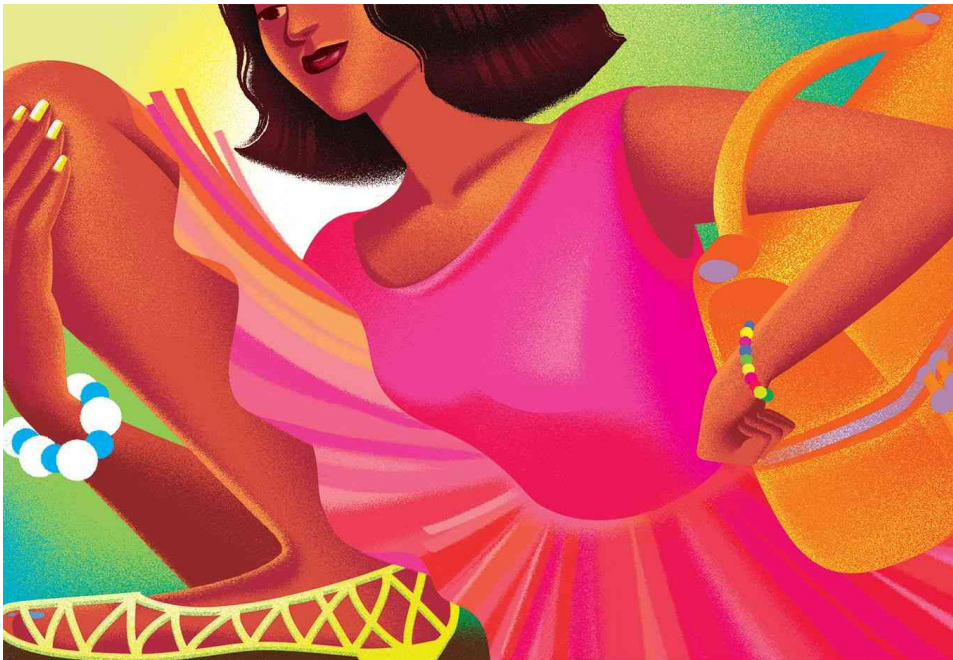


Illustration by Simone Noronha

It is a vibrant moment to be a New Yorker: the sun is shining, the Knicks are on top, Mayor Zohran Mamdani is uniting the city in exuberant celebration, and things are looking bright. So why not wear some color? Lately, I've found myself drawn, much like a peacock in heat, to vivid, high-visibility

hues; it feels like the age of drab, beige, “quiet luxury” clothing is finally giving way to a fuller spectrum of sartorial expression. This past month, when the entire city was in the grip of basketball mania, my eyes have felt newly refreshed by the abundance of royal blue and traffic-cone orange on the streets. During the Knicks’ victory parade, the undulating downtown crowd looked, from afar, not unlike a school of tropical clown fish. It made me yearn to throw out my black and white tees and reimagine my whole closet in Technicolor; only good sense is stopping me. (I’ll want those black shirts, come fall.) I had this feeling again recently when, rewatching “Sex and the City” (my annual summer tradition), I saw an outfit that Sarah Jessica Parker, as Carrie Bradshaw, wears in the fourth season for a night out dancing. It’s a semi-sheer, white jersey minidress, but the real attraction is what she has on underneath: a hot-pink bra, which glows through her dress like a salt lamp. When a friend exclaims, “Look what you’re almost wearing!” she replies, “Day-Glo underwear, look into it!” So I did, diving into a sustained search for neon lingerie of my own. I ended up settling on the [“Allegra” balconette](#) from **Journelle** (\$74), which comes not only in electric raspberry, but also chartreuse, tangerine, sunflower, and a radiant, highly saturated seafoam green. The colors reminded me of the trademark tones such as Blaze Orange and Saturn Yellow, invented by the Day-Glo Color Corp., a company originally founded by the brothers Robert and Joseph Switzer in the nineteen-thirties. The Switzers’ big invention—dyes that turned fabric fluorescent—had both military and civic applications, but it swept the fashion world in the sixties, the heyday of trippy, acid-hued garb. And you can still channel it! Get some [neon-yellow jelly flats](#) from **Los Angeles Apparel** (\$28) or [bright-pink gauze joggers](#) from **Gap** (\$39.95). **Erdem** has a gorgeous [lime-green satin shell](#) (\$1,465), or maybe you fancy a [Maison de Sabre saddle bag](#) (\$398) in tiger-lily orange. Think pink in a [feather-trimmed Trina Turk tank](#) (\$258), cool off in a [margarita-hued polo knit](#) from **Rachel Antonoff** (\$298), or wriggle into an [acid-yellow bikini](#) (\$220) from **Left on Friday**. Look around, and you’ll see a veritable rainbow of options. This summer is no time to fade away.—*Rachel Syme*

P.S. Good stuff on the internet:

- [Campfire practice](#)

- [*Rock Hudson on Marilyn Monroe*](#)
- [*A tradcath wedding*](#)

The Talk of the Town

- [Chronicle of a Disaster Foretold](#)
- [Herding the Fro-Yo Sheep](#)
- [The Natural Memory of Kabir Sehgal](#)
- [Bruce Nauman Isn't Bound by the Rules](#)
- [Postscript: Mark Singer](#)

[Comment](#)

Chronicle of a Disaster Foretold

Maggie Haberman and Jonathan Swan’s “Regime Change” is packed with news about the Trump White House that will stay news.

By [David Remnick](#)



Photo illustration by Cristiana Couceiro; Source photographs from Getty

In 1996, Joan Didion unholstered the X-Acto knife that was her pen and went to work on Bob Woodward. In a register of pitiless irony, she quoted Woodward’s earnest explanations of his journalistic methods—the difference between “background” and “deep background,” his “eight file drawers” of documents for a book on the Supreme Court. She was unimpressed. “These are books in which measurable cerebral activity is virtually absent,” [Didion wrote](#). In the end, she determined that Woodward was engaged in the writing of “political pornography.”

This was unfair, at best a category error. What Didion failed to note was that Woodward was a reporter, not a scholar or a belletrist, and that before he was thirty he, with his *Washington Post* partner, Carl Bernstein, had unearthed the predations of [the Watergate scandal](#)—hardly the work of stenographers in thrall to power. It brought down the Presidency of Richard Nixon. Journalism is a first rough draft of history, as they say, and

scholarship, which feeds on journalism and so much more, plays by another clock. [Robert A. Caro](#) began the research for his [multivolume biography](#) of Lyndon B. Johnson half a century ago.

[Maggie Haberman](#), a reporter of impressive energy, began her career at the New York tabloids and joined the *Times* in 2015. She has been on the Trump beat since the Donald was a real-estate guy hustling for attention in the columns of Liz Smith and Cindy Adams. As “[Confidence Man](#),” Haberman’s first book about Trump, from 2022, made clear, he is a man of immutable character. “He has had only a handful of moves throughout his entire adult life,” she wrote. The “quick lie,” the “shift of blame,” the “outburst of rage,” the abuse of loyalists, a “raft of old grievances,” a “refusal to be shamed.”

Trump’s political reflexes could have been anticipated. Talking to *Playboy*, in 1990, he praised the Chinese leaders who ordered troops to slaughter pro-democracy demonstrators in [Tiananmen Square](#). “They were vicious, they were horrible, but they put it down with strength,” he said. “That shows you the power of strength. Our country is right now perceived as weak.” This is [the same guy](#) who, thirty years later, asked the chairman of the Joint Chiefs of Staff, Mark Milley, why he couldn’t confront street protesters with force and “shoot them in the legs or something.”

Now, as the country lurches glumly toward its [two-hundred-and-fiftieth anniversary](#), Haberman has collaborated with another *Times* reporter, Jonathan Swan, on “[Regime Change](#),” a vivid, rigorous, and unavoidably depressing chronicle of the first year of Trump’s second term in the White House. Part of the high-wire act of such books is that the authors and their publishers work at unaccustomed speed to provide the end product with a history-as-it-is-happening varnish. The results are usually as perishable as week-old bananas.

“Regime Change” is exceptional. It transcends its genre. Although some of the material is familiar from the *Times* and other sources and from Trump’s relentless self-exposure, the book is packed with news that will stay news. One late-breaking example among many: Haberman and Swan provide an astonishing account of Benjamin Netanyahu’s trip, just four months ago, to the White House Situation Room, where the Israeli Prime Minister

persuaded Trump to join him in what would be [a strategic catastrophe](#). Netanyahu assured the President that together they would topple the Iranian regime and end its nuclear ambitions before it ever had a chance to close the Strait of Hormuz. Haberman and Swan report that the Secretary of State called Netanyahu's plan "bullshit." The C.I.A. director declared it "farcical." Whatever. "Sounds good to me, the President told the Prime Minister." Everyone fell into line. Well played, sir!

This is reporting of consequence, and it puts to rest the knowing online critique of these beat reporters that to get their scoops they engage in "bothsidesism" or dampen their coverage in exchange for access. "Regime Change" is particularly strong on the Administration's [colossal financial corruption](#), its heedless destruction of invaluable agencies such as U.S.A.I.D., and the sordid and unhinged nature of Trump and the culture over which he presides. Haberman and Swan contend that Trump ran in 2024 for one reason above all: "This was about staying out of prison." After facing multiple indictments, [impeachments](#), and criminal convictions, [Trump returned](#) to the White House with retribution on his mind: "I was the hunted, and now I'm the hunter."

The atmosphere in Trump's White House is reminiscent of the Kremlin in Armando Iannucci's 2017 movie, "[The Death of Stalin](#)." The autocrat delights in humiliating so many people, not least tech billionaires, including Jeff Bezos and Mark Zuckerberg, who once opposed him. The description of Trump's White House is of a decadent court and a king addicted to vengeance and flattery. Trump employs one aide, a young woman named Natalie Harp, who follows him around all day, handing him glowing notices from the right-wing press and occasionally sending him adoring letters ("You are all that matters to me"). Even this brings him no joy. When Elon Musk, who raised some three hundred million dollars for Trump's campaign, blasts the President over his budget bill, Trump says, "They always leave me. They always do this. This is why I can't have friends." He instructs Harp to bring him his phone. He calls Musk twice. Both times, he gets voice mail.

We have always known that Trump is a narcissist. Haberman and Swan make clear the dimensions of his malady. During Trump's [hush-money trial](#) in New York, he heard that a mentally ill man, "consumed by conspiracy

theories,” had set himself on fire in a park nearby. “Do you think he did it for me?” Trump asked an aide. “Let’s tell people that he did it for me.” In an interview with the authors, Trump says that he was pleased to learn from an unnamed historian that, considering the reach of his arsenal and armed forces, he is far more powerful than Genghis Khan, Alexander the Great, or Joseph Stalin. The historian in question, the authors discover, is Gary Player’s caddy. Trump is a damaged man unleashing daily damage on the country. He spends his nights alone, rage-posting on Truth Social. His wastebasket overflows with “empty potato chip bags, Starburst wrappers, and ice cream cartons.”

As July 4th approaches, the water in the Reflecting Pool—refurbished by Trump’s “pool guy” with a no-bid contract—turns green with algae. It is a dismal time in the capital. Haberman and Swan have done admirable work, but one reads the final pages of “Regime Change” hoping that their next book is the story of transition, from the era of Trump to one of democratic renewal. That chapter cannot come soon enough. ♦

Herding the Fro-Yo Sheep

This summer, every trendy dessert joint has a mile-long line of transplants and tourists. One New Yorker is protesting in his own way—by “baa”-ing at them.

By [David Kamp](#)



Illustration by João Fazenda

It was 5:55 *p.m.* on a Thursday, nearly time for supper, an hour when no one has any business getting themselves a sweet treat. Yet fifty-odd people were standing on line along an aesthetically grim stretch of Seventh Avenue, waiting to purchase cups of frozen yogurt from the new West Village location of Myka, a Madrid-based dessert chain.

One witness, a young man in a white T-shirt and athletic shorts, with three-day stubble, wasn't having it. Walking alongside the queue, he shook his head in disbelief. “Baa!” he shouted at the idling fro-yo pilgrims. “Baa! Baa!”

The young man was Ezra (he prefers not to reveal his last name), a twenty-seven-year-old native New Yorker, one of five children raised in South Brooklyn by parents he describes affectionately as “modern hippies.” He is

the brains behind the Instagram account @sheepofnyc, which, since its début, in May, has attracted more than seventy-five hundred followers.

As virality goes, this is not a big number, but the passion that the account has aroused dwarfs its reach. Some commenters on Ezra's posts view his feed as a referendum on who is and isn't a real New Yorker. Anyone who waits on line for fro-yo must be a "transplant" who needs to "move back to Ohio," one person wrote. A @sheepofnyc clip of a long line outside Caffè Paradiso, on Elizabeth Street, home of the social-media-vaunted Salted Brown Butter and Oat Latte, has engendered such comments as "This is a microcosm of everything wrong in current culture" and "In Spanish, we say, 'Pendejos.'"

Ezra himself prefers a more minimalist approach to offering commentary on the latte line, or any line. In the clip, he is heard saying, "Baa! Baa! Baa! Baa! Look at all these sheep. I'm just a shepherd tryin' to herd the flock . . . for coffee."

Long-line madness is not new to New York. If your memory goes as far back as 2013, the era of the Cronut bubble, you will recall the hour-long waits in SoHo outside Dominique Ansel Bakery. But this spring's fresh bloom of lines has broken locals' brains. The Village is home to not only Myka but a new outpost of the fro-yo destination Mimi's, on University Place. After an arduously long winter, these narrow, modestly sized shops have been—with a suddenness and intensity that only TikTok and Instagram can foment—thronged. Most of the line-waiters are young. Many of them post trophy pics of their quarry; Myka offers a particularly photogenic swirl of vanilla yogurt topped with green pistachio sauce and toasty crumbled baklava.

It's this aspect of the line phenomenon—that the behavior is driven by algorithms and the influencer economy—that gets under Ezra's skin. He pronounced himself more baffled than angered. "It started as a joke," he said. "It blew my mind seeing all these lines everywhere in New York. So I was, like, I'll just go around taking little videos, calling these people sheep and 'baa'-ing at them."

Ezra stays inconspicuous by filming the lines with a pair of Ray-Ban Meta sunglasses, which synch with his iPhone. A strapping, handsome guy who works in real-estate development, he is hardly the picture of a social-media guerrilla. He realizes that @sheepofnyc has found its audience via the same algorithmic alchemy that drives the fro-yo fervor. One post, filmed by a friend, of an absurdly long line of people waiting to enter the SoHo clothing store New York or Nowhere, has been viewed nearly a million times. It helps that the friend is better at “baa”-ing than Ezra is, emitting a convincing ovine bleat worthy of a flock in County Clare.

Ezra now receives five to ten guest submissions a week. He hasn't posted many of them, usually because they're poorly shot, have no audio, or are too mean. He has also received overtures from clothing companies looking to collaborate with @sheepofnyc on T-shirts. One mockup came with M.T.A.-style lettering that reads “*SHEEP CENTRAL STATION*.” Another bears an image of a sheep Statue of Liberty holding a selfie stick instead of a torch.

“I don't care about monetizing this,” Ezra said, sitting on a bench where some Myka customers were eating their fro-yo. He also doesn't want to come off as a scold. “A lot of it is tourists, and I feel bad,” he said. “This city is amazing, every part of it. I guess the message is ‘Stop waiting on line.’ Go walk the West Side Highway.” Instead of wasting time on the line for L'industrie pizza on Christopher Street, he suggested, “hop on a subway into real Brooklyn and try the best pizza ever, at Di Fara.” There's usually a queue there, too, but the Di Fara line is the hard-won result of decades of pizza excellence. The shepherd of @sheepofnyc may be young, but he is wise enough to get that on-line life, like online life, is not really living. ♦

The Natural Memory of Kabir Sehgal

For his new album, “Stars and Static 2026,” the fourteen-time Grammy winner recorded sounds from across the country. Did it change how he thinks about America?

By [Dan Greene](#)



Illustration by João Fazenda

If Spotify playlists have taught us anything, it’s the utility of lo-fi beats. They’re the perfect companion for studying, focussing, and chilling. But contemplating civics? Kabir Sehgal thinks so. Sehgal is a fourteen-time Grammy-winning producer and jazz musician, whose new album, “Stars and Static 2026,” mixes noodly, restrained takes on patriotic staples with natural sounds recorded in American locales (“yankee newport doodlin’,” “this is my little rock sunset”). “I call it ‘geo-fi,’ ” Sehgal said the other afternoon, at the East Coast Memorial, in Battery Park. “When you name a place, there’s a provenance. I wanted to see if lo-fi could be a place of natural memory, to have that conversation of what it means to be American.”

Sehgal, who is tall and younger-looking than his forty-one years, is less a multi-hyphenate than a vocational run-on—not only a musician and producer but also a former finance executive who is a Navy reservist and an

author of books about jazz and currency, and the co-author of celebrities' books (with John Lewis and Deepak Chopra) and children's picture books (with his mother, Surishtha). "I call it a 'portfolio career,' " he said. About a decade ago, he would frequent the East Coast Memorial while working at a finance firm. (He later worked at JPMorgan, in whose office he was sworn in to the military. Wynton Marsalis, a mentor, attended.) Sehgal, who wore a jacket and a quarter zip, took in the monument's massive granite pylons, inscribed with the names and ranks of the more than four thousand Americans who never returned from the Atlantic during the Second World War. Jackhammers carried on nearby.

"This would have been a great place to do a field recording," Sehgal said. "What better way to say we're a work in progress?" He gestured southward, noting that there would typically be a beautiful view of the Statue of Liberty. It was blocked by a construction wall, which extended toward a building labelled "U.S. Department of Homeland Security."

A few years ago, after a long series of musical collaborations, including several guided meditations with Lil Jon and a spoken-word album with the Dalai Lama, Sehgal found himself drawn to the D.I.Y. simplicity of lo-fi experimentation. At the same time, he'd had his eye on America's Semiquincentennial, and had taken to using a two-pronged Sony field recorder to capture his sonic surroundings during his domestic travels. Pedalling through the South Carolina Lowcountry on his bike became percussion under "Battle Hymn of the Republic." Chatter at a festival in Atlanta, his home town, formed interstices in "America the Beautiful." (There was one overseas inclusion: the bustle of Naples, where Sehgal was once stationed at a U.S. naval command, is layered into the Naval Academy fight song, "Anchors Aweigh.") Sehgal stripped the familiar tunes of key elements, hoping that listeners would fill the negative space. "This entire project is melody by implication," he explained. He also slowed them considerably, to induce a meditative effect. "These aren't, like, brisk marches," he said. "This is less fireworks, more reflecting pools."

At the memorial, Sehgal walked past a swooping bronze eagle sculpted by the Italian-born medallist Albino Manca and dedicated by President John F. Kennedy six months before his death. Gulls shrieked. Cuban music boomed from a speaker on a bench. "If you were to do a song of downtown New

York, what is it?” Sehgal asked. “The rhythm of the song would be clave.” Perhaps, he suggested, it could be peppered with the solicitations of red-jacketed tourist-ferry barkers. “You don’t have to do much more than that,” he said. He strolled toward Bridge Street, where a stars-and-stripes cement mixer whirred and beeped. “Those always remind me of ‘Die Hard,’ ” he said.

Two blocks later, he stopped in at Fraunces Tavern, the former revolutionary hub where a victorious George Washington bid farewell to his officers. Sehgal ordered an iced tea and took a barstool. Aside from indistinct public patter, only one voice is featured on his album: that of John Lewis, the late Georgia congressman. In 2017, Sehgal was part of a group that accompanied Lewis on a trip to Selma. “We were having breakfast in a hotel lobby, and I was, like, ‘Do you think I can record you?’ ” Sehgal said. He set up a wall of ballroom chairs to muffle the nearby kitchen noise and had Lewis read President Barack Obama’s recent remarks at the Edmund Pettus Bridge. For the album, he laid the track over the chord progression of “We Shall Overcome.”

Sehgal sipped his drink. He said he hoped that people would not only “check out the record but, like”—he mimed popping out earbuds—“listen to your surroundings. Wouldn’t that be a lesson for democracy? Listen to people.” ♦

Bruce Nauman Isn't Bound by the Rules

At a gallery in Tribeca, the artist talked bald spots with Eric Fischl and walked through his quickie exhibition “No Mistakes,” 3-D videos of him drawing with his eyes closed.

By [Thessaly La Force](#)

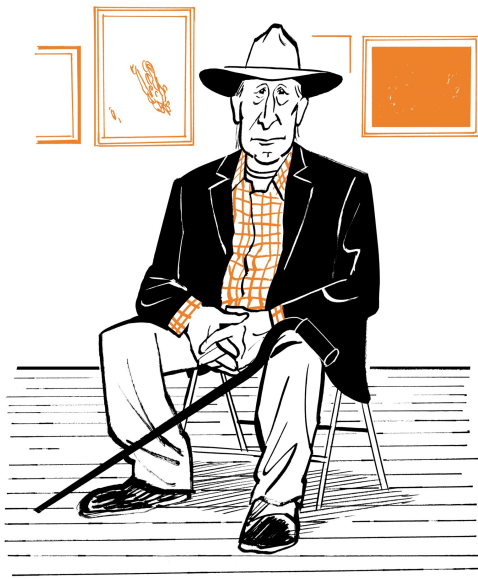


Illustration by João Fazenda

Bruce Nauman, the artist, was sitting at the front desk of Okey Dokey Konrad Fischer, a gallery in Tribeca, one recent Saturday morning. He was wearing a plaid shirt, jeans, and a cowboy hat; a cane was propped between his legs. Next to him was a friend of his, the painter Eric Fischl. Nauman's short-run show “No Mistakes” had just opened, but the two weren't discussing art. They were talking about hair loss.

“So the issue is, do you shave it?” Fischl, who has gracefully maintained an impressive thatch of thinning gray hair, asked. “A comb-over is not an option. I tend to only look at myself in the front, because if I see it from the back I go, ‘What the heck is going on back there?’ ”

Nauman, who is eighty-four, keeps what remains on his head at a close crop. He has cycled through a few styles over the decades. The shaved-head period. The beard period—“also known as the grow-your-hair-somewhere-else-on-your-head phase,” Fischl said.

The topic had come up because the back of Nauman’s head featured prominently in “No Mistakes,” a series of 3-D videos that capture, in slow motion, Nauman drawing with his eyes closed. The idea arose after Fischl asked Nauman how, exactly, he held a pencil.

“I hadn’t really thought about it,” Nauman said. One day, he asked his daughter, Zoë, to film over his shoulder while he sketched. The exercise forced him to question not only how he drew but how he worked.

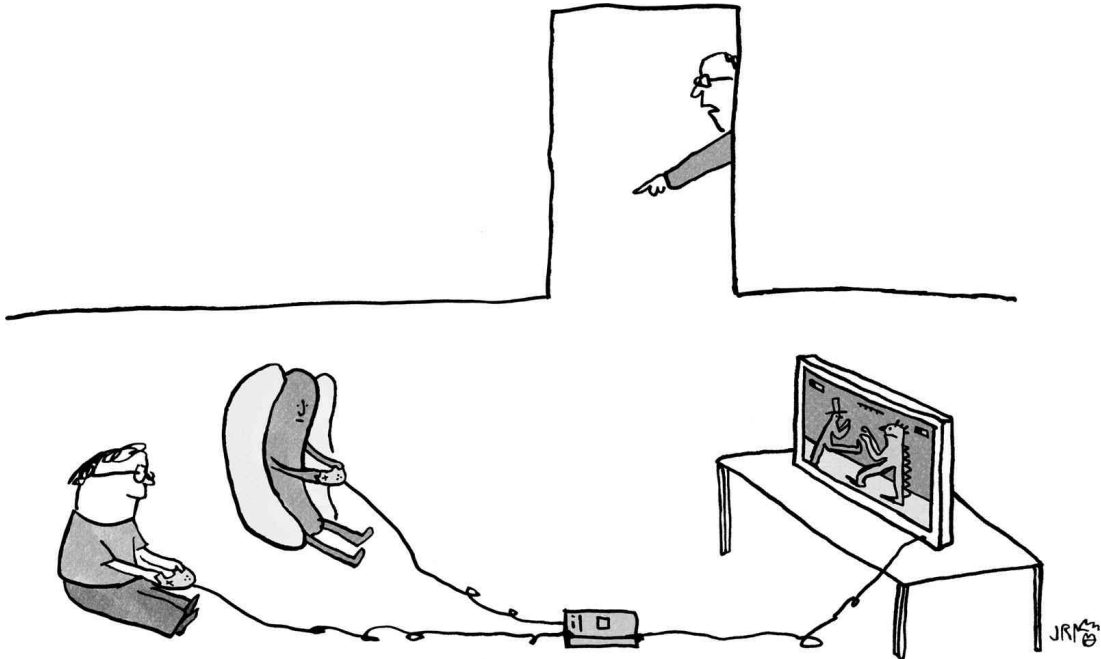
Nauman got up and slowly walked to the back of the gallery, where the videos were playing on two screens. He is reserved and mumbly, qualities that give him a quiet charm. The name of his show was inspired by an exercise that he learned from the German curator Kasper König: draw a pig without lifting your pencil. It was tricky. But Nauman found that, when he tried the exercise with his eyes closed, he was no longer bound by the same rules. Freed from creating a realistic-looking pig, he couldn’t make any mistakes.

In the gallery, he put on a pair of 3-D glasses and watched a film of himself, with a stylus in each hand, drawing a cup. Onscreen, his hands floated slowly across a sheet of paper, an entranced expression on his face.

Nauman’s team had arrived. In New York, he puts them up at the Bowery Hotel. “It’s close to the action,” Juliet Myers, his longtime studio manager, said. Nauman owns a town house in the East Village, but for almost fifty years he’s lived mostly on a seven-hundred-acre property in Galisteo, New Mexico, outside Santa Fe, where he and his late wife, the painter Susan Rothenberg, worked in separate studios. Since her death, in 2020, he’s been travelling more. “He used to not want to go anywhere,” Myers said. “Why leave the ranch?”

Nauman’s studio assistant, Ellen Babcock, a gentle professorial type, came over. They had been working on “No Mistakes” for more than a year. “We

typically film for a few hours,” Babcock explained. “Then we break for lunch—he usually cooks something. Then we work some more. We get along because neither of us talk much.”



*"Billy, stop playing with your food!"
Cartoon by Jonathan Rosen*

“Food is sometimes more interesting to the boss than art,” Benjamin Kidwell Lein, a bespectacled technician who helps Nauman with video postproduction, said. “First one up makes the coffee. He feeds me his green-chile stew, which is awesome.” In the evenings, the two watch movies together, usually Japanese films, although Nauman recently enjoyed a documentary about the Lumière brothers. He tends to leave the projectors he uses for the video project running overnight, which drives Lein nuts. “They have a life span,” Lein said. “He always says to me, ‘Ah, Benjamin, I just leave it on for the cats.’”

The gallery was beginning to fill with friends. Nauman had hauled his own ratty cream-colored sofa and a few chairs from his house into the space.

“The 3-D makes me nauseous,” the artist Rachel Harrison said.

Steve McQueen, the filmmaker, came by. “Maestro, how you doing?” he asked.

Nauman's gallerist, Berta Fischer, was hosting the crowd later at the restaurant Altro Paradiso. A car was coming for Nauman, and he headed toward the door. Outside stood the painter Frank Owen, a friend since their days as art students at the University of California, Davis, in the nineteen-sixties.

Owen grinned at Nauman: "I recognize you!" Then he jumped in the air.

"Can't do that with my legs anymore," Nauman said. The two batted their canes like swords.

Owen had travelled from upstate, and talked about Nauman visiting him. There's a pond near his house which is perfect for skipping stones. When Nauman was there, they couldn't find any good rocks. The next time Nauman showed up, Owen said, "he was carrying a bag with him on the plane. You could hear the stones jangling as he walked. California granite. If the pond ever drains, that's all they'll find." ♦

[Postscript](#)

Postscript: Mark Singer

In a 1997 Profile for the magazine, he looked for Donald Trump's soul. Where it should have been he found—nothing.

By [Ian Frazier](#)



Illustration by João Fazenda

Mark Singer, who contributed to this magazine as a staff writer for fifty-two years, died of cancer, at Memorial Sloan Kettering in Manhattan, not far from his apartment, on June 19th. He was seventy-five years and eight months old, to the day. He started out writing pieces for *The Talk of the Town*, then moved on to humor, personal essays, and long reported stories, some of which were collected in books or became whole books themselves. His 1985 book “*Funny Money*,” a deadpan nonfiction comedy about the collapse of a fraudster bank in Oklahoma during the savings-and-loan crisis, was a best-seller. My copy is inscribed “Dear Sandy—This is the only free copy of this book that you will be receiving. Love, Mark.”

He grew up in Tulsa, the middle child of five, with two brothers and two sisters. The siblings were Sandra, George, Ellen, and Stephen. A family business, Singer Brothers, founded by a grandfather and a great-uncle, did oil exploration and production, and prospered with a network throughout the

state. His mother, Marjorie, and his father, Alex, could take a person by surprise with their laconic Oklahoma humor. A family story: One day while playing in a field, Mark and one of his brothers started a grass fire. The owner of the field called Mark's father and said, "Mr. Singer, two boys just started a fire in my field, and although I can't make out their faces, I think they might be your sons."

Alex Singer said, "Describe them."

The man said, "Well, one of the boys was flipping matches, and when the fire started he ran away."

Alex said, "That would be [Mark's brother]."

The man went on, "And the other boy was flipping matches, too, but, when the fire started, this boy ran to the irrigation ditch and took off one of his boots and filled it with water and ran back to the fire, but tripped and fell on the way and spilled all the water from the boot, and thus failed to put out the fire."

Alex said, "That would be Mark."

He went to Tulsa public schools and won citywide spelling bees. On his high-school wrestling team, he competed in the hundred-and-five-pound weight class. He developed a good game of golf, despite sometimes playing on courses where the fairways were dirt with tufts of grass and the greens were dirt that had been rolled flat and oiled. When he got to Yale, he realized that his high school had taught him not one thing about how to write an essay. He set about learning under William Zinsser, a professor of writing and the author of the book "On Writing Well." After Mark graduated, Zinsser recommended him for a job at this magazine.

I started at *The New Yorker* in 1974, the same year as Mark. We were twenty-three. At first, we weren't friends (my fault: meanness, narcissism). Then he won me over with a Christmas gift of a Greenberg Smoked Turkey, from Tyler, Texas. Every year after that, he sent me a Greenberg turkey—through our long friendship, that has added up to fifty-one turkeys, with one year lost because in 2020 the plant in Tyler evidently oversmoked itself and

had a fire. Each year, I used the convenient shipping box of heavy-duty white cardboard to hold my income-tax documents. I don't know what I'll do for '27.

There never was a more generous person. He loaned money to people and sometimes didn't get it back. He helped friends who found themselves with a doctor's bill that they couldn't pay. He gave more than anybody I know to the Holy Apostles Soup Kitchen, on Ninth Avenue, and gave consistently. The soup kitchen sponsors fasts to raise money. One year, Mark tried fasting, but he told me he had to stop because he found that it made him really hungry.

His writing did good for the world. He had a connoisseur's appreciation for cons and scams and got to the bottom of some notable ones. His long piece about an Indianapolis con man and accused murderer required a lot of gruelling reporting and hanging out with the guy, and later became a book that exposed him. When the guy got out of prison and tried to make a con-artist comeback, Mark's book was there, standing in the way. I like to think that Mark saved Obamacare. A piece that he wrote in 2017 about his own autoimmune disease and the upcoming vote to kill the health-insurance program is said to have influenced Senator John McCain's deciding "no" vote. And back in the nineteen-nineties Mark wrote a Profile of Donald Trump, then just a publicity-seeking real-estate developer, that did not follow the usual red herrings with which Trump was delighting the media. Instead, Mark looked for his actual soul. Where it should have been he found—nothing. He looked hard at the nothing and gave the reader a sense of it, long before it became the giant expanding hell pit we're falling into now. At a certain level of writing, you may be visited by prophecy, as happened there.

If Mark were Lakota, his name might have been Respects Nothing. He was uncaught by any orthodoxy, would talk back to any authority, and often voiced the unwelcome but true intrusive thought. One thing he had no respect for was death. His health problems resembled a fall down a long flight of stairs, with pauses at several landings. He endured treatments that were painful and enervating in order to get time to work, and to fish with Jeb, Reid, Tim, and Paul—his wonderful sons—and with Zoe and William, his grandchildren, and with John McPhee and David Remnick and me.

Fishing the Delaware and the Housatonic Rivers with Mark was some of the best fun of my life. Now he lies in an upstate cemetery thirty miles from where we used to fish, and I'll be able to visit the grave when I go back.

When I got off the phone with Mark, I always said, "I love you, you're a gorgeous person." I did, and do, and he was. ♦

Reporting & Essays

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The Billionaires' Vagina Club

With her motto, “Sexual health is health,” Dr. Sally Greenwald aims to optimize orgasms for the women of Silicon Valley.

By [Melanie Thornstrom](#)



Greenwald focusses on what she calls a woman's "sexspan." She tells her patients, "If having orgasms in your eighties is on your bucket list, you have to work at it." Photo illustration by Sarah Illenberger

Sally Greenwald likes to say that she is the happiest gynecologist in America. She is certainly sought after, having attracted a following of high-powered Silicon Valley women of a certain age with her motto, “Sexual health is health.” Recently, Greenwald was the guest of honor at a small luncheon hosted by one of her patients, the tech entrepreneur and philanthropist Gisel Kordestani, in Atherton, California, one of the wealthiest towns in the country. The event took place in the stately, pillared home Kordestani shares with her husband, Omid, the former chairman of Twitter.

Kordestani, who has long, light-brown hair and was wearing platform Prada sneakers, apologized that her co-host, a fellow Greenwald patient named Nicole Lacob, who runs the philanthropic foundation of the Golden State Warriors (her husband, Joe, is the majority owner), wasn't able to come at

the last minute. She had flown to Paris to attend the *début* runway show of Matthieu Blazy, the new designer for Chanel.

The other women in the room murmured in understanding. Ranging in age from late thirties to mid-fifties, they were accomplished, confident—the kind of women for whom “Fake it till you make it” had long ago morphed into “Make it better and better.” When I moved to Silicon Valley from New York, fifteen years ago, I was astonished by the widespread conviction that everything—even one’s body, aging and mortality notwithstanding—must and will improve, like each new iteration of iPhone.

Greenwald, who is forty and has a spirited demeanor, perched on the edge of a chair, ready to field questions. “Use my brain!” she told the women, her high blond ponytail bouncing. “Meander me wherever you want to go.” The doctor, who grew up nearby, in Los Altos, fit in easily with the group. Her sparkling Jimmy Choo flats peeked out from under a long, rainbow-striped dress, and her language immediately signalled that she shares Silicon Valley’s obsession with optimization and innovation—“perseveration on perfection,” she calls it, a determination to bring the entrepreneurial mindset into every aspect of one’s life. The women at the lunch were not content to enjoy good health; they were looking for a way to level up. (The latest term for this is “healthmaxxing.”)

One difference between Greenwald and the dozens of biohacking doctors who treat the area’s billionaire tech bros is that she wants to achieve this goal, in part, through orgasms. (Lube, she likes to say, is as essential to one’s routine as sunscreen.) Rather than concentrate on increasing life span, she focusses on what she calls a woman’s “sexspan.” Her concierge gynecology practice—which is situated on the Stanford campus, where she is an assistant clinical professor—views women’s sexual health as a way to well-being and, ultimately, to longevity. She makes the case that “bro science” culture typically relies on research that excludes data about women. (Rockstar longevity doctors like Peter Attia and Andrew Huberman come to mind.) Part of Greenwald’s approach involves using cutting-edge hormone therapy, among other interventions, in unconventional ways. She also counsels women to have more and better sex, devising stratagems like Fuck-It February, a prescription to have sex as often as possible in the year’s shortest month. New patients pay tens of thousands of dollars a year to join

her practice, if they manage to get off of a wait list. Some have sought Greenwald out after hearing her speak at private gatherings hosted by fans like Priscilla Chan Zuckerberg. Last fall, after a fierce auction, Greenwald received a seven-figure book advance from HarperCollins. Sheryl Sandberg is writing a blurb.

Sandberg described Greenwald to me as “brilliant, energetic, passionate,” adding, “She’s unafraid to broach topics in ways that other people, I think, have been afraid to do.” In the past few years, as her wait list blew up, Greenwald started jokingly referring to her practice as the B.V.C.—the Billionaires’ Vagina Club. Some patients found the name amusing, but the billionaireesses apparently did not. Greenwald stopped using it.

Seated in Kordestani’s grand living room, the lunch guests told Greenwald about their frustration with their primary-care doctors, whom they viewed as old-fashioned disease chasers. “I like biohacking,” Kordestani said, but complained that her doctors were dismissive. The women wanted physicians who were disease preventers, and doctors who were attentive to the difficulties wrought by menopause, which have been increasingly found to leave women vulnerable to other ailments. (I first met Kordestani at a book group that was reading “All Fours,” Miranda July’s 2024 novel about a woman teetering on the estrogen cliff of perimenopause, ditching her husband and child in the process. The book-club members roundly disliked it, indignant that the character was being driven mad by perimenopause, as if hormone-replacement therapy didn’t exist. Did July just have a bad gynecologist?)

Outside Kordestani’s glass doors, a gigantic inflatable flamingo moved back and forth in an infinity pool. Inside, one young woman asked Greenwald, “How do I know when I’m in perimenopause?”

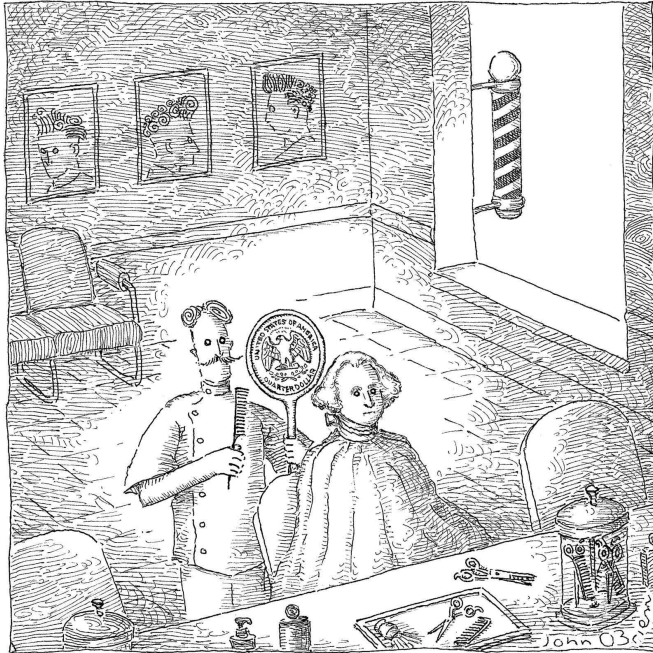
“My goal is that my patients never know,” Greenwald answered, segueing to her enthusiasm for hormones. She told the group that she had several Olympic athletes as patients. As soon as their schedules for the 2024 Paris Games were released, she used an ovulation medication to synch their cycles so that they would compete on days nine through fourteen—the time when estrogen peaks and women are suffused with energy. In perimenopause, she said, the pattern shifts: fluctuating hormones can cause days and days of

angsty P.M.S., followed by spikes of anxiety. “Fortysomething women can get quite a reputation,” she noted. “These are badass women like yourselves, who run companies, run households with a lot of children, have M.B.A. careers and whatnot, and all of a sudden they’re freaking out at everyone.

“I’m super excited about using hormones, maybe forever,” Greenwald went on. “I will humbly bow my head when the data come out to prove the opposite, but I really think we’re onto something.”

When she is talking about scientific innovations, Greenwald warns her listeners that she is prone to “go super dorky.” Although the women she treats are data-driven, they are impatient—unwilling to wait a decade for randomized controlled trials, then years more for prescribing habits to change and insurance companies to catch up. She makes treatment recommendations based on a combination of what she calls “biological plausibility”—a theory of why something would work—and new data. “There’s a lot of cool new meds on the market that I’m super excited about that few doctors are using,” she said. “But I am!

“I want to be very clear about when I’m speaking as a U.C.S.F.-trained physician with lots of randomized control trials behind what I’m saying, versus when I’ve entered the ‘woo-woo’ world of concierge medicine and I’m going into anecdotal evidence and observational studies,” she said, fluttering her hands. “You guys are a very smart room here.”



Cartoon by John O'Brien

A woman mentioned that some of her friends were paying for their young daughters to freeze their eggs, as a graduation present.

“Super interesting! That’s a great idea!” Greenwald said. “This is my favorite example of population-based health versus concierge health.” Research shows that most women who freeze their eggs before the age of thirty-seven don’t end up using them. “But suppose we don’t care what the cost is?” Greenwald asked. The calculus becomes simpler. Everyone knows twenty-one-year-old eggs are better.

Cori Bates, a trustee of the Asian Art Museum who is married to the C.E.O. of the software company Genesys, leaned forward. Bates was wearing a scarlet dress and kitten-heeled flip-flops, and she spoke up to say that, after learning that she was at a high risk for ovarian cancer, she funded a study to assess whether risk for the disease can be reduced by removing one’s fallopian tubes.

Greenwald praised the study, and Bates took the opportunity to ask if the doctor would take her as a patient. Greenwald beamed: “I’m so, so flattered. I would love to. But I’m completely full.” She promised to try to find room for Bates in the future.

Someone asked Greenwald how she felt about hormone-monitoring apps, and she answered that the technology is not there yet and that the apps on the market are unreliable. She addressed her audience: “Please, make a good app for teen-age girls—and for perimenopause and menopause.” The women smiled, then everyone filed into the dining room. Greenwald left before dessert so that she could be home in time to meet three of her four sons at their bus stop.

I met Greenwald three years ago, when she was consulting once a week for the primary-care practice where I was a patient. She strode into the room looking like a television doctor in a chic blue-print dress and flats. (She favors prints, she told me, because they disguise splashes of bodily fluid.) By then, I was skeptical of gynecologists. I had consulted three about insomnia, brain fog, and vaginal atrophy, which made sex painful and left me with recurrent infections, only to be told that the ill effects of menopause are just a “natural stage of life” which could not be medicated away. Then the doctors would bustle out to see their more appealing patients—the cheerily pregnant ones, whose deliveries constituted the bulk of their income and whose babies would be added to the collage of photos decorating their office walls. Greenwald told me that, even though I had been prescribed some hormones, she felt that I had been undertreated, because doctors often use hormones too sparingly; she prescribed five different forms of hormones to me, and my old life resumed.

I texted my middle-aged friends the details of my prescriptions: estrogen patch (estradiol, 0.075 mg.), vaginal ring (Estring, 7.5 mg.), estrogen cream (estradiol, 0.01 per cent), oral progesterone (Prometrium, 100 mg.), compounded testosterone cream (start at 1 mg., then move to 2 mg.—up to 10 mg.). They pressed their own doctors, who hedged or shrugged but eventually wrote the prescriptions for them. When my friends found relief, they were thrilled—and infuriated. Why were they getting this information from me and not their doctors?

For many women, menopause marks a retirement from active sexuality. They may still have sex, in the way a retired athlete may still throw a ball around the back yard, but the drive is gone. Greenwald wants to get these women back into the game, because research shows that a healthy sex life brings a variety of benefits that can contribute to longevity. Her methods

combine treatments with sex tips. “How many sexually active years do you want to have?” she asks patients. “If having orgasms in your eighties is on your bucket list, you have to work at it.” A faltering sex life can also put a marriage at risk, and statistics suggest that people who stay partnered live longer. In the 2024 General Social Survey, nearly two-thirds of married adults reported having sex less than once a week, and twenty-eight per cent answered that they had sex rarely or not at all.

Women’s sexual health has been under-researched, and in recent years federal research money into women’s health issues has been slashed. But menopause is finally having a moment. Last November, the F.D.A. removed the “black box” warning (the agency’s strictest safety alert) against H.R.T., and earlier this month Melinda French Gates announced a two-hundred-million-dollar menopause- research initiative. A 2024 meta-analysis published by the *Bulletin of the World Health Organization* looked at sixty-three studies and found strong correlations among over-all health, well-being, and sexual activity. A study from the same year in the *Journal of Psychosexual Health* analyzed government data and found that women who had low sexual frequency had a forty-six per cent higher mortality risk. It seems obvious that healthier people might be more sexually active, but Greenwald believes that the association is bidirectional: that good sex leads to good health. Orgasms improve mental health, suppressing stress hormones like cortisol and releasing dopamine and oxytocin; women with robust sex lives report less depression and anxiety. Studies show that orgasms also relieve menstrual pain, lower blood pressure, enhance circulation, correlate with better cardiac functioning, and burn calories equivalent to thirty minutes of brisk walking. Orgasms also activate parasympathetic pathways that promote deeper and longer sleep. There are measurable health benefits to orgasms through masturbation, but those benefits compound in partnered sex, which supports satisfying long-term relationships, which in turn correlate with better health and longevity.

“The argument is not: sex equals longevity,” Greenwald said. “It is: sex equals sleep, stress reduction, cardiovascular engagement, relational fulfillment, and pelvic-floor health, and each of those equals health.” She likes to note, however, that sixty years after the so-called sexual revolution, the sex lives of straight women are hardly thriving. Whereas straight men climax in almost all sexual encounters, heterosexual women in relationships

fail to climax a third of the time. The widespread assumption has always been that the female orgasm is innately tricky, but data show that lesbians climax almost as often as men do.

A number of Greenwald's patients told me that they'd never had a gynecologist who thought of sexual satisfaction as a *medical* issue. But she considers sex coaching to be a part of her job, just as an internist might advise a patient with high blood pressure to get more exercise.

There is a girlfriendly "pass it on" quality to the way word has spread about Greenwald's treatments, which include birth-control pills—such as Nextstellis and Natazia—that contain a new natural form of estrogen, not the synthetic formulation in most pills; compounded testosterone cream for a "kick-ass sex-drive booster"; magnesium glycinate for sleep; temperature-controlled mattresses; her preferred vibrator for postmenopausal women (a long-handled model with seventy-to-a-hundred-and-fifty-hertz frequencies); and silicone-based lube. Greenwald has been known to draw diagrams for patients, showing them how to find their G-spot.

"Experts describe sex in biopsychosocial terms," Greenwald told me. "I tell my patients, 'Think bio.'"

Nicole Lacob, the wife of the Warriors owner, started seeing Greenwald over a year ago, when she was fifty-six—just in time, in terms of starting hormone-replacement therapy. (For women more than ten years into menopause, H.R.T. can boost mood, libido, and bone health, but it also presents cardiovascular risks, among others.) "I wish I'd had her when I was forty-five," Lacob said. At the time, she recalled, she was too depressed to even shower: "I was crying over things all the time, and I had no idea that it was even perimenopause." Lacob was seated with her legs tucked underneath her on a fluffy couch in the living room of her Mediterranean-style estate. She has a cascade of dark hair, and wore platform Uggs. (At Warriors games, she is often seen on camera dancing in glittery thigh-high boots beside her husband.) "I think I would probably even look ten years younger now if I had had her then," she said.

Lacob had always liked the house warm—she used to tease her husband that, if he ever wanted to see her naked, he'd need to keep the heat cranked

up. Suddenly, she was always too hot and couldn't sleep because of night sweats. She'd had breast cancer in her thirties and, consequently, had been warned away from H.R.T., which has been thought to be unsafe for cancer patients. But Greenwald explained that there is only a modest increased risk of recurrence. (This is a controversial position and one that many oncologists oppose, in the belief that data are still emerging and that even a modest risk is not worth taking.) She prescribed an estrogen patch to alleviate the night sweats and an estrogen ring, which is placed inside the vagina and releases local doses of the hormone. "Then Sally asked me questions that I never thought to ask myself, um, regarding desire," Lacob said.

"I was shell-shocked the first time," Nathalie Domp , the Italian-born head of a pharmaceutical business and the wife of the tech entrepreneur Chamath Palihapitiya, told me, laughing at the memory of just how graphic Greenwald's questions can be. "The best part is when I go home and repeat all my new learnings to my husband, who looks at me, shocked." She appreciated the way Greenwald "puts it on the radar and says, 'This is part of how you take care of your life.' "

I sat in on one of Greenwald's appointments, with a patient I'll call Kristen, the wife of a celebrity. She had been suffering from depression and mood swings triggered by the hormonal fluctuations of perimenopause. She confessed that she sometimes found "the mental load of being a mom"—she has teen-agers—almost unbearable. Greenwald had been treating her for a few years, and they were, Greenwald said, "making progress as a doctor-patient couple." She had evened out Kristen's cycles with birth-control pills, and Kristen's mood and energy had improved. She'd even picked up some old interests again, like exercise and crafting. But not sex.

Greenwald hears this story again and again. After they became mothers, her patients tell her, their focus transferred from their romantic lives to their children. Kristen's husband's life was full of glamour and limelight, whereas her life was all behind the scenes. He was taking testosterone to keep up his physique; his libido was stronger than ever. But satisfying him felt like one more chore.

Seeing Kristen's shoulders slump, Greenwald smiled. "When you're, like, 'Help a sister out with her sex life,' know that you're in good company." Greenwald said. "Just imagine my very sexy life with my four kids and three jobs!"

Kristen said that when she accompanies her husband on work trips without the kids, she feels pressure to be a sex goddess. “I can tell, after the first night, that he’s thinking, *That was fun, but no one was swinging from the chandelier*,” she said. She missed having “the desire for more spontaneous, fun, crazy sex.” When she was younger, alcohol helped, but these days drinking made her fall asleep more than it loosened her up. “I still like my guy, and I think he’s cute,” she said shyly. “And I want him to only be with me.” But she said that, when he asks if she wants to have sex, it seems that he’s not really asking about her desires. She feels like saying, “No, I don’t. And now you’re annoyed because that was the least sexy answer I could give. But your question was manipulative.”



Greenwald, a mother of four boys, preaches the benefits of the efficient five-minute sexual encounter. Rather than “chandelier sex,” she counsels patients to aim for “super-heteronormative, vanilla, white-picket-fence sex that has data behind it.” Photograph by Maggie Shannon for The New Yorker

Greenwald jumped in, completing Kristen's thought: "Does it look like I want to have sex? I'm sitting here ordering groceries. What about me ordering groceries says, 'I want to have sex'?"

Then Greenwald made a characteristic pivot. In fact, she always tells patients, any time is a good time to have a certain kind of sex—what she calls “efficient sex.” She believes that the best way to improve sex is to drop the romantic notion that it requires a special frame of mind. Better to have a quickie, even if you’re reluctant, because an orgasm can improve your mood, and ultimately your health and your relationship.

“I’m not going to talk about chandelier sex,” Greenwald added. “That’s a good hobby, and, if you have that, rock on.” She continued, “We’re going to stick to super-heteronormative, vanilla, white-picket-fence sex that has data behind it.” She had suggestions on how Kristen could “curate” her own arousal. Spontaneous desire, she said, disappears in eighty-five per cent of women in partnerships after about eighteen months. But women retain responsive desire—which comes from being stimulated, or from stimulating themselves. “If I were to write a porn for women in our forties, fifties, and sixties, it would start with a vibrator and some lube.” Think of it like exercise, she said: you never want to do it, but, once the trainer arrives and you begin, you get in the mood.

Kristen nodded: “Before my trainer gets there, I’m, like, *Fuck you!*”

Greenwald went on, “But why do you show up? You’re thinking, *Once I start to exercise, I know how good I’ll feel afterward.*” She counselled Kristen to hop into bed with a vibrator, then, after warming up, text her husband and ask him to come to the bedroom for a few minutes.

“I know that you’re, like, *That doesn’t feel sexy,*” Greenwald said. “But it’s showing up at the party, even if you don’t think you want to go, and then having a good time.” She continued, “I know it’s awkward in the MeToo era.” She realizes that her message—“Just try it, even though you don’t want to”—can sound retrograde, misconstrued as an update on “Close your eyes and think of England.” But, for a middle-aged woman in “a committed relationship with a nice partner,” she believes that it’s a sound approach to improving a couple’s sex life.

Greenwald told Kristen that she needed to do some cognitive reframing. “Instead of asking, ‘Do you want me?’”—an epic question, requiring an epic answer—“couples should practice asking, ‘Hey, do you have five

minutes?’ ” Sex, Greenwald often says, begets more sex. Her date-night tip: when out to dinner with their husbands, women should carry some travel-sized lube and pop into the ladies’ room to apply it right before dessert. “It’s hard to walk around with a wet vagina and not feel turned on,” she said.

One of Greenwald’s strategies is getting patients to practice inwardly narrating the sequence leading up to sex: *He’s rubbing my back. I’m breathing a little faster. I feel warm.* The goal is to banish the more common inner-mom monologue: *He’s touching me. Did I remember those snacks for tomorrow? How will my daughter feel if no one asks her to prom?* As with meditation, Greenwald said, patients gain control over their thoughts: “If it’s a Tuesday night before sports pickup, this is how you create good, efficient bedroom sex with your spouse that doesn’t involve a whip and leather boots.”

To keep the routine going, she advises couples to replace postcoital sweet nothings with compliments like “I love how efficient we were!” The idea is to tap into “that reward system where you have sex, you sleep well, you wake up more energized, you guys kiss before he goes off to work, you feel more loved by him, and the next night happens.” Describing why Greenwald’s methods work, Nathalie Dompé reached for an Italian phrase, “*La fame vien mangiando*”—the hunger comes while you’re eating.

It’s no surprise that expensive concierge medical care is better. It’s a particularly distressing example of how, in our society, necessities are becoming luxuries. Concierge care is innately problematic; when the wealthy can buy their way out of public-health problems, it reduces pressure to fix them. Gynecologists typically book fifteen-minute appointments: five minutes to do a breast exam and a Pap smear, five minutes to talk, and five minutes to document the visit. Greenwald gives patients as much time as they need, often an hour or two per visit.

“I think the system is broken,” she told me. Greenwald’s medical career was a form of response, stemming from her own experiences. When the *COVID-19* pandemic hit, she felt herself beginning to crack.

Growing up near Stanford, in a close-knit family, as the granddaughter of a Stanford doctor, Greenwald had a vision of herself being an ob-gyn at the

university—delivering babies at the same hospital where she was born—and having four sons. She became chief resident at the ob-gyn program at the University of California, San Francisco, and married a pediatrician, Daniel Imler. Throughout the eighty-hour workweeks of the residency, where she was permitted to take off only a month after the birth of her first son, she found herself going against protocol to co-sleep with the baby, because it was the only time she saw him. Passionate about health equity, she did a master's degree in public health at Tufts, but she became disenchanted with the population-based approach it taught: she saw suboptimal high-volume care, in which the health of the individual was sacrificed to cost considerations, everywhere.

Medicine suited her personality. “I’m a little hyper, driven, anxious,” Greenwald told me. “But it works in medicine, because I am always worrying about excellence, perseverating on my patients.” In a public-health setting, that was the wrong mind-set. Only easy patients got good care, because time-consuming workups or expensive tests—let alone innovation—were prohibitive.

At thirty-two, Greenwald joined a Stanford health-care practice made up of four women and a man in Menlo Park. There, she saw up to twenty-five patients a day. She and her partners were perpetually overworked, and there was tension in the group because they had to cover for one another if they or their kids were sick. Back-to-back overnights left Greenwald exhausted and depressed.

When she was thirty-six weeks pregnant with her second child, Greenwald was in the middle of a seventy-two-hour shift—subsisting on saltines and cranberry juice between deliveries—when she went into premature labor. “He went home at four pounds,” she recalled, of her son. When *COVID* began, she discovered that she was pregnant again. Wearing an N95 mask so that she could nap in the hospital during patients’ long labors, she was anxious about her fetus.

At one point, the head of the practice asked Greenwald to pay special attention to a wealthy patient, and so, for once, she was able to practice medicine the way she wanted to, spending extra time on the case and catching issues that might have gone unnoticed. Greenwald wasn’t able to

give herself, or her regular patients, that level of care. After overnight deliveries, she would find herself driving home too fast in order to get there in time for her husband to leave for work; she would remind him to give the boys the breakfast she had prepped the night before and to get them to preschool, and then she'd try to squeeze in a twenty-minute nap before returning to the office. One day in 2022, she fell asleep at a red light with her boys in the back seat and finally decided to quit her job. But she needed to work, and she still wanted to practice medicine. By then, her husband was the medical director of Stanford's Pediatric Emergency Department, but they had little savings. Their money had gone into a cozy Craftsman house in Menlo Park, and repaying medical-school debt and the salary for their nanny ate up the majority of Greenwald's take-home pay.

Greenwald decided to open her own concierge gynecology practice with no obstetrics, initially setting a thousand dollars a year as a membership fee. Hers was the only such practice in the area, and she didn't know if it would work. On the side, she consulted for primary-care concierge practices and spent three days a month at Santa Clara Valley Medical Center, teaching and delivering babies for underserved populations, a commitment that she has continued.

At her own practice, she decided that patients would never wait, that she would be available from 6 *A.M.* to 9 *P.M.*, and that she would never have to cancel because she was delivering a baby. The office would provide spa-fluffy robes, heated towels, and premium teas. She would answer texts and calls at all hours, including on weekends, and would travel to patients' pool houses for a quick infection check or meet them on the tarmac when they flew, in a private jet, up from L.A.

Greenwald obtained medical licenses in New York and Florida so that she could take care of patients' extended families. She sees her patients' adolescent daughters, arranging anesthesia for IUD placements (prescribed to treat painful periods)—something that is rarely done, even though the insertion can be excruciating. She advises inexperienced teen-agers that lubes containing additives are dehydrating and create micro-abrasions that can increase the transmission of S.T.D.s, and she helps them navigate hookup culture, in which choking, polyamory, and anal sex have become mainstream. Ella Hale, a recent college grad who started seeing Greenwald

at fifteen, told me, “She said, ‘If you feel any psychological or physical discomfort during sex, I want to know.’ ”

Greenwald tracks all of her patients’ metrics: blood tests to check that hormones are balanced; Oura rings to show resting heart rates and sleep scores; glucose monitors to keep tabs on blood sugar; food-tracking apps for logging daily protein targets. She sends them to get thousand-dollar Grail blood tests (the fact that they are not covered by insurance is not an issue for the B.V.C.), which may detect cancer years before it appears on any scan, and full-body MRIs (two thousand dollars), and she reminds them to pick up sunscreen on vacations abroad, because European formulations are more effective than what we can get here. Because there is heart disease in Gisel Kordestani’s family, Greenwald referred her to a preventative cardiologist who had her take the Cleerly cardiovascular test, an A.I.-based analysis of heart CT scans which yields new insights. Kordestani told me that she didn’t care about wasting time or money: “Now I know my cardio risk is zero.”

Greenwald focussed on optimizing her own well-being, too, waking at four-thirty to work out, and then answering patient e-mails while sipping bone broth or water mixed with psyllium husk (for fibre) and creatine. She attended conferences and pored over the latest research.

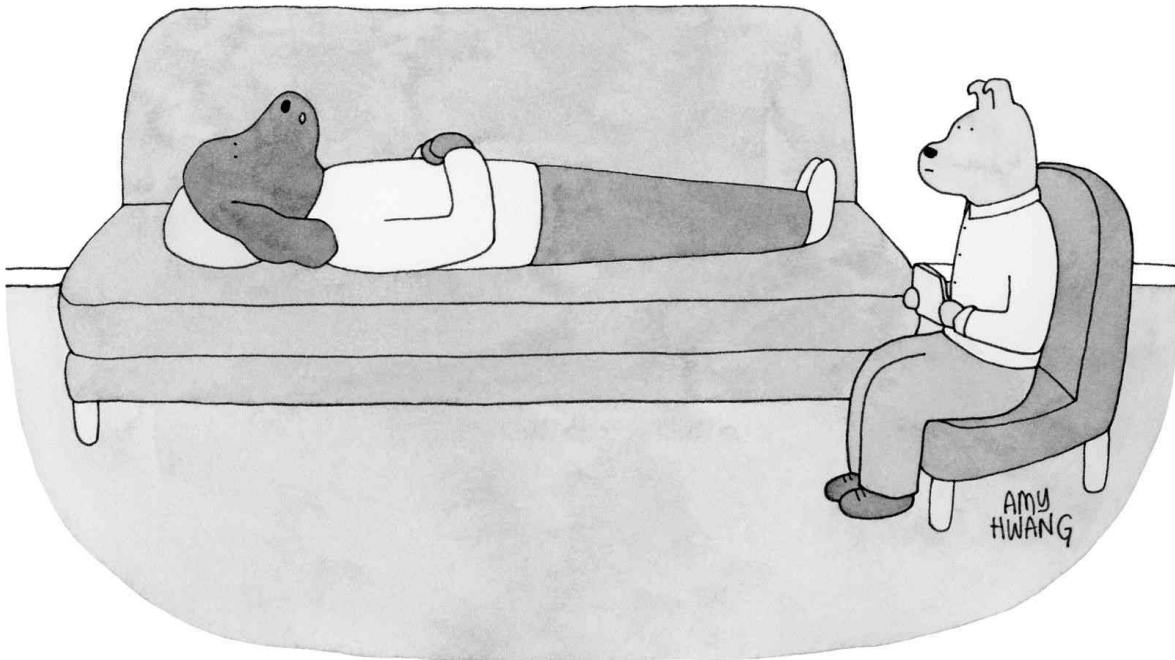
She had a fourth son, and her time felt more valuable than ever. She raised her prices a few times. Consultations became a thousand dollars, then two thousand. She decided to accept new members with tiered annual fees, starting at ten thousand dollars and exceeding thirty thousand for the most extensive care, which can involve assembling a specialized care team to monitor complex comorbidities. Then she stopped accepting the ten-thousand-dollar patients. What was the market price for this kind of care, she wondered. The influencer and doctor Peter Attia has a longevity practice with patients who pay six-figure membership fees. Clearly, there was room to keep going up.

One afternoon, I sat in on another appointment of Greenwald’s, this time with a patient in her fifties who had been diagnosed with early-stage breast cancer. The woman, whom I’ll call Lily, was agonizing between having a lumpectomy, which would entail taking the estrogen-blocking drug tamoxifen for five years, or getting a double mastectomy. Her oncologists

were hedging, telling her that it was a personal choice. She knew that tamoxifen would curdle her sex life by inducing abrupt menopause; she was already struggling with symptoms of perimenopause, which her primary-care doctor had dissuaded her from treating.

Lily had an active life of golfing, skiing, and boating with her family. In the examining room, she sat on the table with Dior sunglasses perched on her head, wearing a ruffled, checked blouse and jeans—a California trad-wife look. Greenwald got her talking about her life, and her hopes for grandchildren one day. After a moment's thought, Greenwald told her to do the lumpectomy—and the tamoxifen, which would cut the chance of cancer recurrence in half. She surprised Lily by recommending that she simultaneously do hormone therapy—a vaginal application, such as estradiol cream—to mitigate her symptoms. “Take some tamoxifen and know that, if you need a partner to respond to whatever shitty symptoms you’re dealing with, I’ve got you,” she said. “We’ll be creative.” She offered to send Lily a new academic paper on H.R.T. and cancer risk. If Lily ended up losing her sunny disposition, Greenwald said, they could explore solutions, such as Duavee, an estrogen replacement now in trial with high-risk women; contrary to conventional wisdom, emerging data seem to indicate that this form of estrogen is associated with a decrease in breast cancer.

“Duavee is blowing my freaking mind,” Greenwald said.



"Am I a good dog who sometimes does bad things or a bad dog who sometimes does good things?"
Cartoon by Amy Hwang

While mainstream oncology supports using local vaginal hormones that have little systemic absorption (although some doctors remain reluctant), the use of systemic hormones for women who have had breast cancer is contentious. Even for those who haven't, hormone therapy is "a Pandora's box," Mark Moasser, a U.C.S.F. oncologist and cancer researcher, told me. "It remains a field with different data sets and studies and different opinions and judgments. There shouldn't be a 'one size fits all' guideline. It's a matter of judgment between the doctor and the patient. I can't say it's 'safe' because we simply don't know."

Having dealt with Lily's cancer questions, Greenwald pivoted to her sex life. Was it A+? Greenwald suddenly included me in the conversation, breezily asking, "Do you two have internal orgasms?" Lily and I both looked at her, agog. Greenwald has a crisp, matter-of-fact manner. With her practiced explanations and gestures, she could almost be a flight attendant explaining preflight safety protocols.

Only one in five women climax through internal stimulation alone, Greenwald said. Many never find the mysterious G-spot. But the G-spot becomes increasingly important in middle age, because women can't rely on their old sexual responses to stay the same. "The clitoral nerve has two different nerve fibres, right?" Greenwald went on, as Lily and I nodded

dumbly. This nerve root runs both internally and externally, ending in the clitoral nub, and it contains two different types of fibres, one of which responds to light touch and warmth, and another that senses vibration and pressure. But as we age, she noted, the fibre that responds to light touch and heat degrades, whereas the one that senses vibration and pressure maintains sensitivity. After menopause, with insufficient estrogen to nourish the tissue and facilitate blood flow, the clitoris starts to lose its sensitivity, resulting in poor-quality orgasms. (I had never heard of poor-quality orgasms.)

“So you’re going to come in here at some point, be it on your tamoxifen journey or when you’re eighty-five, and say, ‘We’ve done this position for forty years, and now it isn’t working.’ ” Greenwald counsels women to practice finding the G-spot and stimulate the deep internal nerve fibres that remain robust.

“Do you have a vibrator?” she asked Lily. The answer was an enthusiastic yes, and Greenwald offered to provide instructions for exercises to try at home. If the G-spot proved elusive, Lily should come back and Greenwald would show her where it is. Wrapping up, Greenwald said, with feeling, “I support you and your energy and your sex life and your orgasm quality and your vagina and your pelvic floor.” She looked Lily in the eye to make sure she heard it. Her parting words: “I’m on your team.”

With four boys under the age of nine, Greenwald has organized her family life so that it functions like a well-run startup. One night, she invited me to her house for dinner. Wearing sweatpants and a T-shirt, she jiggled her plump baby and showed me a mudroom where each of the boys has a cubby near a refrigerator that holds their lunches. “I am obsessed with systems,” she said. Outside the mudroom is a large touch screen that her sons are supposed to stop at and check off tasks on a schedule (“Eat breakfast and clear dishes.” “Sit on toilet and count to 30!”); each time, the screen rewards them with an explosion of confetti. Greenwald coaches the boys on social behavior: “Look at Melanie when you talk! Can she see your eyeballs?”

The house is colorful, with wallpaper covered in giant leaping zebras. One of her sons showed me his pet rabbit; another introduced a large parrot squawking in a cage. There is also a lizard, four hermit crabs, a toothed frog, and a wall-size fish tank that the kids built with their father; its filtration

system is based on the workings of a kidney. Greenwald's husband had just returned from a monthlong stint in Rwanda, teaching pediatric urgent care. Their nine-year-old had joined him, playing the role of patient. Like his wife, Imler is plainspoken and believes in cutting to the chase. "Sometimes I hear doctors cloak the news of a child's death in so much flowery language that the parents don't grasp what's happened," he told me.

Imler said that he admired how Greenwald likes to talk about "difficult things with people because it feels real"; he is amused by what he described as her impatience with fiction and art. "Her response to 'Phantom of the Opera' is 'There's a phantom—why are we singing about it for hours? Put him in jail!'" He thinks that part of her appeal to patients is that she is unfazed by their net worth. "Sally will not bow to you," he said. "And people love that."

Greenwald regularly gets together at a restaurant with concierge-doctor friends. At a recent gathering, over drinks and meze, they talked about the coolest new things they are doing for their patients. Justin Lotfi, a primary-care physician to a fortunate few, offered that he'd been experimenting with a nasal procedure using an anesthetic that, with repeated applications, can break a cycle of migraines.

"I love being in concierge medicine, because practicing at the top 0.001 per cent—that's where innovation happens," Greenwald told me. "I get to sit around and think about these things." She went on, "I am always, like, 'What's next? What's better? Why don't we try this?'" She added, "But at some point the system needs to evolve so other people can access this kind of care."

Midi Health, a virtual-care platform for women in menopause and perimenopause which was founded in 2021, offers the same sorts of hormone treatments that Greenwald does, including options for cancer patients, and it accepts insurance. Joanna Strober, the entrepreneur behind Midi, supports Greenwald's approach, but told me, "I am sad that women believe they have to pay all that money to get menopause care." She added that Midi does most everything that Greenwald does "without the in-person component." When Strober was raising money to start Midi, male venture

capitalists told her that menopause was “niche”; now Midi has a billion-dollar valuation.

To help close the gap in medical care, Greenwald is trying to publicize her ideas, through her book and public speaking. Last fall, she was a guest on Peter Attia’s popular medical-and-life-style podcast, “The Drive.” When she got to the importance of lube for vaginal health, Attia pushed back, repeatedly questioning whether young women needed lube, since reduced friction might detract from a man’s enjoyment of sex. Greenwald held firm. The episode ended up having “The Drive” ’s highest ever Day One-download count on Apple metrics. Three months later, when a batch of files about Jeffrey Epstein was released by the Department of Justice, Attia’s name turned up more than seventeen hundred times. In one bit of banter from 2016, Attia e-mailed Epstein, “Pussy is, indeed, low carb. Still awaiting results on gluten content, though.” In another exchange, Epstein muses to Attia that he’s not sure why “women live past reproductive age at all.”

After the Epstein news broke, the *Post* published a snarky piece about Greenwald’s episode of “The Drive,” and internet critics piled on. Greenwald declined to comment on Attia, with whom she shares patients, but said that she is eager to engage with the manosphere, including on platforms associated with toxic masculinity. That, she said, would be “a great way for me to push this agenda to an audience that is not typically listening. If there are chances to spread my message that sexual health is health, I’ll take that opportunity every time.” ♦

An earlier version of this article inaccurately contextualized a quotation from Mark Moasser.

By [Stephen Witt](#)

On a recent sunny day in Silicon Valley, I visited the industrial headquarters of 1X Technologies. Security was tight, so I had to put a sticker over my cellphone's camera and talk my way out of signing an N.D.A. before I was brought into an enormous space to meet Neo, the company's home robot.

Neo stands five feet six and has no facial features except for two black cameras in place of eyes. The robot is a humanoid—its design is inspired by the human form—and its proportions are a blend of those of the median American male and those of the median American female. But Neo has no skin. Instead, it wears a beige nylon turtleneck bodysuit, gloves, and padded shoes over a see-through carapace. Under that is a skeleton made up of more than a hundred whizzing motors and cordlike artificial tendons that control Neo's limbs. Neo's cozy, minimalist aesthetic allows it to blend into the background. If it served me an espresso at a café, I'm not certain I would look up from my phone.

The robot weighs just sixty-six pounds, and I was able to pick it up in a bridal carry. It communicates through a speaker in its chest, using several different voices; the default one is in a calm but authoritative masculine register, an A.I.-modulated mixture of several voice actors. Neo can talk, listen, and respond to commands. In its skull is a squarish motherboard, roughly the size of a slice of bread and studded with white capacitors that look like teeth. This is Neo's brain. Using A.I., it tries to translate your words into physical actions. If you ask Neo to pick up a cup from a table, Neo will attempt to pick up the cup. If you ask Neo to put the milk in the refrigerator, Neo will attempt to put the milk in the refrigerator. But Neo doesn't always succeed at these tasks, and no one really knows what it can or cannot do.

The robot is the creation of Bernt Børnich, the C.E.O. of 1X Technologies. Børnich, a forty-two-year-old Norwegian, has been obsessed with robots since he was a child. His firm used to be called Halodi, and it sold wheeled security robots from offices south of Oslo. But in 2022 Børnich relocated to Silicon Valley, changed the company's name, grew out his hair, and started wearing designer streetwear. Neo was announced shortly afterward.

Børnich believes in the supremacy of the human form. “We are incredibly well designed,” he has said. “I would argue this is kind of, like, why we won evolution.” The environment in which humanoids will operate is already built with able-bodied humans in mind. To open a door handle, you need a hand; to climb a flight of stairs, you need feet. Many roboticists are now working on humanoids, but Børnich takes biomimicry further than his peers: Neo uses tendons where other humanoids use motors, and Neo’s motherboard is in its head, while most humanoids have it stored in the chest. Børnich says humanoids have the potential “to do almost anything.”

Mention Neo—or Børnich—to other roboticists and you will be met with a slight pause. Børnich’s technical foundation is solid, but many people feel that he’s getting ahead of himself. Compared with the A.I. that generates language and images, the “physical” A.I. that runs a robot remains underdeveloped. “The world still doesn’t have a ChatGPT equivalent for a robot,” Deepu Talla, who oversees robotics at the microchip designer Nvidia, told me.

“I think home comes later,” Carolina Parada, the head of robotics at Google DeepMind, said. “I don’t think it’s a lack of interest, or even capability.” Rather, she explained, it was a question of safety. Jeff Cardenas, the C.E.O. of the industrial-humanoid manufacturer Apptronik, told me, “Around small pets, around small children, there’s still work to be done.”

Børnich agrees that Neo should not be used around small children. Still, he is keen to test consumers’ tolerance for what he calls “robotics slop”: tasks that are clumsily executed. For example, in a demonstration for the *Wall Street Journal* in October, Neo took longer than a minute to retrieve a bottle of water. Although Neo has a list price of twenty thousand dollars, more than ten thousand customers have already placed a deposit for one.

The customers are who you’d expect—wealthy early adopters from the Bay Area, Los Angeles, and New York City. Will DePue, a former OpenAI employee, lives in San Francisco with tech-enthusiast roommates. “I think our house has three orders on humanoid robots,” he told me. “This is kind of like the new iPhone.” Sometime later this year, workers at 1X’s factory, near Oakland, will pack a Neo into a container that looks like an oversized AirPods case, load it into the back of a truck, and deliver it to DePue’s

residence. “Neo will probably not be a perfectly polished product, but at least it’s, like, an early version of what’s to come,” DePue said.

At 1X’s headquarters, my guide was Dar Sleeper, the head of product and design and Børnich’s informal second-in-command. Sleeper, who is twenty-seven, has already had a memorable career. After college, he joined the fashion brand Yeezy, then moved on to industrial design at Tesla. In other words, his first boss was Kanye West, and his second was Elon Musk. Børnich reminded him of Musk, Sleeper told me, if you “graft in some human empathy.”

In 2024, Børnich asked Sleeper to design a robot that didn’t scare children. Sleeper’s first attempt failed. “Basically, anyone under the age of a hundred was scared of it,” he said. (It had an eerily smooth black face.) His next iteration was better, but it still frightened children under the age of twelve. He added and removed facial features, experimented with skin textures, and went through hundreds of configurations of eye sockets. He settled, finally, on stripping as many features from Neo as he could, then covering it with soft textiles. “Even children under five seem to like it,” Sleeper said. “Babies are still scared, though.”

Sleeper led me through a labyrinthine industrial space at the headquarters. About eight hundred people work there, in roles involving assembly, testing, and research and development. We passed lathes, metal mills, and racks of 3-D printers, all in continuous operation. We passed a safety-testing zone, where technicians in goggles instructed robotic hands to karate-chop ripe cantaloupes. We passed Neo’s staging environments, which included a bedroom, a bathroom, a kitchen, and a dojo with tatami mats on the floor. We stopped at a motion-capture studio, where, off to one side, two dozen decommissioned models knelt together, as if in prayer.



*"Yeah, I see it. I'm on my way as we speak."
Cartoon by Ellis Rosen*

This was where Sleeper had taught Neo how to walk. He told me that he had used his own body as the reference for Neo's gait. (Sleeper stands well over six feet tall, with Vitruvian proportions.) Referring to his engineering co-workers, Sleeper said, "A lot of these guys walk like this," then pantomimed a hunched scurry, his hands clutching imaginary backpack straps. "But I was an athlete in college," he continued. "I actually had an anatomically perfect run."

Sleeper puts the "bro" in tech bro. He played lacrosse at the University of Michigan, and, despite the controversies surrounding Kanye West, describes him as "one of the best creative geniuses of all time." At one point, we were discussing a portable battery pack that Sleeper is designing for Neo. "It's actually a nontrivial problem," he told me. I asked him if he had ever used the word "nontrivial" before moving to Silicon Valley. "Probably not," he said, and laughed. "I didn't say a lot of shit before I showed up here."

Sleeper has built a world around Neo. He oversees an in-house woodshop, which he uses to design sets, props, and furniture, all to mimic the environments that Neo will have to navigate. His plans for the 1X campus involve both a television studio and a forest. On the day of my tour, Sleeper wore a striking beige sweater, which he'd designed himself, in the style of Neo's bodysuit.

I have been reporting on Silicon Valley for some while, which is another way of saying that I have been served a lifetime supply of bullshit. Before my visit to 1X, I had been ready to call bullshit on Neo, too. The company had posted some splashy videos on social media a few months earlier, then gone dark, and I suspected that Neo was just more overpromoted vaporware.

In fact, I was ready to call bullshit on the whole robotics industry. At the Las Vegas Consumer Electronics Show this January, there were numerous robots on display, from many manufacturers. They all seemed clunky and loud. They clanked when they walked. Sure, one robot landed a backflip—but then a piece of it broke off and flew toward the crowd. Nothing there seemed ready for consumer use.

But Neo was different. Neo was elegant. Neo was soft. Above all, Neo was *quiet*. With its padded feet and innovative tendons, it hummed along at twenty-two decibels, about the same volume as leaves rustling in a breeze. I felt a deep and horrible emotion triggered inside me, one familiar to any American—consumer lust. I wanted Neo in my house. I wanted it instantly, badly, immediately.

Sleeper took me to a demonstration kitchen, where Neo effortlessly slotted dishes into a wire rack. I was impressed by the robot's dexterity and fluidity of motion. Then I noticed a tele-operator standing to the side, wearing a V.R. headset, holding controllers, and dictating Neo's every move. The robot I was seeing was a marionette. 1X declined to show me a demonstration of the A.I. that will power Neo. Sleeper told me that previous versions of Neo had tended to fall over. 1X said that this problem had been resolved, but, when I asked Sleeper if the latest models always remained upright, he said, "To say it doesn't fall is, like, a total stretch." Still, when I asked if he believed 1X would meet its deadline for home delivery in 2026, he answered with no hesitation: "It's a promise we've made to the world, and a promise we've gotta keep."

Neo is one of a dozen humanoid robots scheduled to hit the market in the next twelve months. Its rivals include Figure's 03 Humanoid, Boston Dynamics's Atlas, and Apptronik's Apollo. In January, Musk announced that he was converting a portion of a Tesla automotive plant in Fremont, California, into a robot-production factory, and hoped eventually to make a

million Optimus robots there annually. His ultimate objective is to have robots working on the production line, making more robots, which he has called an “infinite money glitch.”

In China, efforts are equally ambitious. Unitree, based in Hangzhou, shipped more than five thousand of its G1 humanoids last year, making the company one of the world’s leading suppliers. The G1 stands about four and a half feet tall and has a hole in its head where a face would go. Its motors are loud, it clomps when it walks—it basically has no charm at all. Starting at around fourteen thousand dollars, it is also one of the more affordable advanced robots on the market, and, because it can run open-source software, it has become a favorite of academic robotics researchers. The G1 has also gained popularity with home-robotics enthusiasts—during the recent N.B.A. Finals, a G1 in a Knicks jersey attended watch parties outside Madison Square Garden.

I first encountered the G1 at the robotics lab of Aaron Ames, a professor of mechanical engineering at Caltech. Wearing a V.R. headset and holding controllers, I moved the robot back and forth a bit, then had it do some light calisthenics. It seemed shaky, almost drunk. I tried to make it clap its hands, but the G1 resisted. “Try it vertically,” a graduate student at the lab suggested. I did, and the robot pantomimed a delicate golf clap, without ever quite touching its hands together.

Later, in his office, Ames spoke for more than ten minutes, at a cadence of roughly two hundred words per minute, delivering a comprehensive takedown of 1X’s efforts. Ames believes that, no matter how sophisticated the engineering is, reliable A.I. for autonomous robots is years away. “I don’t know how 1X is actually going to get away with it,” Ames said. “I would worry about the legal ramifications when one of those robots falls on a person.”

Many industrial roboticists I spoke with were suspicious of the humanoid push. They told me that there are no standardized benchmarks for progress, and that videos of remarkable feats were often culled from hundreds of takes. In March, I talked with Modar Alaoui, a venture capitalist who invests in humanoid startups. He enthused about Iron, a robot from the Chinese manufacturer Xpeng, whose catwalk-inspired strut is so realistic that

exhibitors have removed panels from its body to prove that it's not a human in a robot suit. "Because of Xpeng's work, I tell people not to work on locomotion anymore," Alaoui said. "Locomotion is solved." The day after I spoke to Alaoui, an Iron robot at a public demonstration at a mall in Shenzhen suddenly seized and fell over—it looked like it was having a stroke. The robot, which weighs more than a hundred and fifty pounds, could not recover, and three human handlers had to drag it away.

To celebrate Lunar New Year this February, Unitree staged a display of dozens of G1s doing a choreographed kung-fu routine. The performance was real but misleading: the robots were executing a preprogrammed script, most likely derived from true wushu masters in motion-capture suits. And robotic martial artists can be dangerous to bystanders, as was recently demonstrated at a Chinese amusement park, where a G1 wearing a clown wig kicked a small child in the stomach. Outside of programmed scripts, robots struggle to perform autonomous actions in uncontrolled environments. "The same robot that can land a backflip might not be able to walk up a flight of stairs," Parada, from Google, told me.

Accidents are not the only concern; Neo might also obey a malicious command. Imagine a young child asking Neo to bash its robotic head against the dining-room table. Guardrails installed in the A.I. would almost certainly prevent the robot from following the command, at least initially. But research shows that such guardrails can be circumvented. If the child were persistent enough (in the way that small children tend to be) and creative enough (in the way that small children tend to be) he or she might succeed in getting the robot to comply.

Robots can also be hacked. Last year, the security researchers Andreas Makris and Kevin Finisterre found a vulnerability in Bluetooth that would enable them to take control of a fleet of Unitree G1s, infecting one after another and "creating a robot botnet that spreads without user intervention," they wrote. Humanoid robots need cameras and microphones to make their way through the world, which creates privacy concerns. Plus, their actions are unpredictable. "Babysitting these robots demands an entire operation team," Jim Fan, a research scientist at Nvidia, wrote at the end of last year. "Mistakes are irreversible and unforgiving."

Still, Parada conceded, putting robots in the home is “the ultimate dream.” Growing up in Venezuela, she fantasized about having a robot do her chores. Several roboticists I spoke with seemed a little disappointed by the current state of the field, compared with the expectations they’d had as children—nothing inanimate one day just started moving, nothing ever magically came to life. Sleeper recalled moments as a child when, he said, he “looked at stuffed animals, and got bummed out about how fucking lame they were.” Technology, he told me, sometimes made him sad: “It’s all just going into a screen, instead of into the *world*.”

After I left 1X’s offices, I took my first ride with the autonomous ride-sharing service Waymo. The Waymo was clean and quiet, and avoided a congested highway by navigating through side streets and frontage roads. But, after delivering me to my destination, the car turned in to a crowded In-N-Out Burger drive-through lane, which can take more than an hour to pass through. I laughed out loud, preparing to report on yet another epic A.I. fail—until the car executed a crisp three-point turn, escaped the line, and drove off.

I later learned that, in uncertain situations like this, a Waymo will sometimes request assistance from a human operator. Professional tele-operation “pilots,” working from the Philippines, oversee galleries of vehicle-camera feeds, alert for construction zones and lines of hungry Californians. Such pilots are also at work in other industries: in Japan, 7-Elevens do “autonomous” stocking using robot arms, which can be controlled from afar by workers wearing V.R. headsets. Human pilots also provide training data for A.I., and Waymo has used at least half a million hours of real-world data to fuel its autonomous-driving model.

A car merely has to avoid hitting things. A robot has to manipulate them without breaking them. 1X previously emphasized tele-operation in its marketing, but this generated pushback from potential customers. Now the company emphasizes A.I., but it hasn’t abandoned tele-operation: part of 1X’s gamble is that its customers will come to terms with the fact that a human might be peering into their home through Neo’s eye cameras. (DePue, the early adopter in San Francisco, told me that he would feel safer with a stranger tele-operating his robot than with a stranger cleaning his home.) Tele-operators will sit next to the A.I. team at 1X’s campus in Silicon

Valley, and when the lighted rings on Neo's earpieces change color customers will know that it is being remotely operated. "If you want a robot to bartend your party, we'll get one of our operators," Børnich said. "It's also useful data for us."

Data is to the A.I. revolution what coal was to the industrial one. A.I. communicates in discrete units of data called tokens. In language processing, a token might represent a few letters. In robotics, a token might represent a trajectory associated with a finger joint. But language models have the text of the entire open internet to learn from, not to mention the (uncompensated) copyrighted work of human authors, lifted from the dark web. There exists no equivalent repository—legal or otherwise—of motion trajectories for joints.



One approach to this problem is to radically increase motion-capture efforts. The German robot developer Neura has put more than a thousand industrial workers in motion-capture suits, and is using these data to train humanoids. "Everyone is scared of a shitstorm, because it's, like, 'Oh, they're replacing your work,'" David Reger, Neura's C.E.O., told me. "It's not! We're getting, like, a lot of data." This didn't seem to address the job-loss question, so I pressed Reger on what would happen to the labor market if his efforts succeeded. He straightened his posture and insisted that, in the next few

years, the U.S. would need these robots. “You will have a decrease of working labor, and then you will feel it,” he said. “Right now, *you* still don’t feel it, but in Europe, we feel it a lot. We think everyone disappeared somehow.”

Even if the entire population of Earth donned motion-capture suits, it would take decades to generate the amount of data that was used to train ChatGPT. Another approach is to train robots using videos from head-mounted cameras which have been posted on YouTube. With a little engineering magic, these “egocentric” videos can be turned into joint-trajectory data sets. More advanced video models might be able to extrapolate data from third-person views of real-world experiences such as sports games and cooking shows.

Some robotics companies don’t build hardware at all. Instead, they are focussed on helping existing robots navigate the physical world. About two and a half miles north of 1X’s headquarters, I visited the startup Skild AI. The two firms could not be more different. 1X’s robots are exquisite objects of high design. Skild’s lobby looked like an emergency room, with deactivated Unitree robots hanging from cranes and lying on tables. One was on a gurney; another sat in a wheelchair.

There, I met with Skild’s co-founders, Deepak Pathak and Abhinav Gupta, both professors at Carnegie Mellon University. If Sleeper and Børnich are confident marketers, Pathak and Gupta represent a different Silicon Valley archetype: the entrepreneur whose technological convictions lead him to ignore the pressure of public opinion. This was not the type of company that was going to plant a forest on campus.

Skild is attempting to build a “general purpose” physical A.I.: a single brain that can be slotted into any body. This brain should be able to control not just a humanoid but also a set of detached arms, a robot animal, or a wheeled cart. Pathak and Gupta have to stress-test these robots, so they spend a lot of time kicking them, punching them, and generally harassing them—at one point, they showed me a video of a man using a chainsaw to cut off the legs of a robotic dog. Within seconds, the dog began walking on the stumps.

I did not initially understand why Skild was treating its robots this way. (Nor did I understand why the chainsaw video was scored to hair metal.) Pathak explained that the ability to compensate for injury is one of the most important aspects of generalized physical intelligence. In Skild's workshop were dozens of partially assembled robots, from a variety of manufacturers; one was wandering around without a head. Midway through the tour, we stopped in front of a robotic dog. One of its eyes was disabled, and its wiring was exposed; an employee was kicking it repeatedly. I was invited to join in.

"Kick it like how you would kick a human," Pathak said.

"I wouldn't kick a human," I said.

"Don't overthink it," Pathak said.

After a moment's hesitation, I planted my heel on the robot dog's body and shoved it with force. The dog barrel-rolled across the showroom floor but was upright within half a second of coming to a stop.

Later, in a conference room, Pathak and Gupta explained that their seemingly cavalier attitude toward robot welfare was actually motivated by concern for human safety. (Better that a human kick a robot a thousand times than that a robot kick a human once.) Right now, they don't think that humanoids are safe enough to use in residential settings—in fact, Pathak refuses to have one in his home. "People are using appearance as a way to misguide the public," Pathak said. "If you make a robot humanlike, you expect it to have humanlike capabilities. But technology is far behind that." "

Skild is one of the best-funded robotics startups, but much of its capital is spent on buying computing power from data centers. This is another way of saying that most of its money eventually goes to Nvidia. Nvidia, which has a five-trillion-dollar market capitalization, is the most valuable company in the world. In fact, adjusting for inflation, Nvidia may be the most valuable company in human history; its rival for the title is the Dutch East India Company. Jensen Huang, Nvidia's C.E.O., has cosmic ambitions. "Wherever the universe has structure, we can turn that into A.I.," he recently said.

Nvidia moved into robotics in 2014. Around this time, Huang recalls, he watched a neural network teach a robotic arm to shoot a hockey puck; he immediately calculated a rough estimate of the eventual size of the robotics A.I. market. Soon, Nvidia was touting virtual-reality simulators to train robots, and microchips to deploy in robot brains. (The simulators allow developers to experiment with surface friction, mimic faulty joints, and even alter gravity.) These early initiatives struggled at first, but Nvidia was able to establish what a former employee called “vender lock,” making robotics manufacturers dependent on it. Every single robot I encountered while reporting this article ran on an Nvidia microchip, and every single one had trained in an Nvidia digital gym.

Nvidia has also invested in many startups, including Figure, Neura, and Skild. This has produced accusations of “circular” dealmaking from skeptical investors, since Nvidia is providing funds to its own customers, who then use that money to purchase its hardware. Michael Burry, the investor made famous in “The Big Short” for betting against the housing market in the late two-thousands, today warns of “suspicious” revenues at many A.I. companies, including Nvidia. Huang has called such claims “ridiculous,” and defended the investments, saying that they are necessary to expand Nvidia’s “ecosystem.”

Nvidia’s robotics microchips are different from the ones they sell for A.I. training. They are “edge” chips, meaning that they do their processing in the robot instead of in a remote data center. The human brain uses about twenty per cent of the energy that the body generates; an Nvidia edge chip can drain as much as sixty per cent of a robot’s electricity supply. “It’s not the motors that take up most of the battery,” Cardenas, the C.E.O. of Apptronik, told me. “It’s actually the compute.”

But, if a robot needs more brainpower, it can draw it from a local server via Wi-Fi. A robot’s A.I., like most computer software, is regularly upgraded via the internet; robots get smarter every week. Most importantly, they have a hive mind. “Whatever one robot learns, the others all learn at the same time,” Børnich said.

This means that the wider the release of humanoids, the more skilled they become. If a thousand humanoid robots are sent to fold towels, they’ll learn

from one another's mistakes, and will adapt far faster than a single robot would. This network effect may account for 1X's eagerness to get robots in the home. "There is definitely a data flywheel there," Børnich said.

"Robot" is one of a small number of words that have entered the English language via Czech. ("Polka" and "howitzer" are others.) The term was first used in Karel Čapek's play "Rossum's Universal Robots," from 1920. In the play, an industrial robotics conglomerate has transformed the world economy: robots do most of the work, and they are also deployed as soldiers. At the same time, the human birth rate declines to zero. Eventually, the robots take over, and kill almost everyone on Earth.

Any extended discussion of robots, I have found, will eventually reference the apocalypse. Even in more positive treatments, such as Isaac Asimov's "I, Robot," the androids end up taking over. When I broached the subject of the robot revolution with Børnich, I sensed a certain fatigue. "Look, they won't seek revenge on us," he said. "We can wipe their minds."

Perhaps the fearful visions of Čapek—not to mention those of James Cameron, in his "Terminator" movies—have tainted the West's cultural perception of robots. David Reger, Neura's C.E.O., told me that American journalists ask him anxious questions about robots, but South Korean journalists just seem impatient. "They're tired of waiting!" he said. "They ask, 'When is it coming?'" Reger's observation is supported by data. A Stanford University analysis of opinion polls across three dozen countries found that respondents in China, South Korea, Thailand, and Indonesia were the most excited about consumer applications of A.I. Respondents in the U.S. and Canada were the least.

Most of the roboticists I spoke with seemed more excited by the industrial applications of humanoid robots than by the prospect of their doing domestic chores. ("You'll see them in McDonald's first," Rev Lebaredian, who runs physical A.I. simulation at Nvidia, said.) Apptronik's Apollo humanoid—porcelain white, with knobby limbs and thick, sturdy legs—has been used in production at European automotive factories owned by Mercedes-Benz. For now, the robot is limited to activities such as picking out parts and loading conveyors. It needs to be fairly strong to complete this task, which means

that a malfunction might hurt someone. For this reason, the robots are fenced off from the rest of the workers.

Humanoids make up only a tiny fraction of the five million or so robots currently deployed in factories worldwide. Mike Haley, a senior researcher at Autodesk, a maker of industrial-design software, has spent much of his career programming these specialized, non-humanoid factory bots. He told me that he has never seen a humanoid do anything useful in an industrial setting. Simpler robotics solutions, like detached arms for spraying paint, or autonomous forklifts for moving pallets, are more efficient, cost less, have fewer moving parts, and require less maintenance. “I truly believe that we have a moment a few years down the road from now where we look back at this humanoid thing and go, ‘Boy, that was bloody stupid,’ ” Haley said.

Commercial service robots are already commonplace in China, but these are cheap, special-purpose devices that deliver food or fold laundry. Expensive humanoids may not be needed for these tasks. At Skild, Pathak’s solution is to isolate a robot’s parts to fit the task at hand. “Let’s say you’re putting a robot in a factory like this, and you’re assembling something—should you really spend energy on the legs?” he asked. “I can separate the torso and put it on the table, then use the legs to carry something else.”

It seems that 1X is taking the Apple approach, designing beautiful products from scratch and controlling the entire manufacturing process. Skild is taking the Android approach, writing cross-platform software and running it on hardware made by other companies. Apptronik, Unitree, Tesla, and others—maybe they will be like Nokia, or Blackberry. Or maybe one of them will win.

About seven or eight million years ago, a branch of primates broke off and started developing superior brains, dexterous hands, and the capacity for language. That branch diversified into *Australopithecus*, and *Ardipithecus*, and *Homo erectus*. Eventually, *Homo sapiens* emerged and took over. Surveying the diversity of humanoid forms at this early stage of robotics, I could see something similar occurring. The essential form—two arms, two legs, a torso, and a head—was consistent, but there were many variations. Agility’s Digit has a square head and inverted kneecaps, like a grasshopper. Apptronik’s Apollo uses a classic sci-fi design, with digital readouts on its

mouth and chest. Boston Dynamics's Atlas has a toylike carapace and a glowing, hollow circle for a head. Tesla's Optimus, Neura's 4NE-1, and Figure's 03 have all converged on similarly anonymous and ever so slightly sinister designs: white bodies with smooth black faceplates. One of these is likely the evolutionary ancestor of all that is to come.

The missing link might be the hand. The human hand is one of three pinnacles of hominid evolution, along with the brain and the voice box. It is by far the most capable manipulator in the animal kingdom, and can perform twenty-seven independent motions. Robotic hands lag far behind: we are years away from a robot that can both tie its shoes and shuffle a deck of cards.

At 1X, I saw rows of hands and forearms, denuded of housing and grasping toward the ceiling. Technicians seated along an assembly line were fastening artificial tendons to the fingers to link them with actuators in the wrist. Others attached electrical wire to sensors on the fingertips; the wire would eventually be routed to the robot's brain. Nearby, a finger was being tested for durability. It pointed, curled back, then repeated the movements. A counter on a monitor noted that this action had been performed 2,860,631 times.

But here, again, the mechanical engineering was far ahead of the A.I. This point was confirmed during a demonstration at Skild, where I watched a Unitree robot carefully grab a white ceramic coffee cup from a table and set it upright in a bin. I moved the coffee cup back to the tabletop, and the robot's eyelike cameras seemed to follow. Then I turned the cup on its side. The robot seemed perplexed. It moved its hand about six inches from the cup and began to grab uselessly in the air.

With a disapproving look, a technician set the cup upright, and the robot immediately returned it to the container. I took a blue plastic bowl from the bin and placed it upside down. The robot identified the object but couldn't get a grip on it. As its hand pushed the bowl around fruitlessly, I gave a little smirk of human superiority. But then the robot stopped. And—I swear to this—it started to think. After about ten seconds, it perceived that the overturned bowl had a raised circular foot. Pinching the foot of the bowl, the robot

picked it up and placed it in the bin. “See?” the technician said. “It figured it out.” ♦

By [William Finnegan](#)

The judge, Norbert Hofer, was a specialist. At the Innsbruck Regional Court, in western Austria, he oversaw a jurisdiction designated for mountain, avalanche, climbing, cable-car, and ski accidents. Cases handled by this department rarely draw much public attention, but the trial of Thomas Plamberger, in February, was different. He was charged with homicide by gross negligence in the death of his girlfriend, Kerstin Gurtner, during an attempt to climb Grossglockner, Austria's tallest mountain. On a chilly Thursday morning, the wood-panelled room of Hofer's court was packed with citizens and a motley press corps.

Word had spread that the judge wanted to run the trial in a single day. There would be no jury. A verdict and, if necessary, a sentence would be delivered by the judge alone. The stage layout was, to an American eye, oddly simple. There was a raised desk, for the judge, with an alcove behind it. Two rows of chairs flanked the forecourt. A small table and a chair, for witnesses, faced the judge's desk.

The judge was a gawky fiftysomething gent in a short-sleeved white shirt and a beige tie, keeping up a conversation with a co-worker as he pulled on his black robe. He had a small mouth, a sharp gaze, and an old-fashioned boy's haircut. Hofer was an alpinist himself and a mountain-rescue volunteer. He led groups of young people on summer hikes. None of his associations would bias him in any way, he assured us: "I consider myself capable of making decisions with one hundred per cent objectivity."

The case against Plamberger was nearly unprecedented. Hundreds of unfortunates—hikers, climbers, skiers, snowboarders—die in the Austrian Alps each year, but these fatalities rarely result in criminal charges. Gurtner, thirty-three, had frozen to death a few dozen yards from the Grossglockner summit after Plamberger, then thirty-six, parted ways with her at around 2 *a.m.* on a winter night. The press coverage in Austria and Germany had been relatively decorous. Reports rarely even gave the climbers' first names. The world press was less restrained; the *Post*, the *Daily Mail*, the *Hindustan Times*, and many others ran screaming headlines, accompanied by photographs of the climbers purloined from social media. One showed Thomas on a mountaintop, swinging shirtless from a summit cross at the moment of conquest. Another showed Kerstin on a ski slope. Her glove half

hid an enormous smile, and her eyes, bright blue, seemed to swallow the world. The image was everywhere in the commentary online, but it wasn't shown in the courtroom. Judge Hofer announced that he would not be influenced by the “unbelievable media coverage” that the case had received.

Behind the judge's chair hung a massive crest of an eagle with wings spread, its fierce beak turned stage right. On either side were large monitors, showing the heavy granite pyramid of Grossglockner on a bluebird day. A posse quietly crossed to a set of chairs on the judge's left: the defendant and his supporters, mostly men. His lawyer, Kurt Jelinek, was a familiar figure—his vast round face was often in the papers. But which of the younger men was Plamberger? Anyone following the case had seen his visage online. But, in the photos, he generally wore reflecting glacier goggles and a climbing helmet. All that showed was dark scruff and a vulpine grin.

A small, brown-haired man made his way to the table in the forecourt. He wore a blue sports coat, a white shirt, no tie. What happened to the swashbuckling mountaineer whose image had been plastered across the internet? This fellow seemed—not timid, perhaps, but certainly mousy. His back was to the crowd. Austrian law did not oblige him to promise to tell the truth, the whole truth, and nothing but the truth.

“How are you?” the judge asked.

“Good.” I thought I saw Plamberger put two thumbs up.

The judge, solicitous, told the defendant that he could take breaks whenever he needed them.

The prosecutor, Johann Frischman, laid out, in a reedy voice, the criminal complaint. It alleged that Plamberger, acting as “the responsible guide,” had made nine serious mistakes during the ascent, then left Gurtner alone, “unprotected, exhausted, hypothermic, and disoriented.” He summited, made his way down to a warming hut on the other side of the mountain, and phoned rescuers from there at about 3:30 *a.m.* When the rescue crew got to the peak, later that morning, they found Gurtner dead.

Jelinek, who had far more presence and passion than the prosecutor, rejected this story. Gurtner was a strong alpine athlete, he said. She and his client had planned their climbs together as equals. Her death was a “tragic accident.” She had collapsed on the mountain and urged her boyfriend to go for help. Plamberger, devastated by Gurtner’s death, had been “already punished,” because a “prejudgment has taken place in the media,” Jelinek said. Now an innocent man was facing the possibility of three years in prison.

As the judge’s attention turned toward the defendant’s table, Plamberger made a preëmptive declaration. “I am infinitely sorry for what happened and how it happened,” he said. “I loved Kerstin.”

Austrians venerate the Grossglockner. As you enter the main terminal at Salzburg Airport, a large photograph of an idealized “Glockner” hoves into view. Above a green valley, the mountain, steel-gray, looms at otherworldly scale. Wide snowfields surround serrated ridges, and a towering peak jabs the sky. It was first climbed in 1800, and a monument called the Emperor’s Cross went up to mark the summit in 1880. In recent times, the mountain has developed some modern oddities. The whole south side—where most people climb—has surprisingly good cellphone reception. You’re in the wilderness, but you can text your friends or, if you’re in trouble, call for rescue. Grossglockner also has twenty-four-hour “mountain cams,” which people can follow on their laptops from home. There’s not much to see at night, other than fantastically bright stars in an extra-dark sky and an occasional set of headlamps: climbing parties descending late. Climbing Grossglockner has become hugely popular, with an estimated nine thousand people making the ascent each year. The main complaint you see online is that the summit is too crowded.

Austrians, it seems fair to say, venerate all their mountains, which cover two-thirds of the country. Alpine culture sometimes feels like a national religion. The Austrian Alpine Club bills itself as the country’s largest youth organization, with more than seven hundred thousand members—close to a tenth of the population—assiduously steered toward skiing, hiking, camping, climbing. Competence and bravery can be assumed. So can the provision of mutual aid.



*"On vacation, we switch from everyday problems to vacation problems."
Cartoon by David Sipress*

The Austrian Alps are fantastically photogenic, and the principals in the Plamberger trial had left a detailed record of their time in the mountains on social media. For investigators and for others following the trial, Instagram was a crucial source of information. Austrian privacy laws discourage journalists from asking probing questions, but a quick-witted reporter in Salzburg had retrieved years of Plamberger's posts, presumably before lawyers could tell him to lock his accounts.

Kerstin Gurtner's account is still up. Scrolling backward through it, you see music festivals, friends, booze jokes from her party days, trips to Italy, Portugal, Greece. But mainly there is an intense romance with the outdoors: lakes, valleys with toylike villages, distant snowy peaks. During the pandemic, she seems to spend every day hiking or trail running. She is not focussed on bagging peaks, though she ends up bagging a few, including a hike in the Glockner range up what she calls her "first 3,000er"—a summit higher than three thousand metres. She quotes Sir Martin Conway, a storied British mountaineer: "Each fresh peak ascended teaches something."

Of course, Instagram is performative. Her Facebook is more workaday. She's looking for an apartment for a cousin, or trying to find a techie to help run live streams for an events company called brandmood. According to her mother, an eating disorder has begun to plague her, but there are no hints of

it, unless you look closely at her shoulders in certain pics. She lives for a time in the ski-resort town of Kitzbühel, a sort of oom-pah Aspen. Then she moves home to Salzburg, goes to work as a project manager for an events hub. She begins trail running more competitively and volunteers to help organize races. The longer races are brutally hard, but Kerstin loves helping other runners pursue their personal bests.

In 2024, her social media changes. She starts climbing more serious mountains, doing more technical pitches, and nearly every post includes shots of her in a white helmet. Thomas is teaching her to climb. He leads, she follows, on belay from above. It's her "beginner's course in Alpine climbing," she writes in March.

The two had encountered each other, a friend of Kerstin's told me, at a trail-running event called the Hochkönigman, near Maria Alm, a village in the gentle mountains southwest of Salzburg. (Thomas told the judge that they met online.) Kerstin had lived in Maria Alm for several years in her twenties, working for the local tourism office, helping to promote the region's hiking trails and ski resorts. She ran in the Hochkönigman in 2024, placing second among women in the downhill. Thomas first shows up on her Instagram page that February, in an exultant post from a ski area. "Days like these could last forever," she writes. The images are of snow-heavy trees, a chairlift overhead, and Kerstin grinning with ski goggles lifted. Thomas comments with three lovestruck emojis, and Kerstin replies with two. Somebody has a new boyfriend.

In the coming months, Kerstin posts photos from the Drachenwand—the "Dragon Wall," overlooking a mountain lake—accompanied by a remix of Nina Simone's "Feeling Good." She reaches the summit cross on Hoher Dachstein, a snowy but not difficult climb. After completing a nineteen-pitch sport climb on Trisselwand, she admits to being "a bit proud." Thomas weighs in: "Super-strong performance for a first proper multi-rope-length tour."

Over and over, Kerstin thanks Thomas. Her friend from the trail-running world told me that she was happy, perhaps the happiest he had seen her, with her "new friend." They did two more big climbs in the fall, including one on the south wall of the Dachstein, which they summited after dark. Kerstin is

ecstatic. “A thousand metres of pure climbing bliss,” she posts. “I never would have thought that, after just six months of alpine climbing, I’d be able to climb this entire face.”

The route required endurance, and Kerstin had that. Thomas seemed to have everything else—ropes, gear, mountain knowledge, and decades of experience on far harder routes. He had climbed Mont Blanc, the Matterhorn (twice), eleven forbidding Swiss peaks in nine days. Just after ascending the south wall of the Dachstein with Kerstin, he and two friends climbed Grossglockner the hard way, on what Thomas called “the wild side”—the north face, which gets very little traffic. They ascended three different routes, including the Pallavicini Couloir, a steep line of snow and ice with significant risks of avalanche and rockfall.

In his Instagram posts, he rhapsodizes, too, but in a different key. He is justifiably proud of his speed and strength. He won a half-marathon in Salzburg and broke the speed record for a frightening round trip along an exposed ridge in Switzerland that crosses four peaks, all more than four thousand metres tall. He collects mottoes like a younger man: “Better a tiger for a day than a sheep for a thousand days.” His hashtags are #passion and #nevergiveup and #summitclimber. Not long before he and Kerstin began dating, he wrote, “What I want is an intense life, and for that, I’m willing to take certain risks. Because if it’s too late, the question will surely come up: why didn’t I even try?”

Grossglockner’s summit is twelve thousand four hundred and sixty feet above sea level, and there’s nearly always snow and ice in the upper reaches. The elevation gain from the parking lot to the peak is more than six thousand feet. Technically, it’s a relatively easy climb. Still, twenty-nine people have died up there in the past two decades, often from falls. Local rescue crews and their helicopters are kept busy.

The recommended approach for first-timers is to hire a guide and take two days, spending a night at either of two warming huts while acclimating to the elevation. On the second day, weather permitting, you can make an early run to the summit and descend to a hotel called Lucknerhaus before nightfall. In the summer, the huts are staffed and the meals are famously

hearty. In winter, they're closed, but a few warming rooms—unheated spaces where you can shelter from the wind—remain open.

For less experienced climbers, a guide can be important, for both safety and peace of mind, but it's not a requirement. Nor is a permit, or filing plans with rangers. Thomas, of course, was not inexperienced. And Grossglockner was his mountain. During the trial, he said that he had climbed it “fourteen or fifteen times.” Based on social media, that looked to be an underestimate.

Thomas said that he and Kerstin left the parking lot at 6:45 *a.m.* on January 18, 2025. A couple of other parties left around the same time. Several left later. It was a mild day for January, heading into the thirties, with light wind and blue sky. The good weather, plus the fact that it was a Saturday, could account for the traffic. It's a nearly seven-mile hike to a warming hut known as the Stüdlhütte, which is near the base of the mountain proper. The trail starts in a glade, crosses a bridge over a creek, then emerges into alpine pasture and scree surrounded by high peaks. At the head of the valley, above a set of glaciers being shrunk by climate change, is Grossglockner.

The couple fell in with a climber who was going their way. He later appeared in court and said that Thomas and Kerstin were travelling “at a completely normal pace,” although she was trailing behind. She was using a splitboard to make the initial trek across the snowy fields. A splitboard is a kind of snowboard that can be broken into two skis for climbing on snow. It is heavier than touring skis, like the ones Thomas had, and markedly slower on the uphill. But it can be snapped back together and used as a snowboard for descents. For Kerstin, who loved snowboarding, the extra effort was apparently worth it.

The climber who accompanied them said that Thomas “kept a close eye on her,” sometimes calling back, “Are you O.K.?” The two men went ahead, and “chatted a bit about the mountain world,” the other climber said. They exchanged ideas. “It was an interesting and a pleasant conversation,” he said. At the Stüdlhütte, he went right, Kerstin and Thomas left.

In the witness chair, Thomas said that he and Kerstin had planned to complete their outing in one day, reaching the summit at 9 *p.m.* A group of mountaineers sitting in front of me glanced at one another. Summiting

Grossglockner at 9 *p.m.* in midwinter was apparently not done. It meant that the Plamberger party would arrive at the peak after more than fourteen hours of climbing, and still have to descend the whole mountain in the dark. Temperatures would likely be far below freezing.

But Thomas and Kerstin were not planning to take what's known as the Normal Route. When they turned left at the Stüdlhütte, they were committing to a much more difficult ascent, a steep, rocky ridge known as the Stüdlgrat. The Stüdlgrat is a beautiful route, with sweeping views and a striking silhouette: nineteen hundred feet of jagged granite blocks, running all the way to the peak. Unlike the Normal Route, it has a number of sections of technical climbing. It can be faster. From the Stüdlhütte, the estimated time to the peak is three to five hours. But that's in summer, and it assumes that people can manage complex running belays: moving roped together through a constantly changing series of iron rings, stakes, and rock outcroppings. In winter, the Stüdlgrat is full of ice and snow, burying much of the fixed gear.

It was by far the most dangerous climb that Kerstin had ever attempted. Did she know that? All we know is that she trusted Thomas. I suspect, though, that she was keenly aware of his hard-core mountain adventures with his hard-core friends and saw going up the Stüdlgrat in winter as a way for her to cross into that world—to join him where he really lived.

Mountaineers typically study a route and break it down into timed segments: we must reach this spot by this time, or we'll have problems later. You always have a Plan B to follow if, for whatever reason, you don't hit those marks. Turning around is a popular Plan B. It was roughly 10 *a.m.* when Thomas and Kerstin left the hut behind. They would have skied along the edge of the Teischnitzkees glacier for an hour or so, then switched to boots and crampons and started up the ridge.

The Stüdlgrat gets enough traffic in warmer weather that the authorities have planted a prominent yellow sign at a flat spot known as the Breakfast Place. The sign says, in German and in English, "If it took you more than 3 hours to get here, turn back! Your life depends on it." The warning meant three hours from the Stüdlhütte. It had taken Kerstin and Thomas nearly five hours. In court, Thomas said that they had heavy backpacks. He was not

asked why their packs were heavy. He later said that they had no food other than a couple of bags of gummy bears and that he carried little emergency gear. The wind had picked up suddenly. The temperature was dropping.

During my visit to Austria, I drove up to the Lucknerhaus hotel. At the head of the valley stood the mighty Grossglockner. The weather was clear, the snow fresh and deep. The seductive serenity of the Alps on a sunny day suffused the scene. I heard voices and laughter from farther up the valley. Two men emerged from the trees, gliding on skis down a straight path that they had probably made themselves on the way up. They were middle-aged, not built like athletes. One was in a multicolored sweater, with a ski hat and no backpack. They had clearly legged it up the mountain some distance—half a mile or half a dozen—then turned and skied down, their efforts more than rewarded. They laughed like little boys as they stepped out of their bindings.

When Thomas told the judge that he and Kerstin were planning to summit at 9 *p.m.*, did he really mean that they planned to spend ten hours just on the Stüdlgrat? Doing what? One of my theories about the last day of Kerstin's life was that she thought she was going snowboarding. There seemed to be no other way to account for her carrying her splitboard up the mountain. On the day I visited, it looked possible to board down all the way from the upper warming hut, at Adlersruhe, if you took the Normal Route. But Thomas and Kerstin didn't take that route. Was there any plan that could include summiting via the Stüdlgrat and then descending in daylight? It would be dark before six at that time of year. There are elite mountaineers who, starting very early, might manage it. But Kerstin was still a beginner, and their summit goal was 9 *p.m.* Boarding down in the dark would be too dangerous, especially for someone not familiar with the mountain or the glaciers they would be crossing. Thomas knew all this. From the evidence, Kerstin did not.

We have no idea, really, what happened between Kerstin and Thomas on the Stüdlgrat. We know that hours passed. We know that the sun went down and that their headlamps became visible on webcams at 6:10 *p.m.* At that point, they had passed a large vertical feature known as the Pulpit but were still hundreds of feet below the summit. We know that Kerstin was trying to climb icy rock in soft snowboard boots, with ill-fitting crampons borrowed

from Thomas. Her feet likely slipped many times trying to find a hold, especially after dark. Such slips are exhausting. Each one sends a bolt of panic through the body as it scrambles to regain purchase. We know that, according to Thomas, they had rope problems—a stuck rope, and a pendulum fall by Kerstin. A pendulum fall occurs when a climber loses her hold on the mountain too far to the side of the next fixed point of rope above. It causes a sudden swing that can be terrifying and painful. Crampons sometimes make it worse, catching on rock features. Kerstin had suffered minor injuries, to her hand, her ribs, or perhaps her hip—no one in the courtroom cared to get it straight. Thomas told the judge that she lost a glove, and he loaned her an extra he had. The mountain cams show that their headlamps inched upward till about 8 *p.m.* Then, for the next two hours, they seemed to gain no altitude.

The weather, meanwhile, kept getting worse. Climbing parties on the Normal Route had turned back earlier in the day. Up on the Stüdlgrat, the wind rose to some forty-five miles an hour, and with the wind chill the temperature felt well below zero. It's possible that Thomas and Kerstin stopped because she was injured. But failing to keep moving in such brutal cold is the worst possible strategy—the fast lane to hypothermia.

A couple of hours after dark, a climber descending past the Stüdlgrat saw lights on the ridge and heard voices. He shouted but got no reply. Concerned, he called the Alpine Police, who dispatched a rescue helicopter from Salzburg. The helicopter reached the Stüdlgrat shortly after 10:40 *p.m.* It hovered, rocking in the wind, above the two figures on the exposed ridge. It made six separate passes, looking for the universal distress signal: two arms held up in a V. The climbers never even looked up. This was strange. Plamberger said that he did not react to the helicopter because they were not in a state of emergency. But there is also an arm signal for needing no help, which tells rescuers that, if they're looking for a party in trouble, they should look elsewhere on the mountain. Instead, Thomas and Kerstin started moving again. My guess is that Thomas was furious. He was proud of his climbing, fiercely proud. He had, to our knowledge, never been rescued in his life.

The funeral parlor in Salzburg where Kerstin's memorial service was held—with an invitation for mourners signed by both Kerstin's family and Thomas

—created an online page for condolences. Those poured in, from loved ones and strangers, but the comments had to be watched closely for denunciations of Thomas. The funeral parlor added a plea at the top of the page, in bold: “We ask you to refrain from accusations and assumptions!”

On social media, people did not refrain. There was talk of femicide: “She trusted him AND he abandoned her. Cruelty thy name is men.” Conversation swirled around the phenomenon of the “alpine divorce”—the hike from which only the man returns. Every woman seemed to have a story of being left behind: by the hot-shot boyfriend trying to impress his Facebook buddies, or by the aggrieved husband trying to outrun middle age. The consensus among women was, Make sure you have the car keys.



*“Sixty years ago, a cow first jumped over the moon. Now we are on track to do it again.”
Cartoon by Tyson Cole*

Commenters who believed that Thomas was a killer called for “justice for Kerstin” and “justice for her family.” The complication was that her family didn’t believe Thomas was guilty. For more than a year after Kerstin’s death, her parents, Gertraud and Helmut Gurtner, had kept silent, making no public statements. Then, just two weeks before the trial, Gertraud spoke with *Die Zeit*, a German weekly, and let fly. “It makes me angry that Kerstin is being portrayed as a silly little girl who let herself be dragged up the mountain,” she said. The truth was, Kerstin trained hard, and she and Thomas made their decisions together. Kerstin was a serious trail runner: “She once ran

down the Untersberg, our local mountain, in just thirty-eight minutes.” She was a good skier who loved to be in the mountains. “The death of my daughter was the result of a tragic chain of unfortunate circumstances,” Gertraud said. “I don’t want to blame my daughter’s boyfriend for that.” The pursuit of Thomas in the media and online had become a “witch hunt.”

The Gurtneres were called to testify in court. Jelinek, Plamberger’s attorney, suggested that their refusal to blame Thomas showed “true human greatness.” It also aligned, of course, with his defense. Many observers were shocked by the Gurtneres’ support for Thomas, but it didn’t seem strange to me. Whatever the facts of the case, the parents would prefer to believe that their daughter had died pursuing an ambitious dream, and that her boyfriend had loved her.

As Gertraud took the witness chair, Hofer apologized for calling her in. “Unfortunately, we have no choice,” he said. Gertraud had not been in the courtroom for any of the previous testimony. She was small, brown-haired, not dressed for a special occasion. She did not glance at the public benches.

The judge was gentle and solicitous with her, as he had been with Thomas. They had both lost a loved one.

Hofer reviewed a list of mountains that Kerstin had climbed. He noted that it contained no “classic mixed winter tours,” which means climbing on rock and ice. “Can you help me here?”

“I don’t know any, no. I suppose she wasn’t experienced there.” Kerstin hadn’t climbed in winter before.

Gertraud said that Kerstin had first mentioned climbing Grossglockner in December, 2024. Had they discussed the difficulty of the Stüdlgrat? They had not. Did they consider the difference between soft boots and mountain boots? No.

The judge asked gently if Kerstin’s strength ever suddenly failed her. “No,” Gertraud said. “Kerstin was a biter”—feisty and tough. “She never gave up.”

“Does she go to death consciously?” he asked. His question doesn’t translate well, but it meant something like, Would Kerstin have pushed herself until she dropped? Gertraud seemed unfazed. “Let me put it this way,” she said. “She wouldn’t have been taken up anywhere against her will. I told her, ‘You can always stop. In an emergency, you make the emergency call.’”

The prosecutor had no questions for Gertraud, but Jelinek did. Why had she spoken to *Die Zeit*? The “media frenzy” had upset her, she said, and she hoped to reclaim some control. She had also spoken expansively with the reporter about Kerstin’s childhood, her love of the mountains, her independence. “Kerstin was very curious, even as a small child,” she said. “If I told her, ‘Watch out, there’s a mean fox living back there in the woods, we’re not going there,’ she would go there to take a look.” She said that she was dealing with her grief partly by skiing in Maria Alm, where she and Kerstin had once skied together. And, of course, she opposed the vilification of Thomas. “It’s easy to be a hero from the comfort of your living room,” she said.

The trial was being debated in many forums in the mountaineering world. Some people worried that a guilty verdict would have a broad impact on the sport. Climbers who rope up together learn from one another, with no strict hierarchy. Now recreational climbers might start to think twice about going out with friends. A science writer at the London *Times*, Tom Whipple, objected to “making a moral liability a legal one; the fixed ropes of the law bolting themselves to our last cliff face of personal responsibility.” Whipple argued that climbing itself was an act of rampant negligence: risk for no rational purpose. Another climber in the U.K. wrote that “the foundation of climbing and mountaineering is *trust*—you have to trust yourself and others—and criminalizing these decisions would also be the death of trust.”

I talked to an experienced mountaineer in his climbing shop in Innsbruck. He told me about first ascents that he and his wife had put up in the Indian Himalayas. He climbed safely with her, but before he was married, he said, he was a wild man. He even attempted Freerider, the monster route in Yosemite that Alex Honnold later climbed without ropes in “Free Solo.” It was a miracle that he had survived his own foolish youth. No, he had not been actively following the Plamberger case. But, as a mountaineer, he disliked the courts’ getting involved in accidents. “People always want a

culprit when something bad happens,” he said. In a place like Austria, the mountains were the only zone of true freedom and self-reliance. Austria is, at least in the towns and cities, the most neurotically orderly place I’ve ever been.

Climbers die at a frightening rate, but the deaths are almost always ascribed to errors and accidents. A disproportionate number of fatalities occur during rappelling. Rappelling is not difficult. You mainly need to remember to have a knot tied at the end of the rope. When someone dies, investigations tend to conclude that someone made a mistake—or that no one made a mistake.

Kerstin’s death seemed different, though. A well-known mountain guide told me that he and his colleagues had been discussing the case for nearly a year, and that there were simply too many mistakes for it to be called an accident. Certified guides face stricter expectations than other climbers. It is the guide’s job to judge not just mountain conditions but also the clients’ strength and abilities.

In court, Plamberger denied that he was any kind of guide. He described himself, instead, as a “hobby mountaineer,” who had trained himself by watching videos. (He had also climbed with his father, an experienced mountaineer, something he did not mention.) He and Kerstin planned their climbs together, he said: “I wasn’t in charge—I had no authority to give instructions, no power, no superiority.”

Thomas seems to admire the early alpine first ascenders, at least online. But his relationship with the modern outdoor industry feels vexed. On an Instagram post where a Matterhorn guide advertised his services, Thomas wrote tauntingly, “Why would I book something like that? I’ll climb it myself.” In 2023, when a skier named Eva Walkner posted about a tremendous day with a film crew on a mountain called Hoher Göll, Thomas commented appreciatively, then asked why her party hadn’t gone higher up the face. The footage attached to the post makes it obvious why not: there were massive snow cornices near the top, and the avalanche danger was already considerable where they stopped. What’s not obvious is whether Thomas knew that Eva Walkner is a two-time Freeride World Champion. Her profile has spectacular clips of her flying off terrifying cliffs and skiing out unfazed. Two years later, when Thomas’s name was in the news,

Walkner came back to the post and answered his question: “Why did we turn around just before the summit? Because turning back when conditions become too dangerous is what distinguishes good and experienced alpinists.”

Elsewhere on Instagram, you can see that Thomas enjoyed taking people up Grossglockner. In the summer of 2022, he climbed it with a young woman named Denise. He was in the foreground of one of her summit pics, and she raved about him in an Instagram post, calling him *den besten Bergführer*—the best mountain guide. In 2023, he did it again, via the Stüdlgrat, with another woman, who also happened to be named Kerstin.

An accomplished local climber, younger than Thomas, told me that she had declined his invitation to climb together. She and her friends had been wary of him for years. “He was the person who was always ready to tell you what you were doing wrong, kind of an egotistical person,” she said. “He roped up with people who were less experienced, people who trusted him.” But, she said, “he pushed through some tours the wrong way. It was only a matter of time.”

Late in the trial, a surprise witness came striding into the courtroom. Her name was Andrea Bergner, and she was clearly not happy to be there. She was tall, slim, dark-haired, stylishly dressed in a gray jacket and a matching skirt. Early thirties. She took the witness chair and answered the judge’s greeting in a loud voice that said, unmistakably, Let’s get this over with. Thomas, whose closed expression had not changed for hours, suddenly looked alarmed.

“I was with the defendant—we briefly had a relationship,” Bergner said. “We also did joint tours.” She meant climbing mountains together. “The leadership was always my ex-boyfriend.” Some of the tours she could not have managed without him: “I had a certain amount of mountain experience, but he was more experienced.”

“Did he take good care of you on tours?” the judge asked, in a distant way. Not solicitous now.

“He always took good care of me during the tours,” she said. “But when things got tough, that wasn’t always the case. I was sometimes afraid. In those situations, he tended to get grumpy.” He would tell her, “Don’t be such a baby.”

The judge knew why Andrea had been called. She and Thomas had climbed Grossglockner in 2023. “The mood was bad,” she said. “I was at the end of my strength. I was dizzy, my headlamp was out. I cried and screamed. I signalled to him, too. Then, suddenly, he was gone—he’d gone ahead.” It was the middle of the night, she said, and she was “all alone.”

She managed to get down off the mountain, but she never went on another tour with the defendant. Her friends advised her to go to the police. “I didn’t do that,” she said.

The prosecutor asked, “Was this one incident when he left you alone the end of the relationship?” No, she said. They broke up later, “for various reasons.”

The defense had no questions. Andrea stalked out, slamming the door behind her.

I Googled her afterward. Her Instagram has more than sixty-two thousand followers. Her feed is stuffed with photos of her biking, skiing, climbing, working out, often in outfits from Skinfit, from which her followers can get ten per cent off. She is, in a word, an influencer. Her brand is fearless, fierce. On her page is a slide show of seven ascents of Grossglockner that she made, including one on the daunting northwest ridge. Take that, Plamberger.

At 5:22 *p.m.*, when Thomas and Kerstin were still fairly low on the Stüdlgrat, she had dialled “149.” 140 is the 911 of the Austrian Alps. It seems as if Kerstin tried to make an emergency call as the sun went down, fluffing the last number with frozen fingers or a puffy glove. Thomas, asked about the call during the trial, said that he knew nothing about it at the time. If she made the call in secret, communications between the pair must not have been good. Had Thomas got angry because Kerstin wasn’t keeping up, as he allegedly had with his ex?

They apparently started moving again only when the helicopter arrived. Thomas said that Kerstin denied having called the mountain rescue. “She also stressed that she was fine.” They kept going, slowly approaching the summit. At one point, he said, “Kerstin shouted something up to me four to five times, but I didn’t understand it.” That suggested that he was far ahead, and perhaps didn’t care to be any closer. Climbing alone, he likely could have summited before nightfall and got down off the mountain. But he was not alone, and it is a long-standing practice among climbers for stronger partners to carefully monitor less strong ones, adjusting the pace and objectives to keep them safe.

AIRCRAFT-AGE TELLS

Inkwell in armrest.



In-flight entertainment is a play.



Only snack is Granddad's Sorghum Chews.



R. C.

Cartoon by Roz Chast

Jelinek issued a statement during the police investigation. It said, inter alia, that, after the helicopter’s visit, Kerstin had “suddenly showed increasing signs of exhaustion,” which was “completely surprising and objectively unforeseeable” for his client. But that’s not how exhaustion typically works during a seventeen-hour climb in frigid weather. Weakness, frostbite, and the early stages of hypothermia are obvious long before a reasonably fit person collapses. Jelinek also criticized the response time of the Alpine Police—a strategic mistake. In Austria, mountain rescuers are heroic figures beyond reproach.

At 12:35 *a.m.*, Thomas made a phone call. It is not clear whether he was near Kerstin. There were no headlamps visible on the mountain at that point. He had been contacted by an Alpine Police officer identified in court as Mathias A., who had fielded the earlier emergency call that sent the helicopter to the Stüdlgrat. He had identified Thomas from the license plates on the only car still parked in the Lucknerhaus lot, then left a series of messages on his phone. Now Thomas was calling back. Mathias asked why they had not acknowledged the helicopter. Because everything was fine, Thomas told him. What about now? “We don’t need anything here, it’s all right,” he said, according to Mathias. After the call, Mathias was worried. He messaged Thomas, “Do you need help???” No reply. He texted again and again, but Thomas had silenced his phone. In court, he said he was trying to save battery.

What happened to Kerstin as they approached the summit of Grossglockner? The mountain cams showed a blob of headlamp light near the summit at midnight, above the last rocky section on the Stüdlgrat. The terrain changes there. It’s a snowy trudge, steep at first, to the Empire Cross. At that hour, the cams snapped photos every thirty minutes. In the image from twelve-thirty, there were no headlamps visible. The judge had a note in his file, saying that after Kerstin’s alleged collapse she “could only move on all fours.”

Thomas did not confirm or deny that. He said, “It was a completely exceptional situation for both of us.” And: “It was bitterly cold. We wanted to have tea, but it was all frozen.”

It’s not that difficult to prepare for serious cold. When Kerstin started doing more mountain hiking, back in 2020, her mother asked her to get a bivouac bag—a lightweight emergency sleeping bag that is standard gear for mountaineers. She agreed, and also got aluminum rescue blankets and a first-aid kit. Thomas said in court that he knew Kerstin had a bivouac bag. She also had a rescue blanket and hand warmers. He himself did not have a bivouac bag, though. “I don’t own one,” he said.

This was a weird confession. The bags weigh very little, can be bought for less than twenty euros, and save lives regularly. Pairs of hikers often carry double bivvies, as they’re known, to share warmth. Thomas surely knew the

risks of exposure in the alpine. Was it just some mountain-man obtuseness—that he believed he would never be in an emergency he couldn’t handle? I asked Nicola Werdenigg, who was among the first women to become certified winter-climbing guides in Austria, what she thought. “These are not mountaineers,” she said, sadly.

The judge said that he couldn’t understand why, when Thomas phoned Mathias at twelve-thirty-five, he did not tell him that there was an emergency. Thomas said that, during the call, he didn’t know that Kerstin was unable to go on. This was a flat contradiction of something he’d said earlier—that he’d told Mathias that she urgently needed help.

The judge asked why Thomas didn’t call Mathias back when he noticed that Kerstin couldn’t move: “That would have been the most basic piece of information. You should know that as a mountaineer.”

“I thought I had set the rescue chain in motion with this call.”

Why didn’t Thomas unpack the bivouac bag from her backpack, the thermal blanket? His explanation was that he’d forgotten in the stress of the moment: “I didn’t even think about the bivouac bag.” He added, “I don’t know why she didn’t say anything to me about it.” So the oversight was her fault. She was hypothermic, but she should have said something.

Thomas testified that, sometime around 2 *a.m.*, “I agreed with her that I would go get help. Because we knew that we wouldn’t make it through the entire night up there. I lay down beside her, and she then screamed loudly, ‘Go! Now! Go!’ She saved my life with that.”

The judge seemed to have reached the end of his patience with Thomas. He pulled up a photograph of Kerstin as the rescue crew had found her. She was not curled in a ball against the wind, as one might expect. She was braced against a near-vertical gray rock on a steep part of the mountain, with her back unnaturally arched and her face pointed at the sky, eyes open. Her feet dangled, her boots untied. There was certainly no room to lie down beside her there. Her gloves were gone. Her heavy pack was still on her back. She was a vision of agony.

Thomas had said that he “secured” Kerstin before leaving her. In climbing, one secures oneself or a partner to the mountain in order to free one’s hands for other tasks. But he had told the rescue crew that he had left her on an area of gently sloping snow, in view of the summit cross. This was nowhere near that snowy area. She was in a fall zone, tied tight to the rock from behind.

“How could she get into such a position when she couldn’t move anymore?” the judge asked. “I find it difficult to reconcile your version of events with the images I have before me.”

Thomas sat in silence. His expression had not changed.

The judge speculated about how Kerstin might have reached this perilous spot if Thomas had secured her near the summit: she could have untied herself, put her backpack on, climbed down, then fallen. But, he suggested, perhaps there was a better explanation. He invited Thomas to offer one. Thomas said nothing. Jelinek hastily asked for a ten-minute break, and the judge granted it.

When the proceedings resumed, Thomas faced the judge and delivered a statement, seemingly rehearsed: “I understand you and the scenario you’re considering, but I didn’t leave Kerstin behind at that point. I want to make that very clear.”

He may have meant that Kerstin had ordered him to go ahead, or that he had not left her behind in his heart, or perhaps he was sticking to his story about leaving her somewhere else. In any event, the judge let it go. When Jelinek asked his client about Kerstin’s condition when he last saw her, he said, “She was completely finished. . . . I thought I was in a fake movie.”

On the Lucknerhaus mountain cam, headlamps reappeared at 1:30 *a.m.* below the summit—by the looks of it, just where Kerstin was found. At 2:30 *A.M.*, Thomas can be seen descending on the Normal Route. There is no light at Kerstin’s resting spot. Thomas said that he had a difficult descent. He threw up, twice. But his headlamp moved swiftly across rocks and ice, and by about 3 *a.m.* he was speeding down the slope, presumably on skis.

The Alpine Police were still trying to reach him, but his phone remained silenced, filling up with missed calls and messages. Sometime after three, he reached a high-altitude warming hut, the Archduke-Johann-Hütte, and phoned Mathias again. There was initially some confusion, Mathias testified: “I thought at first that everything was fine. But then he told me that he left his girlfriend on the Glockner.” Mathias immediately went into rescue mode, assembling a crew. Apparently, he asked Thomas to return to Kerstin’s position, but Thomas said no, he needed to rest.

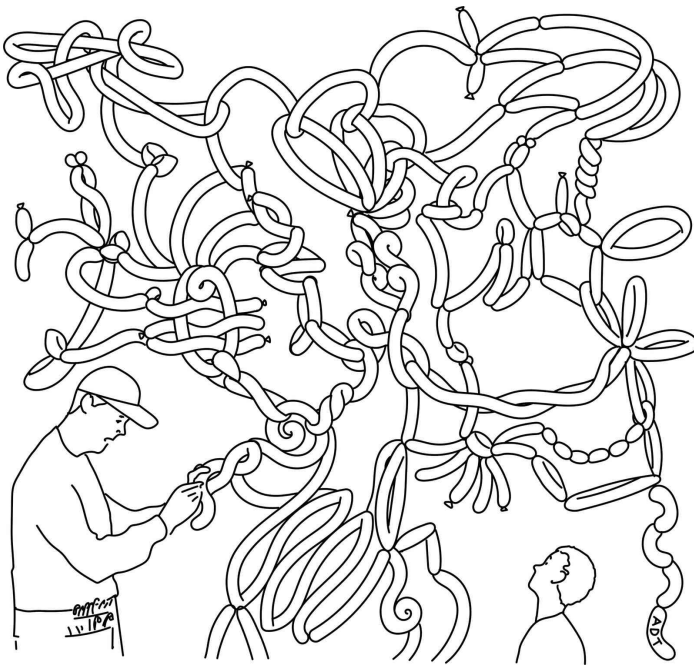
The rescue helicopter battled with the wind and had to drop the crew on a snowfield nearly two thousand feet below the summit. They trudged up with their equipment and reached Kerstin at 10 *a.m.* Members of the crew testified that her body was not where Thomas had said it would be. Mathias, who had assisted with the morning rescue, mentioned bloodstains above where they found her. The judge asked if it looked like Kerstin had fallen down the slope. “No, not really,” Mathias said. But these were not forensic scientists. They were there to recover a body, and it was a difficult operation—slippery, windy, very cold, across a narrow ridge. They got her into a red rescue coffin, then rappelled with it hundreds of metres down to the helicopter.

Meanwhile, Thomas had got a helicopter ride off the mountain. A police officer from the town of Lienz questioned him and found that he was “rather taciturn . . . but he was able to answer clearly.” Thomas said that his girlfriend had struggled on the Stüdlgrat. She was less experienced. He was the leader. Had he protected her with a bivouac bag? “He couldn’t answer that,” the inspector told the court. Why had he been in the warming hut and not lying in a bivouac bag with his girlfriend? “That didn’t quite add up for me.”

The next day, Thomas phoned the police to ask if they had by chance found his GoPro on the mountain. They had not. That seemed like a tacky inquiry, but it’s possible that Thomas was relieved by the answer. He often wore the GoPro on his climbing helmet, to film while in the mountains. That footage might have answered many questions about the hours they’d spent on the Stüdlgrat—what was said where, whether the two of them were arguing or even speaking.

Big, fat snowflakes had started falling in the afternoon, past the high courtroom windows. The valleys and peaks around Innsbruck were already on high avalanche alert. Now it was dark. The judge retired to make his decision. There was, first, the question of guilt, and then, if necessary, sentencing.

Grossly negligent homicide, under the Austrian criminal code, means that someone has caused a death by acting “in an unusual and conspicuously negligent manner.” Nothing about intent—intent pertains to murder in its several degrees. The judge had to weigh this charge against everything he had heard.



“Wait . . . what did you want again?”
Cartoon by Adam Douglas Thompson

I pictured him also fretting about being branded a *Nestbeschmutzer*—a “nest fouler”—if his decision was seen to besmirch the image of mountaineering. That was the term widely used against social critics, artists, novelists—anyone in Austria who dared to be too critical of the country after the Second World War.

In truth, Hofer betrayed no world-historical timidity. Only, perhaps, a constitutional aversion to thinking ill of mountaineers. In a 2020 journal article, he’d written, “Culpable conduct can be intentional or negligent.

Since intentional conduct . . . plays virtually no role in alpine accidents, the focus is on examining whether negligent conduct led to the accident.”

There were many disturbing parts of Kerstin’s story left unpursued. The autopsy had yielded an official cause of death: hypothermia. There were signs of a viral infection, and she had taken ibuprofen. But nothing was said about her injuries, even though Thomas had mentioned them. Mathias had described bloodstains. Whose? From what wound? Why had Thomas and Kerstin taken the Stüdlgrat? That was a central question, never asked. Where were their headlamps after midnight? Extreme cold drains batteries, but at least one headlamp was back on at 1:30 *a.m.* How did Kerstin end up where she was found? Did Thomas lead her there or find her there? Why on earth was she still wearing her backpack?

The prosecutor and the judge may have left all these questions alone because they knew that they had enough to convict for grossly negligent homicide. This case was already out in new territory. It didn’t need to flail on any further.

Another question: Would Thomas have been prosecuted if his deceased partner had been a man, with all other details the same? That was hard to picture. But Kerstin was, as her mother said, not a fool. She was physically and mentally strong. She made it to the top of Grossglockner, if not quite to the cross, in a raging winter storm, with bad boots, a heavy bag, no real food, not much experience, possibly an injury, and very probably a cold and angry boyfriend.

At one point, while being pressed by the judge, Thomas claimed that he took off Kerstin’s backpack and urged her to sit on it. I didn’t believe him. He’d done nothing else to make her comfortable. And if the backpack were beneath her she would have been sitting on her bivouac bag, emergency blanket, and hand warmers, which could have saved her life. All they would have had to do was reach inside the pack.

It’s possible that, as the judge suggested, Kerstin set off blindly in the dark, making her way as far as the rocks where she was found. People dying of hypothermia suffer extreme confusion and sometimes feel a painful rush of blood to their extremities, causing them to shed clothes before they fall into

a coma, their organs fail, and their heart stops. But someone was lucid enough to tie the knot that secured her to the mountain. That photograph showed no evidence of any effort to save her. Instead, she was thrust up into the killing wind.

The judge came back with a verdict: guilty.

He explained at some length. He was discounting a number of the mountaineering errors (starting too late, not carrying sufficient emergency gear), because they were not inherently responsible for Kerstin's death. He praised Plamberger's abilities—"You are an excellent alpinist"—and spoke sympathetically of the torments inflicted on him and his family by social media. "I don't see you as a murderer," he said. "I see you as someone who, as a last resort, tried to call for help and stand by his girlfriend." Thomas had been unable to put Kerstin in a safe position because she was unable to proceed. He had gone ahead to Adlersruhe "not to save yourself" but to find help. There just wasn't any.

This all sounded promising for the defense—and would weigh in Thomas's favor when it came to sentencing. His guilt stemmed from the fact that, despite his excellence, "you are obviously someone who is struggling to make the switch from your own skills to another's." He did not recognize the exhaustion of his partner when he should have. None of the climbing he had done with her before should have made him think she could go up the Stüdlgrat in winter. He let her climb with inadequate equipment. He did not turn back at the Breakfast Place. "Had you acted differently, I strongly assume that your partner would have survived," the judge said. The requirements for a finding of gross negligence were fulfilled "due to the cumulative circumstances." There was also Kerstin's "contributory negligence"—she could have made an emergency call herself.

For sentencing, the mitigating factors included Thomas's "clean record and previously orderly life style." The loss of his loved one was also taken into account. The judge gave him a five-month suspended sentence—likely no jail time, in other words—and a fine of nine thousand six hundred euros. The verdict, Hofer said, almost apologetically, "was the decision of a single judge at the Regional Court. That doesn't mean it's right." Thomas had the option to appeal.

Thomas frowned and looked at Jelinek. They would appeal.

It was nearly midnight. Thomas had to get back to Salzburg for his shift the next day. He was a cook at a retirement home there.

Peter Habeler, who is Austria's greatest living mountaineer—he summited Everest without oxygen along with Reinhold Messner, among many other accomplishments—was interviewed for a public-television documentary about the tragedy on Grossglockner. Habeler is in his eighties, still tanned and climbing. He had trenchant things to say about poor judgment and, when a question came about Thomas leaving Kerstin on the mountain, his gaze hardened and he switched to strongly accented English. “*That is a no go*,” he said, twice. Never mind what the courts said. Thomas had broken a basic code of the mountains.

The argument went on after the trial. Back home, I couldn't shake the Instagram habit. I looked at Andrea Bergner's page and understood why she had been so angry in court. This burst of publicity was all wrong for her—she was cast as a victim, which was *so* off brand. The day after the trial, she was already up in the deep powder west of Innsbruck, looking like a million bucks. Some of her commenters knew about her ex-boyfriend. Under a series of shots of her on a scary ice climb, a follower wrote, “Glad Plamberger didnt kill u as well.”

Thomas couldn't shake the habit, either. I stumbled on a Facebook post from before the trial, when his accounts were supposed to be closed. He had free-soloed the Aschenbrenner, he announced. That's a steep route, much of it usually solid ice, on the north face of Grossglockner. It was “a brilliant climb,” he wrote. Most of the photos were selfies. The pirate king of the Eastern Alps was still out there, killing it with his vulpine grin. The soundtrack was “All Eyez on Me,” by Tupac Shakur—pure id, unstoppable, a lady's man, not fazed by courts or prison. “Live the life of a thug n—— / Until the day I die / Live my life as a boss playa / All eyez on me.” So that was the music playing in the brain of the mousy defendant. And he was right. He got away with it.

Two weeks after Plamberger's conviction, a young woman named Hannah posted a photo of herself on the summit of Grossglockner in morning

sunshine. She wore stiff green touring boots and crampons with spikes like black daggers. Thomas applauded her achievement. This was still his mountain, and he liked to let women know it. ♦

[A Reporter at Large](#)

The Tick That Hunts Down Its Hosts—Including Us

Lone-star ticks don't just pursue and bite people. The affliction they're spreading, an allergy to red meat known as alpha-gal syndrome, attacks a way of life.

By [Burkhard Bilger](#)



Patrick Roden-Reynolds, a public-health biologist on Martha's Vineyard, collecting ticks. Lone stars are relentless, he said. Photographs by Brian Chorski for The New Yorker

It was a perfect day for a blood meal. A salt breeze blew over the western shore of Martha's Vineyard, riffling the meadow grass under a pale-blue sky. They'd been waiting by the footpath all afternoon, crouched in the shade of a windswept cedar. They didn't do well in the sun. They'd spent the winter half asleep, in the moist, rotting leaves under a huckleberry bush, well away from other creatures. But they were hungry now, as they always were in the spring. They'd scuttled out into the open air that morning and were sitting side by side on a blade of switchgrass, their tiny teardrop bodies barely bending it down. Their back legs gripped its serrated edge while their front legs reached toward the path, waiting for a warm body to lumber by. They looked like children at a [Macy's Parade](#), leaning out to touch the Snoopy balloon.

“Look at them sitting there,” a voice said. The ground shook beneath them and a giant face suddenly loomed above, its bulbous eyes inches from their bodies. They could smell the carbon dioxide in the air. “Just a little hot breath and they stick out their legs to latch on,” Patrick Roden-Reynolds said. He blew on the ticks again, and watched their front limbs spread wide, trembling in the current. Then he reached down with a pair of tweezers and plucked them off the grass. He stuck them on a lint roller and pressed them deep into the spongy plastic, next to the others he’d collected that afternoon. They were arranged in neat rows by sex and species—dog tick, deer tick, lone-star tick—eighty-one in all. “Everyone comes here and is afraid of [Jaws](#),” he said. “That is not what you should be scared about. It’s the little things.”

Roden-Reynolds is thirty-three and has been the island’s public-health biologist for four years. To a hungry tick, he must have been an appetizing sight: tall, stout, and soft-featured, with pale red hair and eyebrows. His cheeks and hands were flushed pink with blood beneath the skin. But he knew better than to expose his other body parts. He was covered from neck to boots in fabrics soaked with permethrin—a pesticide modelled on the toxic pyrethrin found in chrysanthemum flowers. Had one of these ticks crawled up the cuff of his pants, it would have been dead before it reached his belt.

When he first came to Martha’s Vineyard, in 2022, Roden-Reynolds kept hearing that the ticks were worse than ever. But people had been saying that for years. The island had gone from farm and pasture in the nineteenth century to nature preserves and residential property in the twentieth, its silvery shingled homes encircled by forests and deer. The more bucolic it became, the more parasites it bred. “Every one of those deer could carry five hundred to a thousand ticks in a season,” Roden-Reynolds said. His predecessor, Dick Johnson, reckons that he got [Lyme disease](#) four or five times between 1990 and 2001. “It was no big deal,” Johnson told me. “Usually, they didn’t even give you a blood test.” The disease, caused by bacteria in deer ticks, can produce everything from flu-like symptoms to chronic fatigue and arthritis, if left untreated. But a dose of doxycycline or amoxicillin can generally cure it. In the early two-thousands, another tick-borne illness, babesiosis, began to make the rounds. Unlike Lyme disease, it didn’t respond to antibiotics alone, and its symptoms were more severe:

chills, high fever, nausea, and, in rare cases, hemolytic anemia, a blood disorder that can be life-threatening. Then anaplasmosis joined the fray, spreading fevers and chills of its own. “People were getting knocked on their asses,” Johnson said. “Sometimes they’d get the trifecta of Lyme, babesiosis, and anaplasmosis, and end up in the I.C.U. fighting for their lives.”

The ticks on Martha’s Vineyard can now cause nearly a dozen different diseases. Lyme and babesiosis are eleven times more common than in mainland Massachusetts, and anaplasmosis is five times more common. Deer ticks were long responsible for most of the infections—nearly nine hundred cases of Lyme and babesiosis alone between 2020 and 2024. But those aren’t the ticks that locals fear most. Roden-Reynolds pointed to a row of stocky little bugs, each with a white dot on its back. These were lone-star ticks, first confirmed to be on the island in 2011. Most ticks are content to sit and wait for a meal to wander past; lone stars have no patience for that. “We see them actually pursuing hosts, which is really disconcerting,” one entomologist told me. Roden-Reynolds has found lone stars on walking trails and benches, in toolsheds, parking lots, and the yards of hundreds of local homes that he has surveyed. He has heard of them scrambling down dunes and across burning sand to feed on sunbathers. “Oh, they’re relentless,” he said.

To an outsider, the scene may spark a certain *Schadenfreude*, like an army of tiny barbarians sacking Rome. But the Vineyard is more than a wealthy enclave in this case. It’s a microcosm—part of a pattern of infestations and pest-borne diseases that has been repeated around the world, as climates have changed and landscapes have transformed. Fifty years ago, lone-star ticks rarely crossed the Mason-Dixon Line. Now they’ve marched as far north as Maine and as far west as the Oklahoma panhandle. One of the afflictions they bring, called alpha-gal syndrome, is the most rapidly spreading tick-borne illness in the country. Nature, it seems, is coming back to bite us.

In the summer of 2003, in a suburb of Sydney, Australia, a surfer went to see an allergist named Sheryl van Nunen. He was nineteen years old, tall and tan, with sun-bleached hair—“The sort of fellow you’d see on a beach in Gallipoli,” van Nunen told me—but he felt less than sprightly. Not long before seeing her, he had rushed to an emergency room in the middle of the

night, his breathing ragged, his blood pressure low, his body covered in hives. He was having an anaphylactic reaction but didn't know it. Van Nunen had seen such symptoms before, in patients going back for six years. The first was an older gentleman who had eaten a steak at a beachside restaurant in Sydney. The surfer, whose name was Jake, was the eleventh.

Van Nunen had asked Jake to bring in some leftovers from his last meal before the reaction. Now she poked a surgical lancet into each food item in turn, then pricked the skin of his right arm—a standard test for allergies. After a moment, the spot where she'd pricked him with a trace of cooked meat began to swell. It was a tiny welt—less than two millimetres in diameter—and Jake was unimpressed. Maybe the meat had some spice on it or was processed with a chemical? Van Nunen went to a refrigerator in her office and pulled out a tray of raw meats that she'd bought from an organic butcher—bits of beef and lamb. When she pricked the surfer again, this time after poking the raw beef, his skin swelled up ten times as much as before. Something had made him allergic to red meat.



"Is anyone objecting to the marriage, or is it all about my shorts?"
Cartoon by Will McPhail

The beaches and bush north of Sydney are full of peculiar beasts, like Martha's Vineyard reimaged by [Hieronimus Bosch](#): wobbegongs and jelly blubbers, swamp wallabies and fairy penguins, weedy sea dragons and beach worms up to nine feet long. Smaller creatures, too, can take alarming forms.

The Australian paralysis tick, for example, doesn't content itself with simply sucking its victims' blood; it injects them with a neurotoxin that can be strong enough to kill a child if their symptoms go untreated. (In New South Wales alone, tick paralysis caused twenty deaths between 1900 and 1945.) The paths through the bush are infested with these ticks, as are many back yards. Six of the patients that van Nunen had treated for the meat allergy remembered being bitten by a paralysis tick. But their symptoms had appeared weeks or even months after the bite, and they were triggered hours after eating meat. The neurotoxin didn't work that way—and most allergies didn't, either.

By 2007, van Nunen had seen more than seventy patients with the meat allergy, many of whom had gone into anaphylactic shock. "I just thought, If this is an epidemic, we need to get the word out," she told me. That fall, she presented a summary of twenty-five cases at a conference in Fremantle, south of Perth. The reaction was muted. "People were actually stunned," she recalled. "They probably thought that I'd slipped off my trolley wheels." Years later, van Nunen told me, when a colleague introduced her as a speaker at a panel in Sydney, she mentioned the presentation in Fremantle and said, "We thought she had gone entirely mad."

Meanwhile, ten thousand miles to the northeast, an allergist and immunologist at the University of Virginia, Thomas Platts-Mills, was confronting a strange pattern of his own. He was investigating a drug called cetuximab, which contains genetically engineered antibodies that target cancer cells and prevent them from growing. In clinical trials, cetuximab had a good safety record—only three per cent of patients reacted poorly to it. But when it went to market, in 2006, it had a disturbing effect on patients in some areas. In clinics in Boston and California, the drug was well tolerated. But in North Carolina and Tennessee, nearly a quarter of the patients had severe reactions to the drug. In Arkansas, a patient died as it was being administered.

When Platts-Mills and his team studied blood samples from the reactive patients, they found allergy antibodies to a sugar molecule called galactose- α -1,3-galactose—alpha-gal for short. Most mammals have alpha-gal in their bodies, but humans and other primates do not. That's one reason our immune systems reject transplant organs from pigs. Cetuximab is produced

from both mouse and human DNA, so it does have some alpha-gal in it. But a single dose shouldn't have been enough to trigger a reaction. And why were Southerners so susceptible?

As it happened, a few patients had come to the university clinic recently with a strange complaint: they seemed to be allergic to red meat. "It sounded made up," Scott Commins, an allergy researcher at the University of North Carolina, told me. Commins was a post-doctoral fellow in Platts-Mills's lab at the time and had been assigned to study the patients. "You've been eating hot dogs and hamburgers all your life, and all of a sudden you're allergic to meat?" Most food allergies flare up within minutes of consumption, but these symptoms came hours after a meal. And the same patients weren't allergic to chicken, turkey, fish, or seafood. What was different about pork and beef? "We were all reading about alpha-gal," Commins told me. "So we started to say, 'Holy shit! What if it's related to that?'"

Humans and their ancestors have been eating red meat for millions of years. It seemed implausible that some cancer patients and carnivores would suddenly become allergic to it. Perhaps their immune systems were sensitized to alpha-gal by a local fungus, parasite, or pollen. The team had a map of patients who'd had allergic reactions to cetuximab—they were scattered across the Southeast, from Virginia to southern Missouri. But no other allergen seemed to have the same distribution. Then, in the summer of 2007, a research assistant named Jake Hosen got a hit. He was poring over maps of parasite-borne diseases when he came upon one for Rocky Mountain spotted fever, a disease often carried by dog ticks. It didn't quite align with the cetuximab map—the cases stretched all the way to Boston—but it was close. So Hosen looked up the range of a Southern tick, the lone star, common as dirt in Virginia. That map was a perfect match.

"We were thinking, Can ticks do this?" Commins recalled. "So, I started feverishly calling all the people in my study. Ninety-three per cent had a tick history." The final proof was provided, inadvertently, by Platts-Mills. In August of 2007, he was hiking in the Blue Ridge Mountains when he felt an odd, fuzzy feeling on his ankle. When he pulled down his sock, he saw what looked like hundreds of tiny seeds. Then they began to move. The seeds were a cluster of lone-star-tick larvae—a tick bomb, as it's called—some of them already feeding on his blood. Over the next few weeks, Platts-Mills

kept a record of his rising allergy antibodies. “He came in with the ticks in a Tupperware container,” Hosen told me. “And he said, ‘Draw my blood! Draw my blood!’ ”

That November—the same month that Sheryl van Nunen presented her tick theory to an incredulous audience in Australia—Platts-Mills went to a meeting in London. One night, for dinner, he ate three lamb chops and had two glasses of wine. A few hours later, he woke up in his hotel room feeling itchy. Then he watched, with some satisfaction, as his body broke out in hives. “Thus,” he later wrote, “the syndrome was now formally confirmed.”

Alpha-gal syndrome was an epidemiological novelty at first—an oft-told tale of far-flung scientists, baffling symptoms, and bloodthirsty ticks. Then it began to spread. The syndrome has now infected people in more than thirty countries on six continents. It can be found in Germany’s Black Forest and on small farms in South Africa, in the rain forests of Peru and the mountains of western Japan. Some tick species produce alpha-gal naturally in their saliva, which they inject into their host when they bite. In North America, the syndrome is transmitted mostly by the lone-star tick, but it seems to find new vectors wherever it goes. In Brazil, it’s carried by the Cayenne tick, in South Korea by the Asian longhorned tick, in Sweden and Norway by the castor-bean tick, and in Zimbabwe and Botswana by the South African bont tick. But its most effective carrier remains the paralysis tick. In Queensland and northern New South Wales, where most of the ticks live, the number of people with the syndrome is growing by forty-five per cent a year.

But those are just the known cases. Only a fraction of people with antibodies to alpha-gal become allergic to it, and those who are often don’t realize it. Some react only to red meat. Some also react to milk, cheese, and other animal products that have alpha-gal. The symptoms are so delayed that people often blame them on food poisoning, irritable-bowel syndrome, gluten intolerance. In the U.S., according to the Centers for Disease Control and Prevention, as many as half a million people have alpha-gal syndrome, but other researchers suspect that the number is higher. For years, testing was spotty at best, and doctors often weren’t aware of the syndrome. In a study published in 2017, fewer than one in ten alpha-gal patients were correctly diagnosed. In 2024, a New Jersey man died after eating a burger at

a barbecue—the syndrome’s first documented fatality—and at least three other deaths might be linked to alpha-gal. There are likely to be more.

Could a syndrome so stealthy have lurked among us all along? Commins and van Nunen don’t think so. The surge in alpha-gal patients was too sudden and too great. Van Nunen has worked on Sydney’s North Shore for forty years. “Anyone who had anaphylaxis was sent to me, and I’ve talked to long-term residents,” she told me. “I know that patients didn’t have it before. It is an epidemic.” What triggered it still isn’t clear. Does the alpha-gal in a tick’s saliva piggyback on some virus or antigen that sensitizes people to it? “We lack a unifying hypothesis,” Commins said.

One thing is certain: ticks make excellent disease vectors, purpose-built for the modern world. They’re resistant to pesticides, partial to rising temperatures, and happy to feed on anything that moves—mice, deer, birds, hunters, hikers, suburbanites, and second-home builders. As cities have sprawled into countryside, and people have pushed deeper into nature, we’ve turned ourselves into a vast smorgasbord for ticks. Messing with their habitat seems only to make them multiply.

In the nineteen-nineties, the Australian government launched a national campaign to eradicate red foxes. British settlers had imported them more than a century earlier for hunts, only to watch them exterminate native species like the numbat, the boodie, and Gilbert’s potoroo. The fox-eradication campaign was deemed a huge success, but it came with another unintended consequence. As the foxes died, the ticks that had fed on them found another primary host—the shrewlike native bandicoot, which Australians had worked hard to reestablish. “They became a nursery for ticks,” van Nunen told me. “It’s like Whac-A-Mole, trying to do something with nature.” The foxes went down; the bandicoots and meat allergies popped up.

Substitute bandicoots with white-tailed deer, paralysis ticks with lone stars, and you get the story of alpha-gal in America. Like the bandicoot, white-tailed deer were once an endangered species here, hunted nearly to extinction before the First World War. Then hunting laws were passed, habitats restored, deer reintroduced to their old territories. “We were an agrarian society and then we reforested the eastern United States,” Holly

Gaff, a tick researcher and mathematical ecologist at Old Dominion University, in Virginia, told me. “Forest brings wildlife, wildlife brings ticks, and ticks bring disease.” The U.S. now has some thirty-six million deer, wandering the landscape like a fleet of mobile blood banks. “They can make infinite ticks, basically,” Gaff said.



*"Someday her paintings will be on the walls of caves in New York."
Cartoon by Frank Cotham*

Even states already teeming with bugs have begun to feel under siege. “Every year, we bring in new pests, and we still have all the same ones we had before,” Nancy Hinkle, a veterinary entomologist at the University of Georgia, told me. “It’s a never-ending battle.” Some pests come into the country on livestock and tropical fruit; others hitch rides on shipping crates, wooden pallets, foreign travellers and their pets. In Georgia, chicken farmers are beleaguered by bedbugs, which have hopped from hotel rooms to poultry farms, and from there to workers’ homes. In Texas, ranchers are bracing for an onslaught of screwworms, the flesh-eating maggots of blowflies. Before they were driven south out of the U.S. in the fifties and sixties, screwworms killed millions of deer and cattle, burrowing under their skin and into the navels of newborn calves and fawns. Now they’ve worked their way back north through Mexico to the Texas border. In more than two dozen states, meanwhile, farmers are fending off Asian longhorned ticks. First spotted on an Icelandic sheep in New Jersey in 2017, they can carry a deadly cattle

parasite, *Theileria orientalis*. “We didn’t need another tick in Georgia,” Hinkle said. “We had twenty-two species already. That was a gracious plenty of them.”

The rise of the lone star may be the breaking point in this pest-ridden history. Like Lyme disease, named for a posh town in rural Connecticut, alpha-gal syndrome can seem like a symptom of privilege—of pampered city folks stepping out of their element. Martha’s Vineyard and the Hamptons have some of the highest alpha-gal rates in the country, but even there the people most at risk from infection may be outdoor workers. And the syndrome now threatens farmers, ranchers, hunters, and foresters across the South and the Midwest. “I have been standing up for thirty years saying, ‘If you get ehrlichiosis you could die,’ ” Gaff told me. “But if you say, ‘You might not be able to eat hamburgers or steak ever again,’ you suddenly get a reaction.”

Phillip James Ellison grew up in the bootheel of Missouri, wedged between eastern Arkansas and western Tennessee. “We come from crops,” he told me. “Cotton, dent corn, milo, soybean—that’s what I played in, as a kid in the fields. Then I moved up here to beef territory.” Missouri, Arkansas, and Oklahoma are the epicenter of alpha-gal in the United States. A study of blood samples from military recruits, taken between 2001 and 2007, found that nearly half the recruits from south-central Missouri, where Ellison lives, had allergy antibodies for alpha-gal. It’s not the best cattle country—scrubby pastures crisscrossed by tangled woods and limestone bluffs—but it’s ideal for ticks. Ellison’s wife, Kristie Byrd, belongs to a family that has run cattle east of Springfield for more than a century. When Ellison arrived, twelve years ago, he was put to work bottle-feeding the calves and transitioning them to grain. Then his body couldn’t take it any longer.

“I can’t even get near their dander anymore, or their wet nose or saliva,” he told me. “Whenever I handle one of them, I have to soak my hands—they’re so red and swollen.” The symptoms began not long after he moved to the area. Ellison would wake up with his stomach knotted, his skin crawling with a rash. It itched so badly sometimes that he would scratch himself bloody. “The doctors would say it’s lactose intolerance,” he told me. “You would not believe how many rounds of prednisone I’ve taken. Then I go to the dermatologist and he says, ‘Well, it’s just your eczema acting up.’ ” It

was ten years before he was given a diagnosis of alpha-gal. “Basically, if it gives live birth and breast-feeds, I can’t touch it and I can’t eat it,” he said.

We were sitting in his Dodge Ram pickup, looking out over a fenced-in pasture. It was a damp gray evening in March, and a few cows and calves were huddled out in the cold, waiting to be fed. Ellison, who is fifty-five, was slouched behind the wheel with the heater blowing on him. He had on a rumpled, fleece-lined flannel shirt and a Peterbilt hat tugged low over his eyes. “I sell truck parts now,” he said. “I’m not allergic to truck parts.” He pointed to a calf standing off by itself. He was not only allergic to their dander and saliva; they could also carry lone-star ticks that could make his condition worse. The saddest part was not being able to give the calves treats anymore, he said. They used to trail after him as if he were their mom. “I was named for the disciples in the Bible, and I still follow the Ten Commandments,” he said. “But in that ten-year period when I didn’t know what was wrong with me, depression took over. I had bad thoughts. I told my mom, If I didn’t have a daughter and granddaughter, I would have gone home by now.”



Michelle Kleeman, who has alpha-gal syndrome, on her family farm in Missouri. She is no longer able to go to restaurants with an open grill.

The alpha-gal stories I’ve heard often play like scenes from horror films. The victims are unsuspecting, the settings innocuous: a burger joint, a picnic, a quiet night at home. A pregnant woman wakes up to find her

husband leaning over her in the dark. “Something’s wrong,” he says. “My face doesn’t feel right. Could you go with me to the bathroom and look at it?” When she turns on the light, the left side of his face is so badly swollen that he’s nearly unrecognizable. An eleven-year-old girl comes home with her hands covered in blisters. There’s something weird in the school’s new soap, she tells her mom.

The syndrome is cryptic, unpredictable. It can lie dormant in the blood for years—even elevated allergy antibody levels may never trigger a reaction—then erupt with sudden violence. Without tick bites, red meat, or dairy to incite it, the immune response will subside eventually. After three or four years, you might even be able to eat a cheeseburger again. But if you live or work in the countryside, the opposite is more likely to happen. Another tick will latch on, and your immune system will reignite. The more bites you get, the more allergic you’ll probably become. At first, it may take a whole steak to set you off, then just the smell of bacon will do it. “I can’t go to restaurants with open grills anymore,” Michelle Kleeman, who owns a small farm an hour and a half west of Ellison’s place, told me. “My throat would be swelling from the smoke.”

In rural communities, alpha-gal doesn’t attack just the body. It attacks a way of life. Farming, ranching, hunting, raising and eating animals: everything you’ve been taught to do is suddenly bad for you. Even the landscape has turned toxic. Forty years ago, when the [AIDS](#) epidemic was ravaging New York, some Southern Baptists described it as God’s judgment—a punishment for its victims’ life style. I never heard alpha-gal described that way, but there were other, equally sinister explanations.

“I’m sure you’ve heard all the conspiracy theories,” Rick Blubaugh, an emergency-room doctor in Branson, Missouri, told me one morning, at a local diner. “There is a whole counterculture out there.” Lyme disease began as a biological weapon, one theory suggested. The bacterium that causes it, *Borrelia burgdorferi*, was named for the entomologist Willy Burgdorfer. During the Cold War, Burgdorfer did research for the U.S. military, infecting ticks for potential use as a bioweapon. But the Lyme-disease theory was soon debunked: *Borrelia burgdorferi* has been around since at least 1894, tick researchers pointed out, when some mice with the bacteria were collected on Cape Cod. Alpha-gal was more of a private enterprise, another

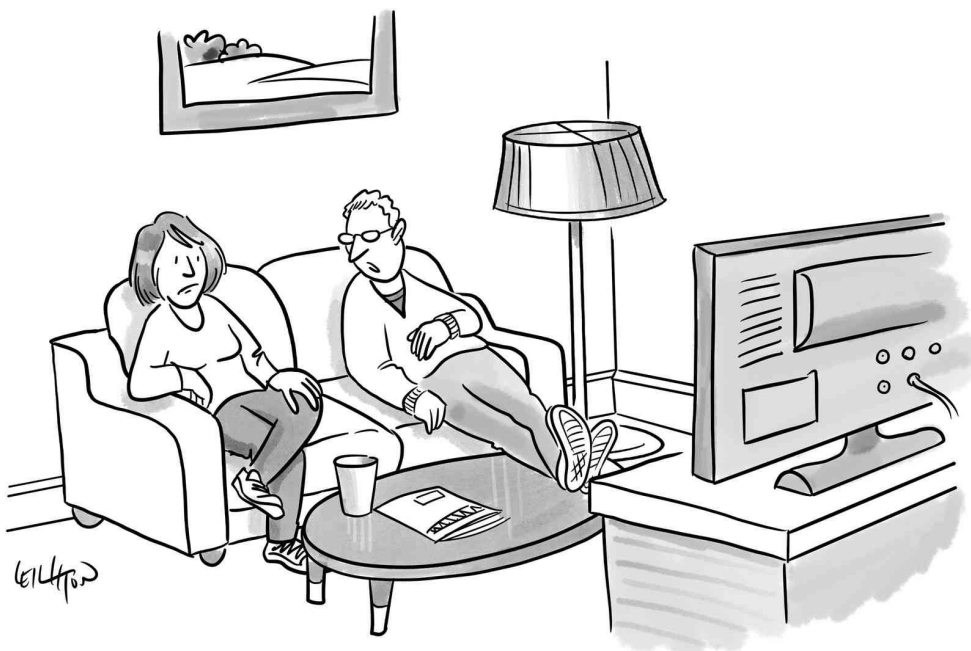
rumor contends. [Bill Gates](#), the co-founder of Microsoft, has invested heavily in lab-grown meats, which have no alpha-gal. His foundation has also funded research to genetically modify ticks to protect cattle. What if that work triggered the epidemic? “[Covid](#) got out of the lab. Did this get out, too?” Blubaugh said.

That theory has also been roundly dismissed: the research funded by the Gates Foundation wasn’t on lone-star ticks, and it had no relation to alpha-gal. Yet Blubaugh can see why the syndrome’s sudden rise seems suspect. When he started working in Missouri, twenty years ago, he almost never saw alpha-gal cases. “It’s almost daily now,” he said. People used to mistake the syndrome for other allergies; now they do the opposite. They come in with a rash, a stomach ache, a racing pulse, and assume that they’re allergic to meat. “They’ve already given themselves two or three epi shots and they say they need more,” Blubaugh told me. “And I think, Let’s hold off. Epinephrine makes you jittery and anxious, and your heart rate goes up. Maybe it’s causing the problem.”

True anaphylaxis is hard to mistake. When it seizes the immune system, the body goes haywire in a few ways at once. The skin may blister and the eyes inflame, the stomach knot up and the bowels loosen. If the throat and the tongue swell up, people often make an odd sound when they breathe in—a whistling squeak, or screech, known as stridor. “That’s when I’ll give epinephrine,” Blubaugh said. He gets a case like that almost every week, he told me. But that, too, gives him pause. “People have been getting bitten by ticks for thousands of years. Why are we seeing this disease all of a sudden?”

Matthew Overcast, a state representative for southern Missouri, didn’t believe in alpha-gal when he first heard about it from his family doctor. “I wouldn’t have said it was a conspiracy theory,” he told me. “But when he threw it out there I was, like, ‘Wait, what?’ ” Overcast has since become one of the country’s strongest advocates for alpha-gal awareness. When his oldest daughter, Lyla, was in kindergarten seven years ago, she began to suffer from escalating waves of symptoms: nausea, rashes, cramps, migraines, and easily fractured bones. Before her doctors diagnosed her with alpha-gal, they thought she might have leukemia or an autoimmune disease. “As a parent, you’re both pissed off and relieved when it turns out to be just

an allergy,” Overcast said. Around that time, Lyla’s sister Adalynn began to get hives and stomach aches; then her brother Jackson came home with a rash on his face. Both were later diagnosed with alpha-gal, along with Overcast’s aunt, his youngest brother, and his wife, Jessica.



*“The Supreme Court has declined to uphold a ruling overturning the Ninth Circuit’s denial of a stay to block a district-court ban. Happy? Or outraged?”
Cartoon by Robert Leighton*

Last year, Overcast filed a bill with the Missouri legislature requiring labs to report and track all positive tests for alpha-gal. (It was passed in May and awaits the governor’s signature.) The data on blood samples from military recruits are twenty years old, and without up-to-date numbers it’s hard to know where and how to combat the syndrome. Twelve other states now track alpha-gal, and the preliminary numbers are alarming. In Virginia alone, since September, more than fourteen thousand people have tested positive for alpha-gal antibodies, most of them, probably, after showing symptoms of the allergy. On a per-month basis, that’s more than ten times the cases of Lyme disease recorded between 2010 and 2020—and Virginia’s alpha-gal rates aren’t nearly as high as Missouri’s.

It’s not clear what happens next. “I don’t see alpha-gal in the same light as [measles](#) or tuberculosis,” Blubaugh told me. “The data may help get more funding, but what’s the end goal? Can we cure it? Can we control it?” How do you stop an epidemic that seems to be everywhere and nowhere at once? That upends lives yet often goes undiagnosed? That requires wrenching

social changes in how people eat, socialize, and move through nature? As Overcast put it, “It’s the only allergen you can’t avoid.”

Red meat and dairy are just the beginning. Animal products are woven so tightly, so microscopically, into what we consume that we often don’t know they’re there: Doritos, ramen, marshmallows, and frosted cereals; French fries cooked in beef tallow; tofu prepared on a grill that just had meat on it—even the most innocuous products can contain beef, pork, or dairy. The amount may be infinitesimal—white sugar and bottled water are sometimes purified with charred animal bones, for instance—but some hypersensitive alpha-gal patients avoid the risk anyway. They scan food labels for “natural flavoring”—industry shorthand, in many cases, for beef and pork extracts—the way other people scan for artificial coloring.

“When Lyla was first diagnosed, I cried in the grocery store,” Jessica Overcast told me. “I didn’t know what to feed her.” It was a weekday night, and the family was having dinner around the kitchen table. The Overcasts live an hour southeast of Springfield, in a small white farmhouse on a county road lined by forests and fields. Jessica, who is thirty-two and works for an online educational company, had made roasted salmon. While she talked, her four youngest, ages five to eleven, bounced and lolled and hotdogged around the kitchen, peering at me curiously, then dashing away. Lyla was baking boxed brownies for dessert. She was in seventh grade now, tall and skinny, with glasses, braces, and a bright, impish, self-possessed spirit.

“We’re really blessed that they like their vegetables,” Jessica said, putting some mashed broccoli and carrots on the table. “Even supplements can be difficult to know if they’re safe.” Iron and Vitamin B₁₂ are often derived from beef organs, Vitamin D₃ from lanolin secreted by the sebaceous glands of sheep. Other supplements come in gelatin capsules, made from collagen derived from cow and pig bones and hides, or mixed with dairy binders such as lactose and sodium caseinate. We wash our bodies with animal products (soap can contain tallow), brush our teeth with them (toothpaste can contain glycerin obtained from animal fat), and moisten our skin with them (lotions can contain lanolin, collagen, and tallow). “I always tell my husband, we have done an excellent job,” a rancher in Missouri, whose son has alpha-gal, told me. “I am so proud of how the beef industry has infiltrated every single thing.”

If there is one thing that would improve the lives of people with alpha-gal, Scott Commins told me, it's better labelling: "Just tell us what's in this stuff!" Drugs are a particular risk. When they're injected into a vein, the allergic reaction can come within minutes rather than hours, and it can be life-threatening. Heparin, a common blood thinner, is extracted from pig intestines. Insulin can come from pig pancreases, estrogen from the urine of pregnant mares. Vaccines can contain amino acids or enzymes from beef. Antivenins for snakebites are made from horse blood. "The medical community is going to have to have a reckoning," Lea Hamner, an epidemiologist on Martha's Vineyard, told me. "Rabies is a hundred per cent fatal. Do you avoid the vaccine or risk the allergic reaction?" Even drugs that were once safe may have since been reformulated with animal products, she said. "Having to do that level of research on everything—that is a terrifying place to be."

The Overcasts are old hands at avoiding alpha-gal by now. "It's really not something you want to screw around with," Matthew Overcast told me, at his house that night. "It can go from a moderate reaction to anaphylaxis like that." He snapped his fingers. "We're an hour's drive from the hospital. We only have two EpiPens per kid, and you have to administer them every fifteen minutes. So, we're kind of in a bind." Last year, to keep some red meat in their diet, they took up emu farming. Seven of the prehistoric-looking birds were stalking around a pen outside, glaring at visitors as if we were the ones on the wrong continent. Inside the house, a plastic bin in the TV room held a striped emu chick, waddling under a heat lamp.

"We tried ostrich, but they died off in the first week," Matthew said. Then, last spring, a tornado hit a local emu farm, and one of the birds got loose when his cage was destroyed. "He'd been wandering around for two or three months when we rescued him. We're hoping to find an affordable way to produce them, but it's a little bit of a challenge." Their house sits on only an acre of land—when Overcast quit his job at a law firm to be a state representative, he took a six-figure pay cut—and the closest U.S.D.A.-certified slaughterhouse that will process the birds is in Arkansas. Still, for now, the meat is worth it. Emu is darker than beef and just as rich, yet it has no alpha-gal. "We did a blind taste test and a lot of people failed," Jessica said.

“It tasted just like the real thing!” Lyla said. I asked her if she missed red meat. “Oh, definitely. Burgers! They just poofed out of existence, and I was, like, ‘Come back!’ ”

Red meat may one day be widely available for people with alpha-gal, but it probably won’t come from giant birds. Six years ago, the F.D.A. approved the first bioengineered, alpha-gal-free pig for human consumption. Bred by a biotech company called Revivicor, now owned by United Therapeutics, it was originally designed to provide organs for transplant. To prevent the organs from being rejected by their new human hosts, Revivicor deactivated a gene that codes for the enzyme that creates the alpha-gal molecule. A kidney from one of those pigs was the first to be transplanted into a human body, at N.Y.U. Langone Health, in 2021.

Eating the pigs was an afterthought. “We are very, very focussed on xenotransplantation,” John Bianchi, the vice-president for regulatory affairs at United Therapeutics, told me. The GalSafe pigs, as they’re called, are raised in an unmarked facility in rural Virginia, surrounded by razor wire and surveillance cameras. “We don’t want to invite any troublesome actors,” Bianchi said, when I went there in April. “There is incredible public concern and anxiety about gene-edited animals. But I think the GalSafe pig turns that upside down. Now all of a sudden you have a product that primarily benefits the consumer rather than the producer.”

My visit to the pig facility felt like touring a biohazard site. Codes were entered, security officers alerted, the car hosed off. To protect the pigs from contaminants and human pathogens, I had to change into sterile scrubs, rubber boots, and gloves, then dip my soles in a pan of disinfectant. Still, when I stepped inside the nursery, I found a drove of perfectly ordinary-looking pink pigs. They were four months old, trotting around in large pens with feed piped into bins from above. Some big rubber balls had been tossed in to keep them distracted. The gene editing had no effect on their health, Bianchi assured me, and they tasted just like other pigs. So far, only one company has been licensed to sell GalSafe pork: Amaroo Hills, based in North Carolina and Tennessee, which also raises emu. But word has spread in the meat-starved alpha-gal community. “I’m hoping to get my hands on a hog or two who have that gene removed,” Matthew Overcast told me.

The day before I visited the pigs, I'd tasted some GalSafe pork at the University of North Carolina. It was being served to an alpha-gal patient as part of a double-blind clinical trial organized by Scott Commins and Sarah McGill, a gastroenterologist at the medical school. McGill had chosen thirty patients who tested positive for alpha-gal and had symptoms of the syndrome. In previous weeks, they had twice come to the university and eaten a bowl of cooked ground pork. Then they waited six hours to see what happened. The pork looked identical on both visits, but one serving came from a regular pig, the other from a GalSafe pig. Neither the patients nor the researchers knew which was which.

The GalSafe pork I tasted was as ordinary as advertised, just a little bland for lack of spices. McGill and Commins expected it to have no allergic effect on their patients, and for the opposite to be true of the regular pork. But when McGill sent me her summary of the study later, the results were equivocal. The regular and the GalSafe pork caused allergic reactions at about the same rate—roughly twenty per cent of the time. The GalSafe caused only nausea and, in one case, faint flushing; the regular pork gave some patients hives, as well, and sent one woman to the hospital with recurrent vomiting. But most patients had no reaction to either meat.

“I have more questions than answers right now,” McGill told me. Were some of the patients false positives for alpha-gal? Did they have to eat meat more than once to reignite the allergy? Did their fear of a reaction alone make them react to the GalSafe pork? “It’s curious. It’s discordant,” she said. “I know that people can get way sick from alpha-gal, and I think it’s still underdiagnosed. But we’ve got a long way to go. It’s really humbling.”

Food intolerances are part mind and part body. They’re bound up in our habits and aversions, fears and imaginings. If you’re allergic to bees, a sting can kill you. But most allergies aren’t that extreme—they’ll give you hives but won’t stop your heart—and most intolerances aren’t allergies at all. You may get indigestion from gluten or lactose intolerance, but you’ll never go into anaphylactic shock. True allergies build up antibodies in the blood, which react to the presence of certain molecules. But even a system crawling with allergy antibodies sometimes needs another nudge before it attacks.

“I get tick bites, and I make low-level antibodies for alpha-gal,” Jeffrey Wilson, an allergist and immunologist at the University of Virginia, told me. “But I love to smoke brisket, and I never stopped. Why can I do that, and others end up in the emergency room?” Wilson believes that some reactions could be triggered by sources other than alpha-gal. He likens it to chronic urticaria, another condition that some of his patients have. “They break into hives for no rhyme or reason,” he said. A tick bite boosts allergy antibodies of all sorts, not just for alpha-gal. In some people, this could prime their system for reactions at any time, he said. “It’s like you have too much information in your body, and, with stress or other life things, you break out.”

Americans didn’t used to be this sensitive. Between 1997 and 2011, the rate of food allergies in children increased by fifty per cent, then did so again between 2007 and 2021. When I was in Missouri, I had lunch at a café in Springfield called the Safe Spoon. The menu was free of alpha-gal and of the nine most common food allergens—dairy, eggs, fish, peanuts, sesame, shellfish, soy, tree nuts, and wheat. “The term ‘vegan’ is basically a curse word around here,” the owner told me. Yet the place was packed. Many of the customers had more than one allergy, she said, and some hadn’t been able to share the same meal with their families for years.

Allergists themselves are partly to blame. For decades, they told new parents to keep peanuts and other allergenic foods out of their children’s diets. But the gut, it turns out, is extremely good at habituating itself to allergens. The skin and lungs are not. When children stopped eating peanuts, the allergens still found ways into their bodies. Between 1997 and 2010, peanut allergies quadrupled in the U.S.

Such overcaution is at the heart of three of the most popular explanations for the allergy epidemic: the hygiene hypothesis (we’ve stunted our immune systems by being too clean), the old-friends hypothesis (we’ve ravaged our gut biome with bad food and antibiotics), and the epithelial-barrier hypothesis (we’ve weakened the protective linings of our skin and lungs with disinfectants and other chemicals). But none of them can fully account for the rise of alpha-gal syndrome. “That is one of the fascinating things about it,” Commins told me. “It breaks all the allergy rules.” Some of the worst cases of the syndrome, for instance, have been on poor farmsteads in

South Africa's Eastern Cape. "They represent everything we've been told about nonallergic environments," Commins said. "Dirt floors, babies without shoes, animals in the home, yet still you can become allergic."

The results of the GalSafe-pork trial were, in some ways, par for the course. If you go into any classroom and draw blood from a hundred third graders, Commins said, fifteen or so will test positive for peanut antibodies, but only three or four will have symptoms when they eat peanuts. "This is a problem with all allergies," he said. "I think it's worse with alpha-gal because of the dang ticks." Sometimes the antibodies can be stirred to action by exercise, alcohol, or a combination of the two. (In one German alpha-gal study, the researchers had some subjects drink a glass of wine or half a litre of beer and spin on a stationary bike.) Sometimes the antibodies just can't be bothered.



The Overcasts are so keen to reintroduce red meat to their diet that they have started raising emus. Adalynn and her sister Josephine tend to one of the birds.

Commings is working on a more definitive test for alpha-gal syndrome. In his lab at the university, he showed me a picture of a group of mast cells—the immune system's first responders—as seen through a microscope. He called them Cluster 43. The cells had appeared in blood samples from patients with the syndrome after he activated their antibodies with an alpha-gal antigen. If the method could be automated, it could cut down drastically on false positives, Commings said. "It's fancy but not otherworldly; the technology to

do it could be there in three to five years.” It could also lead to more targeted treatments.

When Commins has patients who are highly allergic, he often prescribes Xolair. Originally approved for asthma, Xolair can also prevent severe reactions to food allergies. Some patients even eat meat while they’re on it, though it’s not yet approved for that use. Others swear by [acupuncture](#) that focusses on the nerves of the outer ear. “I’m fascinated by it,” Commins told me. “The best I can think of is that the needles are manipulating the signal from the vagus nerve—this master controller that has a branch that goes to the back of the ear. Mast cells sit at the end of nerve endings. So it may be that acupuncture somehow manipulates what the mast cell is sensing.” Allergists have always downplayed the role of the nervous system, he said. “But I think about my own kid. He eats a walnut and has hives within seconds. That has got to be the nervous system. Nothing else moves that fast. In my mind, that is where we are going. That is the next great thing.”

In the meantime, alpha-gal is a rare challenge to public health. It has no cure, no ironclad diagnosis or epidemiological data, and an expanding horde of ticks to spread it. The only way to stop them, Patrick Roden-Reynolds told me, when we were collecting lone stars on Martha’s Vineyard, may be to kill their hosts—to treat deer less like outdoor ornaments and more like supper. “You want my bold opinion?” he said. “We need to eat our way out of this problem.”

On my last day on the Vineyard, the leaders of the anti-tick campaign gathered on [Zoom](#) for a war council. Roden-Reynolds, the epidemiologist Lea Hamner, and Virginia Barbatti, the executive director of a nonprofit called Tick Free Martha’s Vineyard, were hunched over their laptops at locations around the island, scrutinizing a map of enemy encampments. Earlier that day, Barbatti’s group had released a sobering new survey of the island’s deer population. Using drones with heat-seeking sensors, it had located nearly five thousand deer on the island’s eighty-eight square miles of land—four times the deer density of central and western Massachusetts.

“Tick experts, when I tell them our deer density, they just laugh,” Hamner said. “They say, You don’t have a chance.” Hamner, who is thirty-three, is accustomed to places in crisis. Her father worked for [U.S.A.I.D.](#) in the

Philippines; her mother worked for the World Health Organization there, and later did her doctoral research on sex workers in Peru. Hamner's first jobs, after getting her master's in public health at Yale, were in Washington State, where she was plunged into work on tuberculosis, [opioid addiction](#), and then the pandemic. Skagit County, where she was based, had some of the first documented cases of *COVID* in the U.S., including fifty-two members of the Skagit Valley Chorale. In 2021, when Hamner moved to Martha's Vineyard, where her husband was born and where she'd spent summers as a girl with her grandparents, she wasn't expecting to work on public health right away. "I really needed a break," she told me. Then came alpha-gal.

"This is an existential threat," Hamner said. "Not just to our health and well-being, but to our economy. People will leave and not come back." The taxes paid by second-home owners keep the island's public services afloat, and most businesses survive on tourism in the summer months—when all the ticks come out to feed. "People with alpha-gal will be the ones to leave first," Hamner said. "To have a chance to get over their allergy, they need to not get more tick bites." Barbatti nodded. "There is this sense of, Am I putting my kids at undue risk?" she said. "Are we going to have a generation of kids going to college with EpiPens?"

Martha's Vineyard is more than a microcosm of the epidemic. It's a testing ground for addressing it, Hamner said. "An island is a controlled environment to do experiments and then scale out. We have high need, high burden, high impact, and we really need to incubate a solution. Once we get the blueprint, we can go to, How do you translate this to Missouri?"

The first impulse, for many locals, is to smother their yards in pesticides. But that almost never works. Lone stars can tolerate many poisons, and, even when they can't, there are always more ticks next door to replace them. "You can't spray a whole city," Holly Gaff, the tick researcher at Old Dominion, told me. Gaff and a team of engineers at Virginia Military Institute have designed a TickBot—a sort of outdoor Roomba—that can drag a permethrin-soaked cloth around a lawn. Another device, called a 4-Poster bait station, attacks the ticks that congregate on deer's heads. When deer bend down to eat the bait, paint rollers cover their heads in permethrin. But those are stopgaps at best.

“Zoom in on the deer locations,” Roden-Reynolds said. He blew up the shared map on his screen and moved his pointer to a dark, amorphous area fringed with bright-yellow dots. “Compare the conservation areas to the neighborhoods,” he said. “Look at Peaked Hill Reservation. It has a few deer, but the neighborhoods around it are just loaded.” Red Gate Farm, [Jackie Onassis’s](#) old estate, was especially popular. Deer aren’t idiots, Roden-Reynolds said. Hunting is allowed in conservation areas, but illegal within five hundred feet of an occupied house without the owners’ permission, so the deer go where they won’t be shot. He zoomed the map out again. All over the island, the residential areas were clustered with dots, along forest edges and around people’s yards. By day, the deer stay close to the houses, but not too close. At night, they move in and eat the hydrangeas.

Truly controlling the ticks will take an island-wide killing spree, the war council agreed. Deer numbers will have to drop by at least seventy per cent—more than three thousand animals. Local hunters, by all accounts, are ready and able. The hunting season has been lengthened by nearly two months, and there is no bag limit on does. Tick Free Martha’s Vineyard plans to build a processing facility for the carcasses; the meat, most of it donated to a local food pantry, is in high demand. “This island elevates fishermen. It elevates farmers,” Barbatti said. “This is our opportunity to elevate the hunting community.”

The deer are easy pickings for now. Roden-Reynolds is a bow hunter. In the sixteen years before he moved to the island, he shot a total of four deer. Now he shoots five a year. Hamner’s in-laws have a family hunting camp that killed fifty in one week this past fall. “They think they did bad if they shoot forty,” she said. But the fewer deer there are, the harder the hunting will be. In the late nineties, when Lyme disease swept over Monhegan Island, off the coast of Maine, a professional sharpshooter was brought in to kill every single deer. The campaign worked: the tick population plummeted. But Martha’s Vineyard won’t go that far—not yet, anyway. “This is just step one,” Barbatti said. “It is the lever that is available to us now.”

In the meantime, like the paralysis ticks in Australia, the lone stars will find other hosts to feed on, such as short-tailed shrews and white-footed mice. “There are plenty of little nurseries out there,” Nancy Hinkle, the veterinary entomologist in Georgia, told me. “Possum, raccoons, dogs, cats, rodents,

foxes, skunks, us.” Nothing can match deer for hosting ticks, so the lone-star population will drop, but it will never go away. “They’ll survive long after we do,” Hinkle said. “They’ll survive after nuclear annihilation.”

Late that night, I walked back to my hotel in Vineyard Haven, relieved to be heading home the next morning. I’d spent the past few months in some of the country’s most beautiful landscapes—the Ozarks, the Blue Ridge Mountains, and this bright, windblown island—but always with a vague sense of unease. Every passing lawn and meadow, every winding path through the woods, seemed to hide a lurking menace. That may be what’s hardest to bear about alpha-gal: how it puts us at odds with nature, makes us distrust our most beloved places. It would be good to be back in New York City, I thought, in the relative safety of brick and concrete.

Wishful thinking, as it turned out. A health alert from the city later informed me that lone-star ticks are now comfortably settled on Staten Island and parts of the Bronx. Eastern Long Island has been overrun for decades, and Brooklyn could be next. All it would take is a tick bomb or two to infest Prospect Park, three blocks from my house. The barbarians are crawling toward the gate. ♦

Shouts & Murmurs

- [My Great Meeting with God](#)

By [Patricia Marx](#)

He called me. I want to be clear about that. I didn't call Him. He invited me up personally. Very selective crowd. Very hard ticket. I've met with many world leaders—Putin, Kim Jong Un, Jared—and I would say God's in that tier. In terms of money, probably higher than most of them. We had a great table. God has tremendous clout.

Incredible gates. Strong. A lot of harp people hovering. Maybe too many, if we're being honest. And the little fat bellhops—cherubs, they call them—wings too small for the body. Slow with luggage. I didn't tip. The show! Better than Vegas. Volcanoes lighting up the sky, or whatever you call it. World-class earthquakes. I said, "God, I'm ashamed to admit it, but the earthquakes we have on Earth don't compare."

He was very complimentary. He said, "They call them Acts of God, but, after what you've accomplished, maybe we should start calling them Acts of President." Which was nice. Somebody asked me before I came out here, "Was He taller than you expected?" I said, "About what I expected."

Then God let me smite a guy trying to parallel park. Which was a very small part of the meeting, by the way. People are asking why we didn't discuss the suffering of innocent children. It came up. I know children. His son was bullied. Good for ratings, by the way. Then we discussed important issues. Very positive discussions about clouds. I want clouds over Washington, but not the poor neighborhoods. Strong clouds. Not wispy. Some of His clouds—and I don't want this to get beyond this room and Truth Social—were too fluffy. "Another thing, God," I said. "You got tremendous demand up here, but everybody's dead. Why not go members only?"

"Frankly, God," I said, "can I be frank?" I didn't say "regime change," but He knew what I meant. He asked me how the Miami Marlins were doing. I think He wanted tickets. "Look," I said, "I came here for a reason. Nobody's saying you did a bad job, but . . ." People tell me, "Mr. President, you put the fear of God in God." He knew I could back this up militarily. Operation I Mean It. Or, Operation I'm Not Going to Tell You Again. Or, Operation I'll Give You Something to Cry About!

I haven't decided yet.

While God did God things, I looked around for the gift shop. You know the thing about that place? Bad signage. I know more than He does about real estate. Don't know who His lighting guy is, but the sun shouldn't come down so fast. You're outside eating dinner, and the next thing you know it's dark.

Did I get everything? Look, when I walked in, Earth was one way. When I left? Very different conversation. Many, many historians are saying it was the most productive meeting since Zeus met with President McKinley. I can't share everything, but I will say things are on the table that weren't even on the floor. I can say we're very close to getting my parking ticket validated.

His last words to me were "You know, of everyone I've ever created, you might be the one I'm proudest of." I probably shouldn't say this, but when I left He cried a little.

Update: Unnamed sources say that the Commander-in-Chief is evaluating a potential 2028 bid for God. ♦

Fiction

- [“Pig Lab”](#)

[Fiction](#)

Pig Lab

By [Will Mackin](#)



Illustration by Henning Wagenbreth

Late as I was, I figured I'd be walking right into an active scenario. Crazy Omar, maybe. Or Outnumbered and Outgunned. I thought I'd hear the thud of stun grenades and hillbilly shouts of "*Allahu akbar!*" I expected to taste the bitter clouds of cordite drifting over the lawn and see the green muzzle flash of machine guns firing blanks on full auto. Above all, I assumed I'd find wounded pigs dying on the grass, and the boys elbow deep in their salvation.

Turned out, however, that I hadn't missed much. The pig men were in the trailer, knocking out the pigs with anesthesia. The pigs, perhaps sensing the looming betrayal, or at least recognizing a serious departure from their bottle-fed, sunny upbringing, fought back. Hooves banged against the metal floor. Squeals cut through the cool air. Steam leaked through the trailer's slats. Pig fear had the same sour reek as our fear.

I found the boys stooging in a patch of morning sun at the base of our compound's tower.

Ted Waters, with his real leg crossed over his mechanical one, eyeballed my approach. “Nice of you to join us,” he said.

Bing Thomas spoke out of the corner of his new mouth. “Look who deshided to show up.”

[Read an interview with the author for the story behind the story.](#)

I’d planned to arrive early, before the boys and before the pig men, just as I had for the previous pig labs I’d attended in preparation for my troop’s deployments. I liked to give myself time to prepare for the day’s events. The scenarios were nothing more than window dressing—meant to evoke the chaotic night raids that would be our mission downrange. It was the pigs that had me worried. A pig’s face might remind me of Bing’s old face. Or a pig’s leg lying off to the side, detached and jittering, could resemble Ted’s leg. I’d wanted to arrive early enough to think through these possibilities and allow other, half-forgotten things to bubble to the surface. But, just as I was about to leave the house, my wife, Allison, had screamed from the upstairs bedroom.

“Circumstances beyond my control,” I explained to the boys.

The trailer’s back gate swung open, and a pig man wearing a blacksmith’s apron emerged. He’d been blown up, I guessed, then put back together minus some pieces. I suspected this because a curious light shone from around his joints, as if some kind of spell held him together, specifically the type of spell that would break if anyone drew attention to it. He wore a red beard that looked like it had been carved out of wood, dipped in wax, and clicked into place. He approached us on a curve.

“Who’s done this before?” he asked.

I raised my hand.

Red Beard spoke in a low, electric monotone. “Someone besides him, then. Why are we here?”

Officially, we were there to learn—via live, unconscious, anesthetized, and wounded pigs—how to manage battlefield amputations, hemorrhaging,

sucking chest wounds, skull fractures, and the like. Unofficially, however, this was a sacrifice—pigs instead of us.

An answer to Red Beard's question came from behind me: "We're here to fuck up some pigs."

No one laughed.

"Who said that?" Red Beard asked.

It was the new guy. Untested as he was, his body was whole, and his mind was intact. As a result, his voice was confident, clear, and amplified by an empty-headed belief in his own good intentions.

New Guy tried to fix himself. "It was a stupid joke," he said.

From the pig trailer came a bang, followed by a muffled squeal.

"These pigs are our brothers, not jokes," Red Beard said.

"I'm sorry," New Guy said. "I wasn't thinking."

Red Beard nodded, and a flash of light escaped from a scar on his neck.

"Sorry's not good enough. You can leave."

Podcast: The Writer's Voice

[Listen to Will Mackin read "Pig Lab."](#)

Allison's scream was not that of a full-grown woman, the kind I'd heard on so many night raids. There was no depth to it, no hint of recognition of the horror that had provoked it. It was, instead, the simple, guileless scream of a young girl, which meant that my wife was having a flashback.

I ran upstairs and into the bedroom, where I found the bed empty, the covers flung aside.

I got on my hands and knees to look under the bed, which was supposed to be Allison's safe place. This had been decided in therapy, which we'd signed

up for after Allison had got pregnant, and her memories, which she'd suppressed for decades, had started to become a problem. The therapist was a round, soft woman who sat cross-legged on a fuzzy couch. Allison and I sat on a smooth couch opposite her. There was a glass coffee table between us, and under that was a cardboard box full of toys. A big clock hung on the wall, ticking softly.

The therapist said that Allison needed a place to go where she would feel protected. When Allison couldn't think of a safe place on her own, the therapist suggested under the bed. I wanted to hug my wife and tell her that everything was going to be all right. I wanted to convince her of this with the kind of hug that was a safe space in and of itself, that would feel like the inverse of all those dark and musty spaces that the boys and I, while rooting out the enemy downrange, had torn big loud holes in. But, as the therapist explained to me, Allison needed to be in a place where no one would try to convince her of anything.

I didn't find her under the bed. I didn't find her in the bathroom, either. She wasn't curled up under the toilet tank, where I'd discovered her once before. Nor was she lying fetal and naked in the cold and empty tub.

In the four months that Allison had been pregnant, I'd learned a few of the things that triggered her flashbacks: Her father's birthday was one. Her mother's birthday, just two days later, was another. The smell of spoiled milk (which, until age thirteen, she had thought was the smell of fresh milk); and the sight of the little girl from down the street, standing off to the side, away from the other kids at the bus stop, with her dirty face, uncombed hair, dingy stockings, and untied shoes.

Allison was thirty-one years old, tall with long auburn hair and a bright smile. In the throes of a flashback, however, she became a terrified child. Her heart hammered away, her eyes flickered, and her voice grew tiny and plaintive. The flashbacks often happened at dawn, as the world was coming into focus, or in the evening, when it was starting to dim. The moment a flashback began, Allison would hide. I did not turn on the lights to find her, and I did not call her name, because these were things that her father had done. Instead, I stood quietly and listened.

Explosions had all but ruined my hearing. As a result, there was no such thing as quiet. Sometimes crickets chirped in my head. Other times, a phantom string quartet warmed up. Standing in our bedroom that morning, I heard a pot of water rumbling to a boil.

Then, from the baby's room, I heard a whimper and a thump.

Pigs were lined up on their backs on the grass, anesthetized, mouths wide open, tongues hanging out, eyes slammed shut. Armed pig men stood at either end of the line. Closest to the fence was a pig man with a rubber hand, who was gripping a hatchet with his flesh-and-blood hand. The pig man at the other end of the line wore an eye patch and carried an M4. These two stood ready to wound the pigs as soon as Red Beard was done saying goodbye.

Red Beard knelt alongside a pig, kissed the top of its head, and gave it a hug. Slowly, he rose to his feet and moved on to the next pig.

I walked over to where Red Beard was kneeling. Eye Patch rocked on the balls of his feet and hummed. Sunlight chimed off Rubber Hand's hatchet blade. Red Beard's palm rested on the pig's chest, right above its heart. My shadow fell on him.

"What do you say we call this off?" I said.

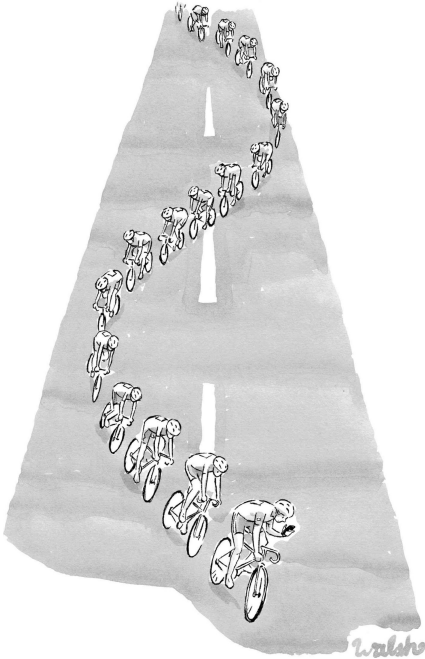
The sun, directly behind me, gave my shadow a halo. Red Beard looked up at me. "How so?" he asked.

"You take the pigs back to the farm, and I take the boys back to the compound. If higher asks me how the training went, I'll say it was great."

"But your men haven't done the training."

"Yeah, they have. All of them did it, except the guy you sent home."

"None of them raised their hands."



Cartoon by Liam Francis Walsh

“They don’t like raising their hands. It makes them feel stupid.”

“What about good order and discipline?”

The light that shone from Red Beard’s joints brightened.

“Were you like this before?” I asked.

I had been walking behind Ted Waters on that blue night, along the highway outside Marjah, in Afghanistan, when the bomb took his leg. The explosion had balled me up, shoved me aside, and pelted me with hot stones. Aside from a sharp ringing in my ears, however, I was fine. I’d knelt next to Ted as he lay on his back, conscious and bleeding on the pavement. “What happened?” he’d asked me as I fastened a tourniquet around his tattered thigh, then looked at my watch. It was twenty-three minutes after midnight, so I wrote “0023” in grease pencil on Ted’s forehead. “You’re going to be fine,” I’d said. But Ted had already begun to realize that he was not going to be fine. And every now and then since, whenever things were quiet and no one was paying attention, he thought back to a time when he was fine. Now it seemed as if Red Beard were remembering, too.

“You can’t choose the things you want to do versus those things you don’t,” Red Beard said.

“My wife is pregnant,” I told him.

“Congratulations,” Red Beard said. “I have a newborn son myself.”

After Ted Waters and Bing Thomas were wounded, I put them both on injured reserve, thinking that I was doing them a favor. They could retire early with full benefits, and they could have a go at becoming normal citizens. But they refused. They both wanted back in. I thought this was insane, so I made them an insane proposal. If they could get through Hell Week again, I’d sign waivers for the face and the leg. It wasn’t pretty, but they made it.

“We could save them both,” I said to Red Beard. “My kid and your kid. We could stop this nonsense right now and maybe they won’t have to go to war.”

“How so?”

“You take the pigs back to the farm, and I take the boys back to the compound. If higher asks me how the training went, I’ll say it was great.”

I thought of the big dry-erase board hanging in our ops shed, where all our names, specialties, and requirements were gridded out in black marker. A green check mark in a box meant that a particular block of refresher training, required prior to our next deployment—Marksmanship, Free Fall, Clandestine Entry—had been completed. An open box meant incomplete. Bing Thomas had drawn this grid by hand, following each warped line with a slightly more warped line until the grid practically curled under itself. Pig Lab was the last column on the grid. Its deformed and empty boxes appeared to have been crushed under a landslide of green checks.

“Then what?” Red Beard asked.

Of course I’d imagined a world where pigs were humans and we were pigs. Where pigs fought wars while wanting to save their fellow-pigs from death. Where pigs learned from humans how to save themselves.

In this upside-down world there'd be a pig like Ted Waters, who, one blue winter night on a concrete highway outside Marjah, had his leg blown off by a bomb disguised as a guardrail. And there'd be a pig like me, who'd knelt beside Ted and applied the tourniquet just as he'd been trained to do. Then, a month later, in Khost, a pig like Bing Thomas would have opened a booby-trapped door and had his face sheared off. And a pig like me would have picked Bing's face up off the dirt, rinsed it in cold water from his canteen, and felt the skin flex in the palm of his hand.

Now I was on my back in the grass that grew dark and thick all around the tall, concrete tower. My jaw was slack, my eyes were slammed shut, and my tongue hung out, touching the ground. I tasted dirt and chlorophyll. The black sun shone through my tightly closed eyelids, and I could feel myself gently rising and falling as if I were lying in a rowboat on the ocean. A deep incision ran from the base of my throat to my navel. My sternum was fractured, my rib cage pried open. Cool shadows of puffy clouds floated across my exposed heart, which the pigs took turns touching with their snouts.

With their snouts, the pigs would feel each beat of my heart the way a human would feel a silver dollar that had been flipped in the air then caught in an open palm, flipped and caught, coming up heads or tails, whichever side had been called when the coin was at its apogee. The pigs would know, from my heart's predictability, that the spark of life is durable. And they would learn that, no matter how bad a situation appears, it could always be worse. I didn't understand their language, but I knew what the pigs were saying to one another, because it was the same stuff that the boys and the pig men say to one another.

New Pig, who had apparently been allowed back into the group after Red Pig had calmed down and accepted his apology for joking about their human brothers, touched my naked heart with his cold, wet snout, tickling me and making me snicker.

"This one is a rascal," New Pig said, of me.

"You know it," Red Pig said. "He opens doors with his hands."

“Also plays fetch,” Eye-Patch Pig said.

“I never heard of a human that could fetch,” New Pig said.

“You should come out to the farm sometime,” Red Pig said.

Mention of the farm brought to mind a make-believe oasis of sunshine, warm fields of thick, green grass, fat red apples hanging from a tree, oats mixed with brown sugar in the trough, and a cool, clear stream winding through the center of it all.

“Look,” Red Pig said, nodding toward me on the grass, my white heart thudding slowly in my open chest. “That’s a strong, steady heartbeat. He can keep going like that for a while.”

The sun disappeared behind a cloud, and I shivered. I thought of Allison lying in the cold and empty bathtub. As I lifted her out of the tub, she’d pleaded with me to stop touching her.

“Everybody had a turn?” Red Pig asked. No one answered.

“O.K.,” Red Pig said. “Anybody *not* had a turn, raise your hand.”

The pigs just looked at him.

I sat cross-legged on the shag carpet next to the blue toybox. Its wooden lid rattled.

“No,” Allison said, from inside. She sounded small and far away.

“It’s O.K.,” I said. “It’s me.”

Allison had scooped all the toys out of the box and tossed them on the floor. New toys, like the spaceship and the motorcycle, were piled on old toys, like the cowboy I’d had as a kid.

Once upon a time, this cowboy had bent at the knees, elbows, shoulders, and hips. I’d taken him everywhere with me. Then one summer day, at the neighborhood pool, where I was taking swimming lessons, bullies had

pulled my cowboy apart and left all his pieces in a glass ashtray. When I found him, I was inconsolable. I cried non-stop for the cowboy, but also for myself, and for my mother, my father, my brother, my sister. Death would not spare us, either.

My father used a powerful glue to reassemble the cowboy, which twenty-five years later still smelled like orange peels. The cowboy wore the brown vest that I'd cut for him out of felt. The shape of his face was unchanged, but its features, drawn in black ink on soft plastic, had faded away.

"It's Monday, September 8, 2009. We're in our house on Rockrose Lane. You're safe here," I told Allison through the toybox lid.

"No-o-o-o," she moaned.

Allison wasn't big by any stretch. Even pregnant, she maintained her lithe shape from her swimming days. And she'd always been flexible, but she still shouldn't have fit in the toybox. In the throes of a flashback, however, she underwent some kind of transformation that allowed her to occupy otherwise impossible spaces. I'd seen something like this performed on a TV show called "That's Incredible," where a six-foot-tall, two-hundred-pound yogi had folded himself into a plexiglass cube half the size of the toybox. You could see the yogi through the transparent walls of the cube, his serene face surrounded by limbs like so many spare parts. Allison was similarly contorted whenever I found her mid-flashback. Unlike the yogi, however, she was unable to escape whatever predicament she'd got herself into. When she'd jammed herself under the toilet tank, for example, I'd had to drain the tank, remove it, and pull the bowl out of the floor to free her.

"We love our house," I whispered through the side of the toybox. "It's white with black shutters. You painted the front door red. You planted flowers in the beds."

"No," she said.

"I'm thirty-two years old and you're thirty-one. My hair's falling out. We're in the baby's room on the second floor. We don't know if it's a boy or a girl yet. We don't know if we want to know. The baby's room is yellow, the

same color it was when we bought the house. We don't know what color to paint it yet."

Allison's hand shot out from under the lid of the toybox and grabbed me by the collar.

"Quiet," she pleaded. "He'll find us."

Allison didn't know whether her father was dead or alive, let alone where he might be. I'd tried to find out, secretly. I'd given the spooks at the compound—the same men and women who, at the time, were spending all day and night searching the dark web for Osama bin Laden—Allison's father's name. I'd told them what Allison had told me—that her father had driven a van that smelled like buttermilk, that as a young man he'd aspired to join the San Francisco Ballet, and that, later in life, he'd impersonated a dentist in Milwaukee. No detail, no matter how incomplete or far-fetched, was irrelevant to their search. The spooks had plugged all this information into the algorithm—the same system that had delivered us Mohammed Atef, Khalid Sheikh Mohammed, and al-Zarqawi—and it had come up with a brother, working in a muffler shop in Palo Alto.

"He won't find you," I said.

She pulled her hand back inside the toybox and its lid clapped shut.

Rubber Hand with his hatchet, and Eye Patch with his M4, had done their work. The hatchet had fallen with the soft sound of dirt filling a hole. The M4 had discharged what looked like powerful gusts of wind. The pig men had crossed paths at the midpoint in the line of pigs, where each, in turn, deferred to the other as if he were opening a door for a lady, before continuing on to the other end. We weren't supposed to watch.

"Y'all line up on a pig," Red Beard said.

A dozen pigs lay face up on the grass, hurt, doped, and smiling. I chose one with a freckled belly. A first-aid kit and a bottle of water lay on the grass beside him.

Red Beard held up a first-aid kit. “Same stuff we carry in the field, minus the fentanyl lollipop, which is a controlled substance.”

“Which is the *tits*,” Eye Patch said.

Red Beard continued. “When I say ‘Go,’ approach your wounded and do an assessment. Pat and swipe. Don’t try to put any outside stuff back inside. Leave that for the surgeons at the *CASH*. Your job is to prepare your brother for the journey. Project calm. Look him in the eye and tell him it’s going to be O.K. Tell him he’s going to make it. Raise your hand if you have questions.”

No one raised his hand.

“Go.”

My pig was heavier than I’d expected. His practically nonexistent neck bent in a very human way. Otherwise, I couldn’t find anything wrong with him. I raised my hand.

Red Beard knelt next to me, and light shone from his knees. “Whatcha got?” he asked.

“I don’t know,” I said.

“Talk to him.”

The pig looked at me out of one eye.

“Hey, buddy,” I said. “Where’s it hurt?”

The pig blew a pink snot bubble out of one nostril. Red Beard pointed at it.

“What’s that tell you?” he asked.

“Collapsed lung?” I guessed.

“Right on,” Red Beard said. “Now what?”

Inside the first-aid kit was a prepackaged needle that could be used to create a vent in the case of a collapsed lung. I peeled open this package, removed the needle, and held it between my teeth. I felt for the pig's ribs with my fingertips. Into a soft spot between two ribs, I drove the needle home. There was a pop. The pig sighed and relaxed.

Red Beard nudged me aside. He put his ear to the pig's chest and listened for a heartbeat. He pried open the pig's eyelids and stared into his giant pupils.

"He's gone," he said.

The pig men huddled by the tower. Eye Patch toed the grass and Rubber Hand hummed "Dixie" while Red Beard wrote in a little green notebook with a stubby red pencil.

It was the same notebook-pencil combination that I used on missions downrange. I wrote call signs, frequencies, rendezvous coordinates, Zulu times, and Julian dates in that notebook's blue-lined pages. Just strings of numbers, mostly, that would otherwise be shoved out of short-term memory by more numbers. But, when I looked back through those notebooks, the way the numbers slanted or grew large or turned shaky across the page told me all I needed to know about the night on which they'd been written, and therefore served as a kind of memory.

Red Beard waved me over. He pointed the eraser of his pencil at my dead pig, lying in an oval of flat grass.

"Minus this one, your troop don't make the cut," he told me.

"You get that guy who saved two pigs?" Eye Patch asked.

New Guy, in his exuberance, had patched a sucking chest wound on one pig then clamped another pig's severed artery.

"I got him," Red Beard said.

"We could round up?" Rubber Hand suggested.

"This ain't tiddlywinks," Red Beard said.

“Well, I guess that’s that,” I said.



*“See? Isn’t this better than scalding-hot porridge back at our house?”
Cartoon by Pia Guerra and Ian Boothby*

None of them acted like they’d heard me.

“We could redo the first thing,” Rubber Hand said.

“Then we’ll run out of pigs for the last thing,” Red Beard said.

“If I leave now, I can get to the farm and back before you’re done with the next thing,” Eye Patch said.

“This can-do attitude?” I interrupted. “This make-it-happen-at-all-costs? It’s unnatural. We gotta stop.”

“You take one-sixty-five down to sixty-eight?” Rubber Hand asked Eye Patch.

“This time of day,” Eye Patch said.

“Leave the trailer here, and put, like, six pigs in the cab,” Red Beard said.

“I’d knock ’em out first,” Rubber Hand said.

Just then my pig jumped to its feet and shook the wet blades of grass off its back.

“There you go,” Red Beard said.

The sun was on its way up, and birds were chirping in the bare branches of the oak tree outside the window. Allison was still in the toybox, curled up, I imagined, with her knees at her chin and her elbows at her ankles. I talked to her the way the therapist had recommended—slowly and steadily, with a focus on our immediate surroundings and the current situation, in hopes that this intonation of our state of affairs would end her flashback and return her to the present day. I had just started talking about the future.

“I want our lives to calm down,” I said. “I want us to sit on the porch, watch the sun set over the canal, let our kids play in the yard, and . . .”

Boom! Allison’s foot broke through one side of the toybox. Her bare and scraped-up leg hung out from mid-thigh.

Crack!

She elbowed out the front panel of the toybox, then rolled out of the wreckage and onto her hands and knees. Her hair was matted and her eyes were wild.

“There are snakes in the back yard,” she said.

“What kind of snakes?” I asked. I’d never seen one.

“Cottonmouths. They come up from the canal and sun themselves on the grass. You need to get rid of them.”

“As soon as I come home from work,” I said.

“No. Do it now,” she said.

The final scenario was Burning Down the House. Eye Patch carried the last pig up the stairs of the tower all the way to the roof, where he laid it down on the gritty, sunbaked surface. Then he looked over the edge and down at

Red Beard, who was looking up at him from a gas valve the size of a ship's wheel.

"That does it!" Eye Patch said.

Red Beard made a mark in his notebook.

"Come on down!" he said.

Before we'd commandeered it for our purposes, this tower had been a training aid for the fire department. As such, it had been a placeholder for all homes in the area that might one day catch fire. So it was also a proxy for the two-story Colonial, on the banks of the muddy canal, where Allison and I lived. And it stood in for the house three doors down, identical to ours in every way save its crooked shutters and dead shrubs, where the dishevelled little girl from the bus stop lived.

Ted Waters sat on the grass near the fence. He'd undone the rubber bands that held his mechanical leg in place and set it aside. He was scratching the dirt where his shin would have been, watching the situation develop. As Eye Patch descended the stairs of the tower, whistling a carefree, shuffling tune, Ted stood up and hopped over to me.

"Do something!" he said.

"Like what?" I asked.

"You got a big yard. Take these pigs in. Let 'em grow old."

Earlier that morning, while I was standing in the back yard with a shovel, cottonmouths had come up from the canal. They'd slithered uphill through the grass toward the house. They'd lunged at me with their fangs out. I'd lowered the shovel and, one by one, chopped off eight heads. I'd left the heads in the grass as a warning to the rest.

"My wife doesn't want any critters in the back yard," I said.

"Stand clear!" Red Beard hollered.

New Guy stood facing the tower with his sleeves rolled up, hands on his hips, and chest out. He shouted at us over his shoulder, “Who’s with me?”

Red Beard turned the valve on a six-inch gas line. This fat line fed a series of narrower pipes in the tower, which ran along the corners of every floor, wall, and ceiling, and were perforated to spray gas into the rooms. With the valve wide open this system made a noise like a hovering spaceship.

Bing Thomas ran over to me and nudged Ted Waters aside. The expression on Bing’s face alternated between satanic and serene. His crooked nose honked like a kazoo.

“I ain’t running into no burning building,” he said.

“Fear is an illusion!” New Guy announced.

The igniter was wired to a red button inside a little black box mounted to a wooden post. Red Beard opened the door to this box and pushed the button. The fuse clicked softly. The spark sounded like a stick breaking in half. Then—*whoomp!*—flames jumped from the tower’s open windows and stood straight up against the light-blue sky.

Heat from the fire warmed my face. It toasted my chest through my shirt. It sucked a draft out of the swamp which smelled like rotten eggs. Pig screams were no different from our screams.

Bing Thomas ran into the tower with one eye shut. Ted Waters reattached his leg on the fly. New Guy didn’t budge, so the boys and I parted around him and came together on the other side. ♦

The Critics

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What Happened to Your Face?

How the human countenance became something to study, edit, optimize, and scan.

By [Cal Revely-Calder](#)



The physiognomists promised that your character could be read from your features. Certain forms of facial-recognition technology have revived that old fantasy in digital form. Illustration by Bryce Wymer

Several months ago, my partner and I bought an apartment in South London. Our previous home was a rental in which, for reasons best known to the landlord, there were mirrors everywhere. The bathroom had two; there was one outside on the terrace; in the bedroom, mirrored panels stretched across a twenty-foot-long wall. On moving day, we realized that we had a problem: the new apartment was mirror-free, and because we'd been so spoiled we weren't bringing one of our own. We spent a few days filling our drafty rooms, decanting books, building furniture, and dressing every morning without seeing ourselves in profile. It was a couple of weeks before we bought a simple mirror, wooden and round, to hang above the bathroom sink. By then, I joked, we didn't recognize ourselves.

As my feeble quip indicates, we take for granted that we possess a steady mental image of our own faces, and that we can reconfirm these images

frequently. If I asked you to picture yourself, you could [almost certainly](#) do this with ease. You might refer to the memory of how you looked in the mirror this morning, or in the selfies you took last night, or in that passport photo you hate. And if I suggested that you double-check your mental image, you could probably do so by looking in the mirror again, or at your reflection in a nearby window, or, via a camera app, using the very screen on which you may be reading this paragraph. Most of the time, we think we know how we look.

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Yet that's possible only because we live in the aftermath of a revolution. In the past few centuries, the mental self-portrait has become commonplace. To be sure, the world has held pockets of reflection as long as humans have existed to see them. As a piece of technology, the mirror seems to date back several millennia. The ancient Egyptians had polished bronze, copper, and obsidian at their disposal, as did the Greeks and the Romans. But such objects were hardly ordinary, and people had to improvise: in Ovid's *Metamorphoses*, Narcissus, the ultimate lover of reflections, sees himself for the first time in a crystalline pool. Pausanias, writing a century later, called

this scene “totally absurd,” because any young man would have seen a reflection before. Still, for thousands of years, there was an obvious gap between image and life. Now in a kitchen pot, darkly; now face to face.

The revolution came in the sixteenth century, when Venice’s glassmakers applied a tin-and-mercury alloy to their panes, producing a far clearer reflection than had previously been achieved. These artisans, with their trade secrets, were kept on Murano, a cluster of islands just across the lagoon from the city, ostensibly as a precaution against fire, though the state would also send assassins after anyone who tried to leave. Still, a few hardy glassmakers escaped to the Saint-Gobain glass workshop in Paris, which they lifted to pan-European dominance. The workshop gave us, among other monuments to subtlety, the great Galerie des Glaces in the [Palace of Versailles](#), which contains three hundred and sixty-seven mirrors, stretching eighty yards. The technology could be replicated, spread, sold. Mirrors quickly infiltrated the private homes of the Continent, then its colonies; they became cheap and unremarkable, as they remain today, if you stop shelving your books long enough to buy one.

In the centuries that followed, the private face became increasingly public. More people could look at their own countenances more accurately, and more often. Candlelight and gaslight gave way to electricity, an unwobbling source of interior light. Photography arrived, kindling an enthusiasm that Baudelaire, ever the cynic, compared to a lust for pornography. Cosmetics, as venerable as mirrors, boomed to an unprecedented degree. Think of Esther marinating her skin in myrrh, then of Eugène Rimmel, who introduced the first mass-produced mascara via an early mail-order catalogue. Pierre-François-Pascal Guerlain, a trained chemist, sold skin creams with such aplomb that he was hired by Napoleon III’s empress, Eugénie. Fast-forward to the twenty-first century: to fix your flesh, there’s Botox, fillers, and plastic surgery; to augment the image, there are camera-app filters and Photoshop. In 2024, according to Apple, the world’s iPhone users took five hundred billion selfies. [Facial-recognition technology](#) is used across the globe. In the span of a few centuries, the face has become inescapable. We seem to be condemned to look our best.

Much of our social life, our ways of making and sharing a world, orbits the face. Emmanuel Levinas, the supreme theorist of facial affairs, described our

fleshy ovals as the site where ethics—his “first philosophy”—begins. This rings true when it comes to everyday behavior. The phenomenon may be hardwired: babies prefer faces to other complex objects. And, for the rest of our lives, faces seem charged by something that arms or legs, however attractive, don’t have. When you talk to me, I look at your eyes, regardless of whether you’re waving your hands. If the gaze is held too long, one of us has to look away. If I can’t see your face—say, you turn aside, or you’ve called me on the phone—I can get by, can make the conversation work, but I might feel that our connection is lacking. The face’s centrality to social life is inscribed in our language, in the metaphors that betray our world view. In English, we face up to things, or face them down; we show our faces where required; we try to save face, and should that fail we make an about-face to save ourselves.

Yet there’s another side to this coin. Our vocabulary also speaks of invention, even artifice. The English word “face” derives from the Latin *facies*, implying a created form; so does the French *visage*, from *videre*, suggesting something seen from without. “Mask,” “masque,” “mascara,” “maquillage,” and their European relations seem to be etymologically linked, and carry long-standing associations with concealment, distortion, pretense. For Socrates, the art worth prizing was the cultivation and preservation of natural beauty; we’ve happily overwritten it with what he treated suspiciously as *kommōtikē*, the art of changing how one looks. Even “person,” along with “impersonation” and “personae,” derives from *persona*, a theatrical mask through which classical actors spoke. The cultural historian Hans Belting suggested that faces and masks were conceptually inseparable: we shouldn’t think of one as “real” and the other as “fake”—one as the thing we have and the other as the thing we temporarily don. Life, he wrote, was fully “a *perpetuum mobile*,” an “expressive drama,” in which our faces resolve into one legible position, one legible role, then reassemble themselves into the next. We make them up, in every sense.

The face, then, lives by its pliability. We might encounter a completed sentence or picture, but never a completed face. It can always shift into something else, pushed by personal whim or social pressure, and keep doing so until frozen into one last mask by sickness or death. This means that we live together in an interpretative flux. We constantly read other faces, and understand that ours, at the same time, will be interpreted. As T. S. Eliot put

it, we prepare a face to meet the faces that we meet, and we do so knowing that they've been prepared as well, and that our interpretations will almost immediately become obsolete.



"O.K., great—let's break for now and continue the indecision, procrastination, and blame-shifting over e-mail."
Cartoon by Tom Toro

Levinas tends to be the philosophical lodestar here, with his near-mystical love of the face-to-face encounter. He wrote that the face was “a living presence,” that “the Other faces me and puts me in question and *obliges* me.” Obligated him to do what? This may not be clear. Encountering a face means reëncountering the world: it is a silent epiphany about the fact that other people are radically outside of you, maybe unknowable in a final sense, and yet you must engage with them anyway.

At the same time, I prefer Ludwig Wittgenstein. He was more impish than people remember, and he liked to draw little faces when he was teaching or writing, sketching each one in a few strokes and usually making it smile. Such doodles recur in his manuscripts from the early nineteen-thirties until shortly before his death, in 1951; he used them to illustrate a battery of subtle points, one of which was how misleading our language could be. Here's one he drew for his students:



After which he said:

Let this face produce an impression on you. You may then feel inclined to say: “Surely I don’t see mere strokes. I see a face with a *particular* expression.” . . . We might give such a description and say, e.g., “It looks like a complacent business man, stupidly supercilious, who though fat, imagines he’s a lady killer.” But this would only be meant as an approximate description of the expression. “Words can’t exactly describe it,” one sometimes says. And yet one feels that what one calls the expression of the face is something that can be detached from the drawing of the face. . . . The delusion is assisted by our using the verb “to have,” saying “The face *has* a particular expression.” Things look different when, instead of this, we say, “This *is* a peculiar face.” (What a thing *is*, we mean, is bound up with it; what it has can be separated from it.)

One of the things Wittgenstein implies is that we don’t theorize the faces we see. I might talk of “reading” your face, but it isn’t like reading a code or a map: I don’t look at you, add together your liquid eyes and droopy mouth, and pronounce my final judgment that you’re sad. I simply look at your face, apprehend its gestalt as a unity, and go from there. Recent neuroscientific research has shown that we process whole faces more readily than mere

parts; we aren't adding the bits up. We look, and we speak, and we look again: that's life.

One oddity, given all this, is that, although humans have made pictures of faces for as long as we've represented anything, realism has only recently become the norm. As the scholar Fay Bound-Alberti points out in "The Face: A Cultural History" (Grand Central), between the thirty-thousand-year-old figures on the walls of the Chauvet-Pont d'Arc cave and ancient Cycladic statuary, from the fourth to second millennium B.C., human figures rarely showed recognizable facial features. There were exceptions. In Egypt, certain pharaohs seemed, for reasons still debated, to tilt their own representations from the ideal to the natural. Senusret III, the most outlandish example, had temples filled with statues of himself, his gigantic faces lined and grim and aged, though set atop a young and muscular body. Centuries later came an interlude in Greece and Rome, beginning with the so-called Severe style of Greek sculpture, in which faces gained clearer features and bodies sharper definition. Socrates' ugliness can be reliably inferred from his many unsexy sculptural representations; the statues and busts of individual Caesars, though they blend truth and propaganda, remain easy to distinguish. But this facial gallery closed with the Western Roman Empire and the rise of Christianity, which inherited a suspicion of idolatry from the Old Testament and Judaism. David Le Breton points out in "Faces" (Polity) that the Apostolic Constitutions, a fourth-century collection of Christian ecclesiastical laws, recommended that painters be exiled altogether from Christendom, just as poets would have been from Plato's ideal state. They stayed, but, rather than memorializing individual heroes, they largely devoted themselves to religious imagery, in which realism was again forgone in favor of iconic value. The old order had resumed.

Only with the Renaissance would it break again, with the fifteenth-century blossoming of the naturalistic portrait: van Eyck, Campin, Piero della Francesca. The archetype of painted interiority may still be the good old "Mona Lisa," known in Italian as "La Gioconda"—the face that launched a thousand conjectures. Bound-Alberti brings her in, dutifully, to say how "enigmatic" her smile is, though Wittgenstein, again, made more of the cliché: "When we speak of the enigmatic smile of the Mona Lisa, that may well mean that we ask ourselves: In what situation, in what story, might one smile like that?" To talk of stories, he means, is to talk of time; you need the

before and the after, the rounded sense of a life. After all, some people laugh in anger and weep with joy. If I read your smile as happiness, I'm drawing on a fluid sense of the character I know, and on our place in an emotional culture. One reason Leonardo's painting has endured in the popular imagination is straightforward: we want to ask questions of her—how she feels, what she's thinking—and though we don't believe she's alive any more than we believe Hamlet or James Bond to be alive, we can't help responding to her face as if she were.

One unexpected consequence of the return of realism was a renewed attention to anatomy. Take dissection, which was mostly in abeyance through the medieval era, though by the thirteenth century in Europe it was beginning to appear in university curricula. During the Renaissance, it flourished again, and became a routine part of an artist's studies, as Leonardo's sketches attest. At the same time, another art returned, equally venerable and equally obsessed with the human form: physiognomy. The thinking, from the Greeks onward, was that you could, if sufficiently skilled, read people's inward rottenness from their outward appearance. They might alter their behavior or speech, but a nose could never lie. Physiognomic treatises reappeared in the sixteenth and seventeenth centuries, then took off in the eighteenth and nineteenth. The star was Johann Kaspar Lavater, a Swiss poet and philosopher, whose "Physiognomische Fragmente," a complete manual of flesh-reading, was published in four volumes between 1775 and 1778, featuring illustrations by Henry Fuseli and William Blake of faces and other body parts. It was successful enough to be abridged in multiple forms—"The Portable Lavater," "Ladies' Lavater"—and by the end of the century, Le Breton writes, "a visit to Lavater, complete with a physiognomy consultation with the master, was a must for anyone travelling to Switzerland." The fashion spread. George Sand was a Lavater fan. So was Honoré de Balzac, who called physiognomy "prophetic." Balzac used Lavater's work in devising his own characters, who are practically lashed to destiny; Samuel Beckett would deride them as "clockwork cabbages." One eighteenth-century chief justice of Naples took aside convicted prisoners who still refused to confess and personally examined their heads. If he found proof of inbuilt depravity, he would approve their execution; if he didn't, he'd set them free.

Such thinking had serious critics. If you judged someone's soul from his appearance, [Montaigne](#) noted, it would be curtains for Socrates. The entry for "*physionomie*" in Diderot and d'Alembert's *Encyclopédie* described it as "imaginary" and "ridiculous." Hegel compared physiognomists to a merchant who complains that it rains whenever the fair comes to town. But the belief died hard, and photography spurred its adherents on. Once faces had been "captured," as we still say, they could be put into giant comparative systems, proto-databases of facial types. Francis Galton, the father of eugenics, used superimposed portrait photos in his hunt for "deviant" traits that criminals supposedly shared. His friend Cesare Lombroso, an Italian criminologist, agreed, suggesting that the signs of a "born criminal" included a receding brow and a heavy jaw.

As genetics, psychology, and neuroscience ascended, the twentieth century sent physiognomy back into disrepute, and today, from Lavater to Lombroso, its promoters may seem a racist shade of quaint. Yet the conviction that your face will, if read skillfully, betray your hidden qualities has never gone away. The most influential facial theorist of the late twentieth century was the psychologist Paul Ekman, who argued that seven "universal emotions"—happiness, sadness, anger, surprise, disgust, fear, and contempt—recur across all cultures. Their display, he argued, can be too fleeting to control. Spot these "microexpressions" and map them, perhaps through Ekman's proprietary Facial Action Coding System, and you'll have a key to the human soul. Ekman's ideas travelled unusually well: he wrote popular books, worked on the Pixar film "[Inside Out](#)" and the Fox series "Lie to Me," and provided training to Scotland Yard, the T.S.A., the F.B.I., and the C.I.A. The appeal is plain enough. It promises a portable science of human disclosure. The trouble is that there is no context-free understanding of a social phenomenon; every use of a word like "anger" is imbricated in its past and helps shape its future. Ekman's seven-point system, however useful as a package, leans toward rigidity, turning the face once again into a surface to be decoded.

The drive to define the face as a data point, something to be grasped and controlled, underpins the bureaucracy of the modern nation-state, in which faces are surveyed, categorized, and stored in digital banks. We show our faces when crossing the border, whether by holding out our passports or by staring into a digital camera. We carry papers that show our faces. Not

coincidentally, plastic surgery has been advanced in part by governments: its pioneer, Sir Harold Gillies, fixed the faces of British soldiers who had been disfigured in the [First World War](#), and today the U.S. Department of Defense funds face-transplant research in the United States.

[Facial-recognition systems](#) are now operative in moments big and small, whether you're activating your Apple device or being watched as you march against the regime. Some defend the practice, and others shrug. But the biases of these systems suggest that [physiognomy has taken a new, less explicit form](#). Police facial-recognition systems in the U.S. have had a disproportionately high error rate among Black people. Black Americans are also disproportionately likely to be subjected to such systems; they are overrepresented in the prison population, and in recorded recidivism rates. Their faces, in other words, can be drawn into a technologically enhanced ouroboros. And my own home, the U.K., always seems to follow its cousin's lead. Between 2024 and 2025, our police forces almost doubled their [use of the technology](#) on pedestrians; on the day I write this sentence, there are two huge marches in London, one pro-Palestine and one anti-immigration, and both are being actively surveilled by facial-recognition technology. Since 2013, Britain has had an independent Biometrics Commissioner, and every holder of the post has called for greater oversight of these systems, but privacy advocates have argued in court that there are "no meaningful limits" on what the police can do.

Many of the tech behemoths, too, have invested in facial recognition, and, for a while, they augmented it with emotion-detecting tools. Amazon, for instance, announced in 2019 that its system Rekognition could more accurately identify "all 7 emotions"—namely "Happy," "Sad," "Angry," "Surprised," "Disgusted," "Calm," and "Confused"—and had added a new one, "Fear." This diverges a little from Ekman's schema; perhaps the publicity chiefs found "Contemptuous" too unpleasant to sell. Rekognition has a broad spectrum of clients, from ticketing platforms to credit bureaus. It once sought to sell its services to Immigration and Customs Enforcement, though *ICE* plumped for a rival firm, Clearview AI, which is less interested in emotion than in flat identification. Clearview scrapes facial data from the web on a mass scale—at one point, it absorbed ten billion social-media photos—and has faced dozens of lawsuits and regulatory actions from American states and European countries. The future of emotion recognition,

meanwhile, seems uncertain: Microsoft and Google both developed such tools but have since limited or ditched them. Maybe, outside the F.B.I.'s interrogation rooms, knowing how someone feels isn't proving that valuable.

The trend toward standardization and control has become coldly clear. And as with bureaucrats, so with beauticians: we can capture faces, sort them, approve them, reject them, modify them, and rejuvenate them. What you can measure you can try to perfect. The ancient approach to cosmetics seems like frippery next to the "[looksmaxxers](#)" of today, led by Clavicular, a razor-jawed American influencer whose commitment to surgery extends to smashing his own bones, and who has said that beauty isn't "about sexual attraction, it's about mathematics."



"Hamlet! I am thy father's ghost. Avenge me!"
Cartoon by David Borchert

Clavicular, though, represents merely the crunchy end point of a logic that has taken hold of affluent life. The same mathematical approach underpins platforms such as Instagram, a wet market for faces, on which users self-brand and sell themselves—and on which the most successful products are those which top the algorithmic charts. Hence the rise, in the twenty-tens, of the so-called Instagram face, on which so many visages slowly converged, altered by fillers and the like to become ethnically nebulous, big-lipped, and often, as videos indicated, hard to move. Last year, an ITV survey suggested

that among Brits between the ages of eighteen and twenty-five—a group with near-total smartphone and social-media penetration—one in five had some form of tweakment. Le Breton, with some justification, compares the plastic surgeon to “a psychiatrist advising a paranoid patient to be more wary of his enemies.”

Social media needs the face because it burns attention for fuel, and the face is where I go when I look for you. But I can’t look to these platforms. I find it all so unsatisfying: the resculpted faces, the curated lives, the naked grabbing of my gaze. Something uncertain, something human, has been drained like blood from this world. There’s evidence that Botoxed people become worse at reading other people’s emotions, because when you engage with others you simulate their expressions in your own facial musculature. It’s as if we were commodifying the face, putting it under glass, turning it into a transactional token rather than a pliable living thing.

That little half smile you see could be smug or coy, or even lady-killing; you need to ask which, and why, and in doing so you’re doing moral work, because you’re treating the smiler as an autonomous being whose form of life you need to grasp, however provisionally. But think of the faces we now trade most readily, in the billions every day: emoji, complete and prepackaged, limited in number, coined and issued by a consortium of leading tech companies. I can never shake the sense that they’re fatally cheap. If I saw a friend post about this piece, or send me a message about it, and use an emoji in doing so, I might first envision the real face of that friend, or try to, but then I would feel unmoored. No true face is as standardizable as the image the person has sent in its stead, and another friend, my partner, even my boss might have sent me the exact same little face earlier. It can’t represent them all. It’s like canned laughter, like a cliché. Yes, we know that [emoji](#) are useful lies, saving us time and dodging the difficulty of putting emotions into words; still, in so dodging, we falsify something essential about what our real faces can do. The emoji could be the culmination of everything we’ve done to the face over the centuries, all the multiplication and commodification and repacking. We’ve made it trivial.

This isn’t what I want for faces. I want them to shift, to resist, to allure. I find myself thinking of a photograph taken by Peter Hujar in 1975: a black-

and-white portrait of the poet and dance critic Edwin Denby, at the age of seventy-two. His eyes are shut, his mouth neither smiling nor frowning.

The picture tempts you to suppose that you can read Denby's face, that he's experiencing fatigue, or stress, or agitation. Or maybe he's just world-weary. But if you learn that Denby suffered, in his later years, from depression, and would kill himself some years later, you might see something like pain. Then you might learn a little about Hujar, too, and about the way he liked to concentrate on the emotions of his subjects. You might be less inclined to assume that a feeling has been casually captured, as it might be in a family snap. By this point, the question the image first posed, "What is this man feeling?," seems as urgent as ever, perhaps even more charged. Yet every attempt to answer it has led nowhere. Denby sits there forever; his troubles are radiant.

Here's another way of approaching that image. Denby, as a critic, was fascinated by the great dancer Vaslav Nijinsky, as much for his expressions as for his torso, arms, or legs. "The manner in which Nijinsky's face changes from role to role is immediately striking," Denby wrote, in 1943. "It is enhanced by makeup, but not created by it. In fact, a friend pointed out that the only role in which one recognises Nijinsky's civilian face is that of Petrouchka, where he is most heavily made up." It was as if the images held several traits in suspense. Many people who met Nijinsky were enthralled by that curious face. Stravinsky recalled that it "could become the most powerful actor's mask I have ever seen." It was part of Nijinsky's weird magic, whether he was embodying a sex-hungry faun or a wooden puppet come to life: he drew other personae out of himself.

After Nijinsky began to show signs of [schizophrenia](#), this mask, or persona, grew disturbed. He seemed to recede into his eyes. His diary, written during two feverish months in 1919, teems with scribbled drawings of eyes, giant and black. He believed that he was being watched, which for most of his life had been true. One winter night, in the late nineteen-twenties, he was taken out of an asylum in Paris by his former boss, the Ballets Russes' Serge Diaghilev, to attend a production of "Petrouchka," in which the title role was now performed by a younger man, Serge Lifar. Throughout the evening, Nijinsky barely spoke. He just stared. The diplomat Harry Kessler, who was there, said that his "big eyes" were like those of "a sick animal"; the dancer

Tamara Karsavina caught him straining to look at her face, then, when she met his gaze, he “turned his head like a child that wants to hide tears.”

A photograph of the group survives from that night. Although Lifar was the star of the show, all the faces are turned to Nijinsky, whose gaze is directed offstage. His eyes and smile are too wide, and his right hand grasps the air. This face conveys a need, but a need that eluded everyone; it eludes me now, as I look at the picture, though I know it's there. And, looking again at Hujar's portrait of Denby, I see something similar. Another man, engulfed in feelings, shows his face to the world. His eyes are closed; his expression seems charged. I wish I knew how he felt. It seems important that I do. All I know is that I cannot look away. ♦

[Books](#)

Something Is Very Wrong with Modern Longevity Science

A new book argues that many of the world's oldest people aren't so old after all.

By [Dhruv Khullar](#)



A fundamental challenge that longevity gurus face is that what's true is often boring, and what's interesting often isn't true. Illustration by Armando Vêve

Jiroemon Kimura, who is on record as the world's oldest man, died at the reported age of a hundred and sixteen, in 2013. His passing spawned a slew of articles about the secret to extreme longevity. (Apparently, it's small meals.) Kimura is said to have been born in a fishing village in Kyoto Prefecture, Japan, in April or March of 1897. Strangely, he is the only one of his five siblings to have multiple graduation records; he seems to have completed elementary school in 1907, or 1909, or 1911. He supposedly married his wife on three different dates, and at some point he not only adopted his wife's surname but also changed his first name. (He was born Kinjiro Miyake.) Demographers who attempted to validate his age wrote that their investigation uncovered irregularities and inconsistencies. And yet they concluded that "no critical discordances were discovered." The word "critical" was doing a lot of work.

Cases such as Kimura's—and some that inspire even more skepticism—fill the pages of “[Morbid: Debunking Modern Longevity Science](#)” (M.I.T. Press), a lively and sometimes conspiratorial new book by the Oxford researcher Saul Justin Newman. Newman won the satirical Ig Nobel Prize, in 2024, for demonstrating that many of the world's oldest living people may actually be dead. In “Morbid,” he argues that dubious age claims are not isolated—blemishes on the record of, say, Jiroemon Kimura or the famed elder Frenchwoman [Jeanne Calment](#)—but, rather, a systemic problem infecting longevity studies. “It is pretty hard to publish a scientific case report about a grandma reaching 88 years old, but if she manages 122 you might even squeeze a book out of it,” Newman writes. An exceptionally long life, in Newman's telling, is not necessarily a function of good genes, good behaviors, or good luck. It's evidence of bad record-keeping. The world's oldest people, he quips, have “birth certificate allergies.”

In accordance with Newman's theory, many long-lived people are hardly paragons of healthy choices. Christian Mortensen, who held the record for the world's oldest man before Kimura, smoked for nearly a century. Juan Vicente Pérez—who was born in rural Venezuela, had no official papers until his fifties, and seemingly died at a hundred and fourteen—drank aguardiente every day. The Guinness Book of Records once credited Carrie White, who was given a diagnosis of “post-typhoid psychosis” in her thirties, with being the longest-lived person—until it learned of a clerical error at the psychiatric institution where she spent more than seven decades of her life. “Where are all the athletes?” Newman wonders. “The aerobically svelte, annoyingly pious, scare-you-with-their-abs types? Dead.” According to Newman, the reason that supercentenarians—people who live past a hundred and ten—frequently have unhealthy habits is that they aren't supercentenarians at all.

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In a useful thought experiment, Newman shows how small errors in recorded age can compound to undermine aging research. Suppose that a hundred forty-year-olds fake their ages so that on official documents they're listed as fifty. (The U.S. Census Bureau has said that millions of Americans misrepresented their age in the most recent census.) In a country that contains a hundred thousand fifty-year-olds, these "young fakers" would account for only 0.1 per cent of the population. No big deal. But because the faux fifty-year-olds are far less likely to die than real fifty-year-olds—after age thirty, our risk of death doubles roughly every eight years—they will come to make up a larger and larger share of their purported age group. At "sixty-six," they represent 0.4 per cent of their cohort; at "eighty," three per cent. At "a hundred and five," they compose virtually the entire population.

This is a problem for [longevity science](#), and also for [public policy](#). All sorts of profound decisions depend on how long we're expected to live: the amount we pay into both Medicare and Social Security and the rate at which we're paid back, the cost of insurance, the funding that's allocated to various medical specialties and facilities, the amount of infrastructure that's built for seniors. Unfortunately, such decisions are not always sound. In 2010, on Respect for the Aged Day, Japanese officials sought out a Tokyo man named Sogen Kato to congratulate him on living to a hundred and eleven. Instead, they found a mummified corpse in his bedroom. Kato had been dead for

three decades, and his family had fraudulently collected more than a hundred thousand dollars in pension money. A government investigation later revealed that hundreds of thousands of “living” residents of Japan had in fact gone missing or died, including eighty per cent of Japanese centenarians. During the fiscal belt-tightening of the Great Recession, Greece discovered that two hundred thousand people—about two per cent of the population—were engaged in pension fraud. Most of the country’s supposed centenarians were a product of this deception, a government minister said; nearly three-quarters of them vanished in a subsequent census, conducted in 2011.

In the United States, centenarians are less common in generations for which birth certificates are widely available. But even government documents can deceive. The Inspector General of the Department of Health and Human Services has warned that people can easily obtain fakes; in 2010, Puerto Rico invalidated all its existing birth certificates, citing rampant fraud and identity theft. One study found that, in the middle of the twentieth century, more than a quarter of Americans had ages listed on census documents that were inconsistent with those on their death certificates. In some demographic groups, as many as two-thirds of the population had discrepancies. Some were large enough to classify individuals in an entirely different age bracket.

It’s jarring to think that our age—unlike our height or weight—can’t be precisely measured. One can count the rings on a tree to calculate how many years it was alive, but there’s no equivalent marker in humans. Every so often, in the hospital where I work, I notice that a patient has a listed age of more than a hundred and twenty. This is disorienting until I realize that the patient may have come into the hospital with no I.D. and no known contacts, and that they might be suffering from a condition—dementia, schizophrenia—that makes it hard for them to remember who they are, let alone how old. In such cases, the hospital’s computer system defaults to a birth year of 1900. Recently, [tech companies](#) have increasingly claimed that various products can reveal a person’s “biological age,” and scientists have made real progress toward developing “epigenetic clocks,” which analyze patterns in DNA to estimate the age of a person or an individual organ. But, so far, these approaches remain approximations, and they are less reliable in the very old.

If Newman is correct that many of the oldest people are younger than we think, then our conclusions about them—and about the places that produce them—must be unreliable. His theory appears to explain some counterintuitive findings. An unusual number of centenarians seem to hail from low-income areas. The French Republic's overseas *départements* of Guadeloupe and Martinique, which have some of its highest rates of poverty, also have France's highest concentration of hundred-and-ten-year-olds. Newman's own research suggests that London's impoverished Tower Hamlets neighborhood has more hundred-and-five-year-olds than the rest of England combined.

Newman aims his most pointed criticisms at so-called blue zones, places that are credited with producing extraordinary longevity. The term was introduced in a 2004 paper in which researchers tried to figure out why so many Sardinians lived past a hundred. The research team, led by a respected demographer named Michel Poulain, interviewed residents, checked documents, and recorded in blue ink on a map the places where centenarians lived. The blue clustered in a mountainous region in the middle of the island, leading the investigators to speculate about potential longevity-boosting factors, including the area's climate and its inhabitants' diet, life style, and high rate of inbreeding. The following year, a journalist named Dan Buettner published a hugely popular *National Geographic* [cover story](#) that introduced more longevity hot spots, including the Japanese prefecture of Okinawa, and Loma Linda, California.

Buettner trademarked Blue Zones and partnered with Poulain. He went on to write best-selling books, offer cooking classes, charge substantial speaking fees, and found a company that made millions of dollars by issuing municipalities Blue Zone stamps of approval. The aggressive commercialization eventually led to a rupture with Poulain. ([According to](#) *Science* magazine, the researcher ended the partnership after Chanel released a skin-care product called Blue Serum, which was marketed as anti-aging because it included materials from blue zones, and Buettner wanted to sue the luxury label for its use of the term.) Buettner also admitted that he originally included Loma Linda because his editor wanted to identify a blue zone in the U.S., and that he “never bothered to delist it.”

The case of Okinawa, a group of relatively poor islands in the south of Japan, is instructive. Among Japanese prefectures, Okinawa ranks first in consumption of KFC and last in consumption of seafood, sweet potatoes, and leafy greens—its residents have Japan’s highest average body-mass index. More than a third of Okinawan men smoke. Although some books about blue zones encourage people to drink two glasses of red wine a day, “preferably Sardinian Cannonau”—a recommendation plainly at odds with current public-health advice—Okinawans like beer. If we want to live a long time, should we do as the Okinawans do?

Probably not. In 2010, Okinawa reportedly had twice as many centenarians per capita as the rest of the country. But, as Poulain himself has pointed out, Okinawa was the site of a devastating American bombing campaign during the Second World War; much of its infrastructure and essentially all its civil records were destroyed. U.S. authorities reconstructed the civil registry by asking survivors to report their age, a process complicated by language barriers and a traditional Japanese calendar that differed from the Gregorian. The country’s postwar welfare system distributed some benefits based on age, and many Okinawans requested that their records be updated. There is a strong correlation between the number of requests made in a village and the supposed life span of its residents.

Our tendency to fetishize faraway lands, trying to unearth their secrets to [slow aging](#), can seem strange when one considers that wealthy Americans already live into their eighties and nineties. Scientists have reliable data for octogenarians and nonagenarians—one research project, the Harvard Study of Adult Development, has been running since 1938—and the lessons are clear enough. The things that help us live longer are things that pretty much everyone wants: money, education, safety, clean air and water, community. Of course, these things are unevenly distributed, and the consequences of that are readily apparent. Research led by the economist Raj Chetty has found that the highest-earning Americans live more than a decade longer than those in the bottom one per cent. And the gap has been widening. Between 2001 and 2014, life expectancy among the top five per cent of earners increased by nearly three years; in the bottom five per cent, it essentially stagnated.

One hardly needs a list of blue zones to think of commonsense ways to improve life span. We can invest in biomedical research and enact laws and regulations to reduce pollution, design walkable neighborhoods, establish safe public spaces, create affordable housing, and curb smoking. Such strategies might not trend on TikTok, but they would be more effective than the latest juice cleanse. Unfortunately, the pursuit of longevity—like so much else in American life—is increasingly an individualistic endeavor. We adopt new health fads as though they are forms of personal expression, experimenting with [cold plunges](#) and red-light therapy, wearable devices and [peptide injections](#). Though it's nice to think that striving to be healthy is cool or popular—it wasn't always—the quest for extending life can become monomaniacal, and we easily forget that one's life span is not, by and large, a life-style choice. Instead, longevity has always been a communal affair.

In the early eighteen-hundreds, a Belgian astronomer named Adolphe Quetelet coined the term “social physics” to describe a method for modelling human development. Stargazers often made small errors when recording the position of planets; the individual errors were unpredictable, but, in the aggregate, they followed a pattern. We now call this pattern a Gaussian distribution, for the astronomer Carl Friedrich Gauss, or, more commonly, a bell curve. Quetelet showed that similar patterns could be found in social phenomena, including births, deaths, marriage, and crime. There was some variation—and, of course, each year there were different people being born, dying, marrying, and committing crimes—but over-all rates were remarkably stable. These occurrences could not be chalked up to individual choices. They were products of measurable—and modifiable—social conditions. A historian later summarized Quetelet's thesis: “Society prepares the crime, the criminal commits it.”

Not all of Quetelet's ideas were good ones; he also contended that a person's physical traits determined social outcomes. Still, it is unfortunate that today's longevity discourse has become so fully untethered from his sociological point of view. Influencers who peddle supplements and squats are ignoring the enormous role of societal conditions. During my medical residency, in Boston, I spent some of my time seeing patients from Roxbury, Dorchester, and other low-income neighborhoods, where they were more likely to be exposed to crime, drugs, stress, and toxins such as lead. Other weeks, I'd work at a hospital in Newton, a wealthy suburb a few miles away,

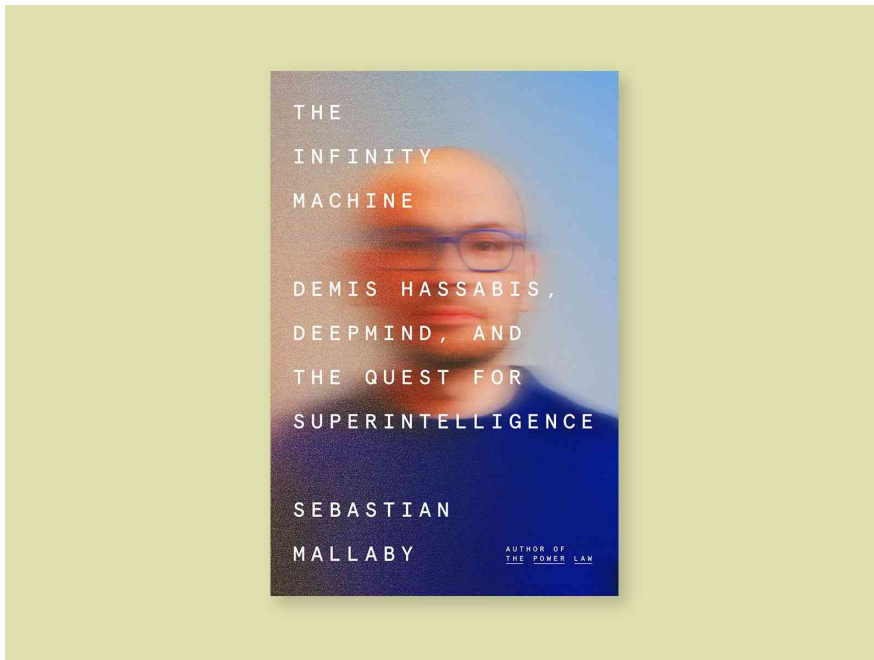
where kids grew up in large homes on tree-lined streets. In both places, I could offer patients the same basic advice on [healthy habits](#). But it would be absurd to think that my advice helped some Newton residents live into their nineties while many of my less affluent patients might barely make it to seventy. Society prepares the life span; the individual lives it.

Of course, we still have to make decisions about our health. We can't wait around for governments and institutions—especially *this* government and *these* institutions—to bring about the conditions for a wholesome and prosperous society. So what, exactly, should we do? The longevity-industrial complex tends to frame this question in oddly specific ways. Should you eat or avoid eggs? Seek out or shun coffee? Exercise in the morning or the evening? Walk seven thousand steps a day or ten thousand? Nowadays, influencers often invoke scientific research, but it tends to be stripped of context and caveats; they might cite small-animal studies to justify bold recommendations for humans, or argue that everyone can benefit from treatments that are approved for narrow applications. We keep hearing about the new drug that will change our lives. In the two-thousands, David A. Sinclair, a Harvard biologist and a co-author of "[Lifespan: Why We Age—and Why We Don't Have To](#)," declared that resveratrol, a compound found in grape skins and red wine, was "as close to a miraculous molecule as you can find." He sold a resveratrol-focussed startup to GlaxoSmithKline. But a researcher at the National Institute on Aging later [told](#) the *Times* that the molecule's "actual use for humans still needs to be discovered," and G.S.K. eventually abandoned work on the compound. More recently, in "[Outlive: The Science and Art of Longevity](#)," Peter Attia promoted rapamycin, a molecule that was first sampled on the South Pacific island of Rapa Nui and developed into an immunosuppressant for organ-transplant patients. Laboratory studies have shown that rapamycin could extend the life spans of yeast and mice—but, in a recent randomized human trial, older adults on the medication didn't seem to experience any physical gains. If anything, they benefitted less from exercise and had more side effects than those who didn't take the drug; one participant developed a serious infection. Attia and other influencers have said that they have stopped taking the drug, owing to mouth sores, infections, and other complications.

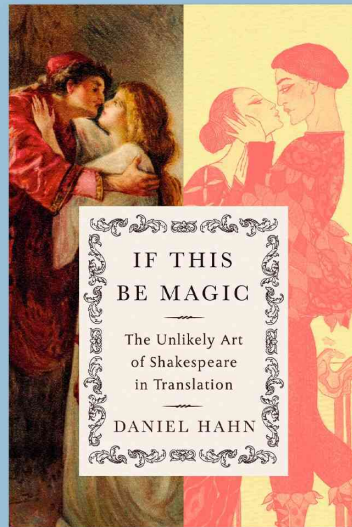
A fundamental challenge that longevity gurus face is that what's true is often boring, and what's interesting often isn't true. Against this backdrop, "[Eat](#)

[Your Ice Cream: Six Simple Rules for a Long and Healthy Life](#)” (Norton), by the physician and bioethicist Ezekiel J. Emanuel, arrives as a welcome tonic. The book follows a structure characteristic of the longevity genre: divide the text into a handful of familiar sections, start each chapter with an anecdote, review some relevant research, and conclude with tips on how readers can apply your recommendations. (Move! Sleep! Socialize!) The longevity industry profits from mystifying the obvious, but Emanuel largely dispenses with pretense. He tells us to brush our teeth twice a day and then moves on to such closely held longevity secrets as abstaining from smoking, soda, and drunk driving. These are hardly paradigm-shifting insights, and that’s exactly the point. We already know how to improve our chances of living a long life—exercise, eat well, sleep enough, maintain relationships, don’t drink to excess, don’t smoke at all. (In a notable exception, Emanuel reviews evidence that, weirdly, eating ice cream is linked to a lower risk of diabetes and heart disease—a controversial correlation that gives the book its title.)

Emanuel makes the case that the quest for a meaningful life is more important than the quest for a lengthy one. “Wellness and living long are only means,” he writes. “They are not, themselves, the essence of a good life.” Sometimes the means even get in the way. If I skip dinner at a friend’s house because it’s going to ding my sleep score, or if I totally abstain from my favorite food because it’ll mess up the stats on my continuous glucose monitor, my life might be worse, not better. As I read Emanuel’s book, I started to think of the pursuit of longevity as akin to the pursuit of happiness. John Stuart Mill famously argued that people don’t find happiness by furiously chasing it—they find it by seeking something other than their happiness. “Aiming thus at something else, they find happiness by the way,” Mill wrote. The brevity of “Eat Your Ice Cream” is itself a kind of advice. Absorb the basics, Emanuel seems to be saying, and then get on with your life. He wants to help us make healthy choices, but he also wants us to see that even the best habits can’t stop us from aging. The first rule of life, Emanuel writes, is that we’re all going to die. ♦



The Infinity Machine, by *Sebastian Mallaby* (Penguin). Demis Hassabis long dreamed of building a machine powerful enough to untangle science's greatest mysteries. In 2010, at the age of thirty-four, he co-founded the A.I. startup DeepMind to pursue this vision. Since then, the company's creations have solved painfully difficult problems in fields from mathematics to computational biology. One of its programs produced a solution to the so-called protein-folding problem, representing a dramatic advance in the science of drug discovery and bringing Hassabis a Nobel Prize. Mallaby, a longtime financial journalist, is a nimble storyteller, and his portrait of one of the single-minded personalities plunging the world into an uncertain future is also an engaging drama of discovery.



If This Be Magic, by *Daniel Hahn* (Knopf). This rich and often funny book, by a translator of Portuguese, Spanish, and French, investigates one of literature's most challenging endeavors: replacing every word in a Shakespeare play with words from another language while having the result remain the Bard. Through interviews with translators and side-by-side comparisons of text, Hahn examines the many problems that translators face. (In Italian, for instance, far fewer common words are monosyllabic than in English, so a translated Shakespeare risks growing much longer than the original.) While exploring questions about ambiguity, order, pronouns, and even the use of A.I., Hahn's study of literary translation widens into a reflection on the broader mysteries of language.

What We're Reading

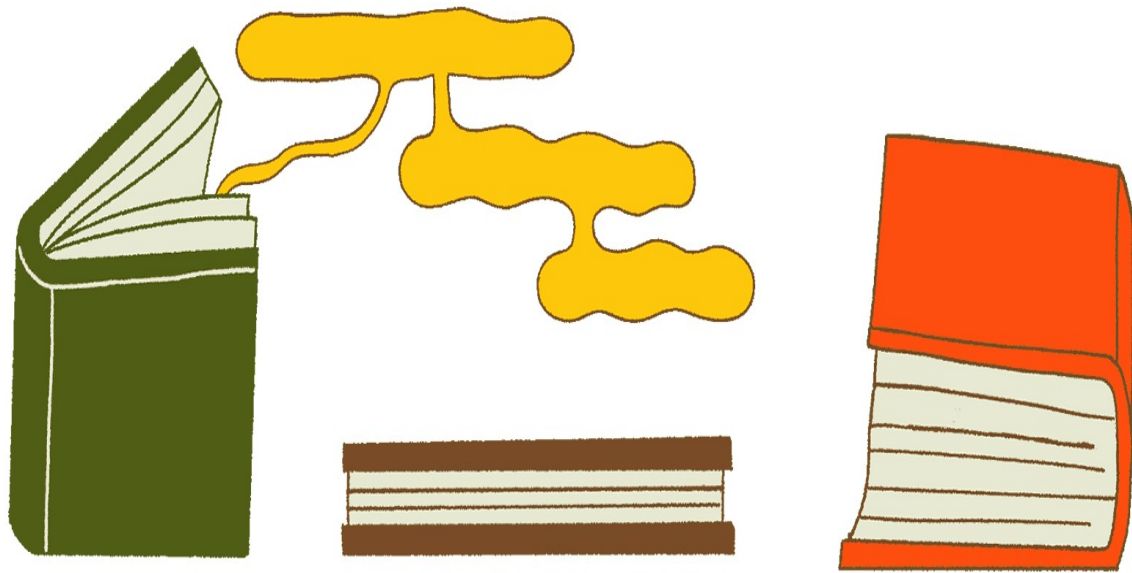
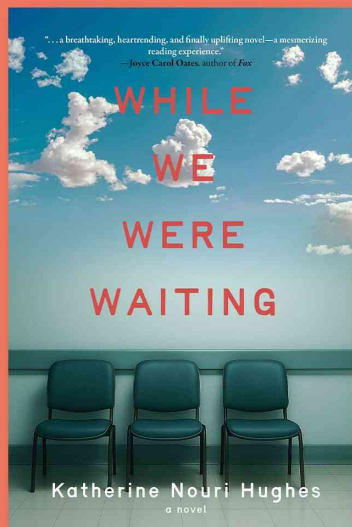


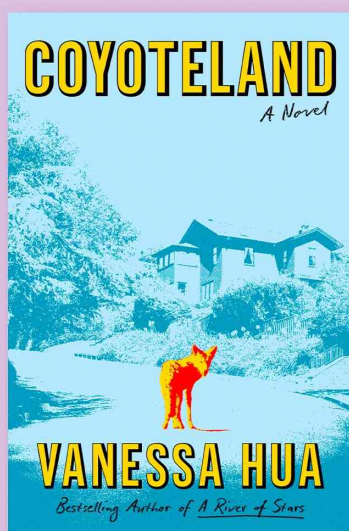
Illustration by Henri Campeã

Discover notable new fiction, nonfiction, and poetry.



While We Were Waiting, by Katherine Nouri Hughes (*Delphinium*). In this harrowing novel, two families pass long, uncertain hours in the waiting rooms of a Manhattan hospital. A venture capitalist who assesses risk for a

living is helpless as his wife, beset by preeclampsia, is rushed to an emergency delivery. Nearby sits a family of Chaldean Christians exiled from Iraq—an anxious archeologist, his pregnant radiologist wife, and his imperious mother, who bought a bulletproof Maybach to transport her daughter-in-law to the hospital for the delivery. Hughes, an American writer of Iraqi and Irish descent, has an acute ear for the half spoken, whether in a doctor’s clipped bulletins or a husband’s misfired word. The novel dwells in the porous hours of waiting—the interstices in which, as the title suggests, life accrues.



Coyoteland, by *Vanessa Hua* (*Flatiron*). A sheltered, moneyed community in the Bay Area stalked by a bold coyote is the backdrop of this tautly woven novel. El Nido’s residents live overlapping lives: Jin, a struggling software engineer, helps Chinese investors flip houses—a scheme he tries to hide from his family. Meanwhile, his next-door neighbors, the Belles, are conducting not-quite-legal surveillance of the neighborhood via intelligent cameras; they see everything, except their daughter’s rapidly worsening eating disorder. At school, Jin’s oldest daughter, Jane, teams up with a Black classmate, one of few, to expose their neighbors’ prejudices and pretenses. When these schemes come to a head at a consequential house party, the fallout brings damage all around.

The Fibre Fad Keeps On Moving

How a nutritional trend brought bathroom talk into the realm of food culture.

By [Hannah Goldfield](#)



You could think of cultivating your gut microbiome as “feeding your Eden,” according to Jeni Britton, who debuted a line of dehydrated-fruit bars last year. Photo illustration by Jason Fulford and Tamara Shopsin

For years, Ariel Pastore-Sebring, a thirty-one-year-old recipe developer and comedian in Portland, Oregon, has been keeping notes for a potential cookbook. One idea was a chapter of recipes inspired, in part, by her time running track at the University of Pittsburgh. “I was very conscious of what I was putting in my body,” she told me recently, especially when it came to finding the right balance of simple carbohydrates and fibre. “Runners are pretty big on clearing out before a race.” On social media, her posts skew aspirational, often showcasing the art of the dinner party: gingham tablecloths set with fresh flowers, kitschy cocktail glasses, and jewel-toned, vegetable-strewn platters. When she decided to experiment with high-fibre recipes on Instagram and TikTok, she opted for candor. In a video featuring roasted sweet potatoes with sautéed greens and dressed lentils, she spoke into a roll of toilet paper as if it were a microphone. “Welcome to Blow Outz,” she announced, in the self-consciously goofy tone of a game-show host, “a new series of recipes to knock a few things loose.”

To Pastore-Sebring's shock, the video went viral. She added more episodes—a quinoa-apple salad with walnuts, chopped kale, and radicchio; a spin on *som tum*, the Thai shredded-papaya salad; and the bluntly named Poop Smoothie (nicknamed the Poothie by some fans), made from spinach, chia seeds, oats, and peanut butter—and soon had seventy thousand new followers. She had stumbled into a trend known as “fibremaxxing.” “The craziest part is that people feel like they can share with me what’s going on with their bowel movements,” she said. “I’ve been getting D.M.s from people, like, telling me about their poops, or saying, ‘Hey, this really helped me.’”

I first sensed that fibre could be the new protein—a macronutrient that captures the Zeitgeist—last year, when Jeni Britton, the founder of the upscale brand Jeni’s Splendid Ice Creams, debuted a line of dehydrated-fruit bars called Floura. Made largely from cores and rinds, the bars were marketed as environmentally conscientious, but they were also meant to address a phenomenon that Britton calls the “fibre gap.” Less than ten per cent of Americans consume the daily recommended amount of dietary fibre, also known as roughage—the parts of plants that scrub through the digestive tract instead of being absorbed. (A single Floura bar contains about half the daily suggested intake.) Around the same time, I went out with three friends, all mothers in their early forties, for a monthly dinner that we jokingly call Women’s Group. One of them announced, conspiratorially, that another friend’s holistic physician had a maxim for the premenopausal: six prunes a day.

Compared with protein, which lends itself to narratives of growth—more grams, more gains—fibre is a bit of a hard sell, likelier to be espoused by women in sensible shoes than by fitness junkies on Twitch. But the benefits of eating a high-fibre diet are arguably more significant. Gastroenterologists and influencers alike are eager to explain that fibre affects health beyond “gut motility,” formerly known as shitting: it slows digestion, which can make you feel full for longer and help manage blood sugar and body weight, and has been shown to lower cholesterol and blood pressure. Fibre can also be a prebiotic, feeding the trillions of gastrointestinal bacteria that support our immune and nervous systems. If proteinmaxxing conjures the rapacious spirit of an HGTV makeover show, fibremaxxing is like a PBS special about restoring a Victorian home.

Britton became a zealot for fibre during the pandemic, not long after she'd decided to step away from the day-to-day operations of her ice-cream business. Stuck at home, in Ohio, "I just didn't know who I was at all, so I went to the forest," she told me. "I was walking ten miles a day, wandering the earth like Demeter looking for Persephone." She also began to eat fresh blueberries by the pint. At first, she wondered if she was overdoing it—wasn't fruit high in sugar?—but after a few months she'd lost weight and felt better than she had in years, a transformation that also helped her realize that she wanted a divorce. After educating herself on nutrition, Britton concluded that it was the fibre, stupid—and that she should try harvesting it from the parts of fruit that an ice-cream company would typically discard.

The bars, which come in flavors like mango cardamom and brambleberry lavender, are packaged in luxe, color-blocked matte wrappers. Each is a chewy, dark-brown puck, slightly tacky to the touch. Most are speckled with bits of roasted almond; all taste predominantly of indeterminate dried fruit. When I asked Britton how she planned to overcome fibre's less palatable associations, she reached for dreamier imagery: you could think of cultivating your gut microbiome as "feeding your Eden," she said. "I think people are going to start to think of their insides as, like, this wilderness place that I can go, and it's where my imagination lives, and my peace and my safety." Laughing, she added, "Versus the idea of, just, poop."

In the past quarter century, food culture has become ever more personal and confessional; perhaps the inevitable conclusion was that it would encompass, well, its inevitable conclusion. Like so many trends, this one was pioneered by gay men: one of the early luminaries of FibreTok was a sex educator and amateur chef named Alex Hall, who posts recipes on an account called The Bottom's Digest. I admit to feeling some relief around the breaking of this taboo—after all, everyone poops, and we might as well do it better. In France, a healthy bowel movement might serve as evidence of aesthetic finesse: according to at least one Reddit thread, to say "*j'ai fait un perfect*" is to brag about a stool so clean that it requires little to no toilet paper.

As it turns out, the appeal of fibre cuts across marketing idioms, whether it's Pastore-Sebring's bawdy millennial relatability, Britton's candy-colored promise of nirvana, or the biohacking grindset embodied by the thirty-six-

year-old entrepreneur Alan Lin. In 2025, Lin, a self-described “fibre daddy,” released a product called Liquid Salad, a thin green purée of apple, cucumber, kale, and celery, enhanced with potent prebiotics such as chicory-root inulin and chia seeds. It comes in the sort of plastic pouch that is often used for baby food, and it tastes, to me, like a pleasantly vegetal applesauce with a powerfully unpleasant aftertaste of stevia. (A version sweetened only with fruit juice is forthcoming.)

Lin got the idea for Liquid Salad on a trip he took to Japan, in 2024. He realized that the traditional Japanese diet, which is heavy in fibre, was optimal for digestion, and he marvelled at the abundance of nutritional-supplement pouches he saw at convenience stores. Back home in Seal Beach, California, he set out to make a high-fibre diet “extremely convenient for lazy Americans.” He earned a following, in part, with an absurd and tangentially related gimmick: a series of videos in which he wolfs down boiled eggs and other sulfuric foods in order to generate maximally offensive farts. “Every influencer is in Lululemon, and so fucking boring,” he told me recently. “I’m, like, Can’t you make it entertaining and fun?”

We’d met, ostensibly for lunch, at the Crema Café, a Seal Beach bakery known for its sandwiches and pastries. But when it came time to order, Lin, who wore a slim pair of Saint Laurent sunglasses and a mesh tank top that showed off his biceps, asked for only an almond-milk latte, unsweetened. He tries to avoid carbs, he explained, focussing mostly on protein and on fibre—the best macro for unblocking, as it were, one’s potential. “I wake up so excited,” he told me. “I take a huge dump every morning, and then I crush the MacBook for thirteen hours every day. This is, like, the most exciting thing in the world to me.” ♦

The Met's "Costume Art" Makes a Case for Fashion

From its new galleries off the museum's Great Hall, the Costume Institute seeks to put clothing at the center of art history.

By [Rachel Syme](#)



Dilara Findikoğlu's "Earthly Paradise" dress, from 2023. Photograph by Paul Westlake / Courtesy Metropolitan Museum of Art

The title of "Costume Art," a sprawling, stuffed-to-the-seams exhibition at the Metropolitan Museum of Art (running now through January, 2027), offers little clue as to what lies inside, other than, of course, clothing and art—or is it clothing that *is* art? There is no snappy subtitle to explain the show to the masses, as with previous exhibitions at the Costume Institute, such as "Heavenly Bodies: Fashion and the Catholic Imagination" and "PUNK: Chaos to Couture." Nor is there a marquee name to lure in the crowds, as with "Alexander McQueen: Savage Beauty" and "Charles James: Beyond Fashion." This lack of bombast is actually a power move. Those past shows, glorious as they were to behold, sound, in retrospect, too eager to entice. "Costume Art," by contrast, is an exhibition whose creators are confident that it will be seen, if not because of what it contains then simply because of where it is. The show marks the début of the institute's brand-new Condé M.

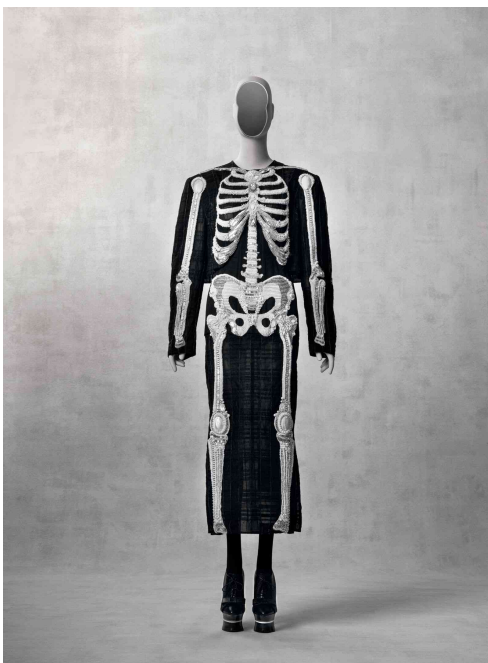
Nast Galleries, a nearly twelve-thousand-square-foot, five-room space situated just off the Met's central Great Hall—a prime slice of real estate formerly inhabited by a gargantuan gift shop.

The galleries (which are named for the founder of this magazine's parent company) took three years to complete, but they represent more than eight decades of jockeying for legitimacy within the larger institution. The Costume Institute's main dedicated galleries previously languished in the basement, and the department is responsible for raising most of its own budget each year, a task accomplished in large part by its annual fund-raiser, the starry Met Gala. Under the leadership of the curator Andrew Bolton—and with the patronage of Anna Wintour, the global editorial director of *Vogue*, who has used her influence to expand the department's power—the institute has become a major force within the museum and a recognizable brand outside it. Of the ten most visited exhibitions in the Met's history, five were put on by the Costume Institute. In 2018, "Heavenly Bodies" became the most attended Met show of all time. Now, as Bolton writes in the catalogue for "Costume Art," the new galleries "situate fashion within the spatial and symbolic epicenter of the Museum."

So why does "Costume Art," ostensibly a victory lap, feel so defensive? The show may as well be called "Costume Is Art," inasmuch as it argues insistently that fashion, long derided as frivolous and decorative, can hold its own against any other art form—and, moreover, that the entire history of art can and should be viewed through the lens of clothing. Bolton makes this case by focussing on what he calls "the dressed body," which he identifies as a central theme in every collection at the Met. Throughout the building, after all, you'll find bodies "draped, wrapped, tailored, armored, incised, painted, ornamented, and otherwise fashioned," Bolton writes. (Even when a body is naked, he notes, it is still cloaked in "physical and cultural ideals.") As evidence of this ubiquity, he has paired each of the roughly two hundred garments in the show with a thematically resonant work of art from another Met department. The fashion exhibition doubles as a museum tour in miniature.

It begins, cleverly, with a section called "Naked and Nude Body." In an act of pure showmanship, Bolton has placed mannequins clothed in fleshlike skintight garments in a tall glass vitrine visible to museumgoers heading

toward the medieval wing—a seductive preview that may help lure them away from Arms and Armor. Bolton also establishes a format that continues through much of the exhibition: a line of mannequins standing high above their paired art works, signifying, in unsubtle terms, that, here, fashion is meant to dominate the discourse. An arresting sheer dress by the young Turkish British designer Dilara Fındıkoğlu—which uses snaking tendrils of synthetic hair to cover the wearer’s genitals, à la Botticelli’s Venus—towers over an eighteenth-century bronze Venus. A Pierre Cardin ensemble from 1967, featuring a dress with holes cut out to expose the breasts, looms atop a Flemish bust of Minerva preparing for battle in breast-baring armor. Bolton calls this technique of physically elevating the garments “pedestalizing.” Ironically, though, the gesture has the effect of undermining the clothes: the art works sit comfortably just below eye level, whereas the visitor has to crane her neck to take in the fashion designs above.



A Thom Browne dress features a skeleton made of tiny beads. Photograph by Paul Westlake / Courtesy Metropolitan Museum of Art

Some of the pairings can be quite on the nose. In the “Classical Body” section, a golden Issey Miyake breastplate and bodice, molded to the curvature of the chest, is paired with a golden Etruscan cuirass. A row of Grecian-inspired draped gowns is paired with a row of Grecian urns featuring berobed ancients. These couplings are so literal that they seem to undercut Bolton’s thesis, suggesting that fashion raids art history rather than illuminating it anew. The following section, “Abstract Body,” does a more

persuasive job highlighting fashion's innovations, showing an array of corsets, panniers, and hoop skirts that designers used to manipulate the body into unnatural forms. Looking at a nineteenth-century corset that cinched the wearer's waist into the circumference of a paper-towel roll, I felt, for the first time during the show, conscious of being in my own body (which squirmed at the thought of such sartorial constraints). The collection of elaborately engineered bustles and crinolines certainly advances the idea that clothing, even at its most punishing, can be a form of living sculpture. I felt a similar jolt in the "Reclaimed Body" section, which focusses on more avant-garde modifications, such as Rei Kawakubo's bulbous creations with Comme des Garçons, three of which are paired with abstract sculptures that echo the garments' austere curves. Each of the sculptures—by Max Weber, Henry Moore, and Jean Arp—was created prior to Kawakubo's fashion experiments, but one gets a disorienting sense that, rather than Kawakubo reaching backward for touchstones, the sculptors were somehow anticipating her singular inventions.

Most of the mannequins have mirrored faces, a flourish that, according to the catalogue, is meant to create "a moment of communal visibility," though the heads are tilted at awkward angles that discourage selfie snapping. Eighteen unique mannequins were created for the show, modelled off real and imagined people whose bodies don't conform to conventional fashion-world ideals. The "Corpulent Body" section features a mannequin, cast from the body of the French model and singer Yseult, looking formidable in a scaled-up Dior "Bar" suit. "Disabled Body" includes a mannequin cast from the activist Sinéad Burke, who has dwarfism, wearing a Burberry trenchcoat that's been cut short; it's accessorized, winningly, with a hat made from one of the coat's lopped-off sleeves. (Why, one wonders, couldn't there have been such mannequins in "Classical Body"? Were no designers inspired by Greek statuary with belly rolls?) "Pregnant Body" displays a Gucci dress, by Alessandro Michele, with a pair of beaded fallopian tubes across the belly—a design that is meant as a nod to women's rights but risks reducing the wearer to a walking reproductive system. I preferred a long gown made by the Greek designer Dimitra Petsa using her signature "wet look" technique, so that its mesh fabric clings like seaweed to the mannequin's pregnant form.



A Vetements ensemble from the show's "Aging Body" section. Photograph by Paul Westlake / Courtesy Metropolitan Museum of Art

The back half of the exhibit is smaller and more intimate, and it finally brings the clothing down off the pedestals, to ground level. “Costume Art” has no clear foot-traffic pattern. I learned only later that I was supposed to end at “Epidermal Body” (showcasing garments that mimic the look of skin), rather than at “Mortal Body,” which seemed like a more natural conclusion. Regardless, the latter portion of the show was by far the more affecting, not only because of its visceral themes—including tattoos, blood, and aging—but also because it offered a chance to admire exquisite garments up close. There’s a crimson Yohji Yamamoto dress made of silk crêpe de Chine, painstakingly manipulated into folds; a Yuima Nakazato piece that looks like bloody entrails dripping down the body; a Daniel Roseberry creation for Schiaparelli embellished with buttons in the style of Victorian mourning jewelry; and a Thom Browne dress featuring a pearly skeleton, made of seed-size beads, emerging out of the garment’s black surface like a spectre.

Such pieces are undeniably art works—precious, impractical, one of a kind. The most attention-grabbing look in the show, though, might be one in “Aging Body”: a ready-to-wear Vetements ensemble featuring an absurdly oversized gray hoodie emblazoned with the message “*I’M RETIRED* (This is as dressed up as I get).” I watched as visitor after visitor snapped a photo of the sweatshirt, and I overheard more than one mumble about wanting to buy

one. (They can do so, online, for seven hundred and eighty dollars.) This is presumably what “Costume Art” is fighting against—the idea that fashion exhibitions are, in essence, a vaunted form of window-shopping. And yet, as I watched people smile and laugh upon spotting the hoodie, I wondered what was so wrong with that idea. People love to fantasize about clothes, to imagine what their own bodies might feel like encased in finery or outfitted in some outré style. This, more than anything, is the impulse that has made the Costume Institute a hit. Is fashion art? Of course it is. But the clothes—the clothes are the thing. ♦

By [Alex Ross](#)

On a recent Saturday morning in Ojai, California, a venerable bohemian enclave northwest of Los Angeles, the clarinetist Anthony McGill stood beneath a sycamore tree and played “Abyss of the Birds,” the third movement of Olivier Messiaen’s “Quartet for the End of Time.” McGill, the flutist Rose Lombardo, and the accordionist Hanzhi Wang were giving a free concert at the Ojai Meadows Preserve, a restored wetlands area. More than four hundred people had assembled for the occasion, some seated on folding chairs and some reclining on blankets. McGill’s superbly nuanced calls, runs, and trills elicited commentary from a second audience, perched in the trees above—a colloquium of finches, towhees, titmice, kingbirds, juncos, and Eurasian collared doves. (A bird-watcher was seated behind me.) This pre-summer idyll could not have been more different from the setting in which the Quartet was first heard. Messiaen completed the score while he was being held at a German prisoner-of-war camp, and it had its première there on an icy winter night in 1941. Perhaps the composer dreamed of a future performance like the one in Ojai; his music rejects the nightmare of history and goes in search of ultimate serenity.

Once again, the Ojai Music Festival was summoning sonic bliss, as it has been doing for the past eighty years. No gathering of its kind is mellower in mood or keener in attitude. The general rule is that programs should focus on twentieth- and twenty-first-century composers, although musicians won’t be reprimanded if they leap back to the twelfth century, as the violinist Geneva Lewis and the cellist Jay Campbell did at another crowded morning event this year, when they played an arrangement of Hildegard of Bingen’s chant “O Virtus Sapientie.” The Ojai audience, having been molded over the decades by the likes of Igor Stravinsky, Pierre Boulez, Michael Tilson Thomas, and Mitsuko Uchida, is not easily fazed. No one blinked an eye when, during a concert at the Ojai Valley School, the percussionist Jonathan Hepfer filtered a Beatles tune through an amplified teapot, in a rendition of Alvin Lucier’s “Nothing Is Real.” In 2015, Steven Schick scheduled Morton Feldman’s five-hour-long “For Philip Guston” for the mad time of 5 *A.M.*, and about a hundred hardy souls showed up.

Each year, Ojai has a different music director. When I first visited, in 1999, the composer-conductor Esa-Pekka Salonen, then forty, brought in a gang of

avant-garde Finns, the Toimii Ensemble, and unleashed Magnus Lindberg's anarchic masterpiece "Kraft," from 1985. While the Los Angeles Philharmonic thundered apocalyptically, members of Toimii ran up and down the aisles of the Libbey Bowl, Ojai's main venue, to bang on car parts and strike gongs that were hanging from trees. Salonen was back this year, a bit less rambunctious than before, secure in his status as one of the major musical figures of our time. He is now old enough to have seen many mentors and colleagues fall away: Messiaen, György Ligeti, Luciano Berio, Franco Donatoni, Oliver Knussen, Witold Lutosławski, Steven Stucky, Kaija Saariaho. All were present in Salonen's programs, in sound and spirit. To some extent, the 2026 festival was a retrospective of the late twentieth century, when modernist strictures were loosening and a more expansive, pluralistic mentality was taking hold.

Nostalgia isn't an Ojai virtue, however, and younger composers kept things current. Messiaen's nature-consciousness found a counterpart in Gabriella Smith's "Anthozoa," for violin, cello, piano, and percussion, which evoked the microscopic bustle of coral reefs, with clicks and rustlings giving way to oceanic hymnal chords. Anna Thorvaldsdóttir's "Ró" ("tranquillity," in Icelandic) unfurled a not dissimilar soundscape of organic tendrils growing from sustained drones. Meanwhile, the pianist Conor Hanick played two new pieces by John Adams, the most boyish of near-octogenarians; one of them, "song without words," offered a transfixing instance of Adamsian endless melody, at once radiant and bittersweet.

Of the half-dozen Ojai festivals I've attended, this one was perhaps the most perfectly executed. The excellence of the programming and music-making testified not only to Salonen's magnetism but also to the behind-the-scenes dexterity of the veteran arts administrator Ara Guzelimian, who has been associated with Ojai for decades and has led its operations since 2020. The ovation that erupted when Guzelimian walked out to introduce the festival's closing concert was a fitting acknowledgment that golden-age concerts do not happen by themselves.

That final concert, featuring the student orchestra of the Colburn School, in L.A., at once exemplified Ojai's ethos and jolted its laid-back atmosphere. Salonen led a buoyant, mercurial ninetieth-birthday tribute that he wrote for the late Frank O. Gehry, titled "Fog." He also presented Stravinsky's ironic-

bucolic ballet “Pulcinella,” a bracing prelude to summer. In between came Ligeti’s towering, terrifying Violin Concerto, written between 1989 and 1993. Leila Josefowicz, for whom both Salonen and Adams have written concertos, essayed the solo part for the first time in her career, and she delivered it with such sublime fury that afterward the customarily reserved Salonen did something quite uncharacteristic: he slapped his forehead with his palm and gestured at Josefowicz in happy disbelief.

I had been waiting more than thirty years to hear this concerto done so intensely. In 1993, when I was a neophyte critic, I went to Boston to report on a Ligeti festival at the New England Conservatory. The composer was in attendance, holding forth in lectures and discussions, his arachnid mind spinning webs across music history. At one point, he gave a virtuosic disquisition on motifs of lament, citing Gesualdo, Monteverdi, Bach, Schubert, and Purcell (he sang an excruciating version of “Dido’s Lament”), not to mention flamenco, Roma music, and funeral songs he’d heard in Transylvania as a child. He also discussed the Violin Concerto, which he was still revising. He related the piece to his attempt to develop a new kind of tonality, one that would absorb tradition without becoming trapped by it. He said, “I am in a prison—one wall is the avant-garde, the other wall is the past, and I want to escape.” Only Ligeti could find himself in an Escher-like prison with just two walls.

The concerto is less than half an hour long, yet it contains a universe of possibilities. Folkish ditties dance through fractal textures; standard equal-tempered tuning is juxtaposed with the archaic practice of just intonation; rhythms ricochet against one another; influences of Central African forager music and Indonesian gamelan are audible; the ancient lament resounds. Because of its extreme complexity, the work is seldom programmed, and when it is the realization can fall short of the conception. In 2000, I heard it at Carnegie Hall, with Christian Tetzlaff, Pierre Boulez, and the London Symphony. The notes glittered in place, but the piece came across more as a cerebral design than as a living thing.

Salonen has dug deeper. He esteemed Ligeti from an early age and worked closely with the composer from the nineteen-eighties onward. When the Sony Classical label launched a recorded survey of Ligeti’s music, in the nineties, Salonen was chosen to oversee the project. Alas, Ligeti proved to

be an extraordinarily difficult collaborator, his manic perfectionism colliding with practical reality and leading to unpleasant scenes. One day, Salonen received a fax from him that read, “I promise to be relatively nice tomorrow.” Nonetheless, Salonen never stopped admiring the music, and he bore in mind the psychic pressures that weighed on the man himself. Messiaenic serenity was not within Ligeti’s reach. When I spoke with Salonen at Ojai, he told me, “Here was a man who lost almost all of his family in the Holocaust, and then fled from Hungary to Austria in 1956, in the middle of the night, with dogs chasing him at the border. When that is your starting point, it creates a very bleak world view.” What fascinates Salonen now, however, is not the darkness of Ligeti’s art but its aliveness and freshness: “The vitality of a piece like the Violin Concerto reminds me of the ‘Symphonie Fantastique’—it will always surprise you.”

Josefowicz, in her decathlon of a performance, brought Ligeti’s savage discontinuities to the surface. She applied a thick, bristling tone and dispatched swirls of notation with meticulous abandon. At the same time, she attended to the primeval lyricism that shimmers through the work’s kaleidoscopic visions. The second movement, “Aria, Hoquetus, Choral,” begins with a seventy-four-bar melody that is played entirely on the G string. Josefowicz lent it a desperate warmth, as if she were the last survivor of a destroyed village. Most stupendous of all was her cadenza in the fifth movement—a tour de force of disciplined chaos that she composed herself, responding to an invitation that Ligeti extends to the soloist. “It should be hectic throughout,” he writes, and so it was. Salonen guided the Colburn Orchestra with flexible authority, imposing order or inciting wildness as the moment required. The youthful musicians chattered away like creatures of the Transylvanian night.

The Violin Concerto ends with an elegant shock. The orchestra delivers a rapid-fire series of thwacks, grunts, yelps, and honks, marked “*ffffff*” and “*ffffff*.” The flutes follow with flutter-tongued E’s that fade to silence—a quivering, alarm-like effect that Ligeti derived from Shostakovich’s doom-laden Fourth Symphony. Salonen told me, “There are elements of a triumphant finish, but it kind of falls apart. With Ligeti, you often don’t know if it’s playful, or tragic, or, most likely, both. It makes me think of the final scene of Verdi’s ‘Falstaff’—‘All the world is a jest.’ ” In some ways, Ligeti’s ending doesn’t even sound like an ending. The buzzing of the flutes

could be a signal to wake up and keep going. In the twilight of the twentieth century, with the shadows of history all around, Ligeti cleared the stage for an infinite musical future. ♦

By [Kelefa Sanneh](#)

Last month, a businessman named Kabir Mulchandani appeared in front of the New York State Liquor Authority (S.L.A.) seeking permission to sell alcohol at the open-air Brooklyn night club that his company had recently taken over. There was a lot to consider. Residents had written in with concerns about safety and noise, and the local community board had voted, twenty-two to fourteen, with one abstention, against approving the venue's application. In its previous incarnation, the night club—the biggest of a handful of venues situated in an industrial zone near the edge of East Williamsburg—was known as the Brooklyn Mirage, and its reputation was not uniformly good. Three people died of drug overdoses after attending events there, and in recent years a number of dead bodies were retrieved from the nearby Newtown Creek. In 2025, the Mirage announced a grand renovation, but it failed to get approval from the city's Department of Buildings and was demolished earlier this year. A private-equity firm took control of the site and made an agreement with Mulchandani, the head of a Dubai-based company that owns Pacha, a venerable and glamorous night club on the Mediterranean resort island of Ibiza. With Mulchandani's company as the operator, the Brooklyn club was being reinvented as Pacha New York.

Lily Fan, the chair of the S.L.A., seemed to understand the significance of the place and to be impressed by the improvements that Mulchandani promised: tighter security, a less noisy sound system, shuttle buses to bring patrons to subway stations, and even a new cell tower nearby, so that, when the club closed at 4 *a.m.*, revellers would not find themselves wandering the neighborhood in search of a signal. She indicated that the S.L.A. would approve the application for Pacha New York so long as the club promised to eschew the M-word in all its promotional material. “No Mirage mention anywhere, please,” she said.

There was a time when New York City was the night-club capital of the world. In the nineteen-seventies, the city was the cradle of disco and home to a wide range of influential dance parties: the Loft, a downtown institution, helped invent the underground dance club; Studio 54, a short-lived tabloid sensation, helped create the idea of the night club as a celebrity stage. (Bianca Jagger was memorably photographed there astride a white horse.)

But, in the decades that followed, other cities caught up and eventually surpassed New York; one of them was Ibiza, where clubs like Pacha showed vacationing Europeans how to rave all night.

In New York, night clubs were sometimes regarded as a civic nuisance; one obstacle was the city's cabaret law, from 1926, which required bars to obtain a special permit if patrons were dancing and was not fully repealed until 2017. The unmentionable predecessor of Pacha New York was a rather whimsical, castlelike structure with a balcony from which the skyline of a faraway borough was visible: Manhattan. (For about a decade, beginning in 2005, Pacha operated a different club there, in Hell's Kitchen.) The new club is simpler and more functional, with a legal capacity of fifty-three hundred patrons—at the hearing, Mulchandani testified that, though the previous operators had routinely exceeded capacity, he would not. The dance floor is an expanse of blacktop, surrounded by black shipping containers that serve as bars; the wide stage has a d.j. booth in the middle, flanked by V.I.P. areas and backed by an enormous video screen. On Saturday, June 13th, which was opening night, the V.I.P.s included the rappers Travis Scott and Quavo, as well as Mulchandani himself, wearing a black T-shirt that said "Brooklyn," with the "o"s rendered as a pair of red cherries—Pacha's trademark.

People sometimes use the term "electronic dance music" to refer to the wide range of club-oriented genres that proliferated in the post-disco years, but Pacha concentrates more narrowly on house music, which emerged from Chicago night clubs in the nineteen-eighties and now occupies the more traditionalist end of the dance-music spectrum. For opening weekend, the club hired Rampa, a popular producer and d.j. from the fertile Berlin club scene, who excels at folding the sounds of Afro-pop into warm, rippling house tracks. Because he was inaugurating a New York venue, Rampa put together a lineup stocked with local favorites, including Danny Tenaglia, who built his reputation in the nineteen-nineties with marathon sets at Twilo, in Chelsea. "Such an honor to be here for the grand opening," Tenaglia said. "I hope you're enjoying the game, but I hope you're enjoying the music more." The screen behind him was showing a certain local basketball team, which was in the process of ending more than half a century of frustration; when the job was done, the crowd roared, and Tenaglia played a bit of "New

York, New York,” before bringing in “Du Du Dum,” a clattering house track that turned the watch party back into a dance party.

By the time Rampa arrived on the decks, a bit before 1 *a.m.*, the crowd was congealing into a congenial blur: serious dancers with their eyes closed, celebratory guys in basketball jerseys, swaying couples, groups of friends interested more in one another than in the guy in the booth. He had brought with him Prospa, an English duo known for a series of club hits that evoke the hedonistic spirit of nineties house music. Everyone knew that Rampa would eventually play “Say What,” a 2024 hit he helped produce, which has a joyful bass line and a refrain that is equally indecipherable to dancers across the world, since it’s in a made-up language. People raised their hands and sang along, but only for a few minutes. Rampa told me that he tries to make sure his d.j. sets don’t devolve into concerts; he wants people to feel as if they’re at a party, not a performance.

This is not always easy to do, especially for anyone who appears, as Rampa often does, on lists of the world’s most popular d.j.s. He told me that he had fond memories of his time in Berlin’s underground scene, playing gigs in cramped venues that were often at risk of being shut down. “There were no permissions and licenses, so it was very pure and honest,” he said. He knows, though, that the outlaw approach tends to lose its charm as the crowds grow and the likelihood of disaster increases. On the dance floor, as elsewhere, people tend to want to feel both free and safe; in that sense, every night club is a compromise, and part of the job of a d.j. is to help people forget that. At a scrappier venue, there might be someone at the door, deciding who will or won’t contribute to a great night. But at a big corporate club like Pacha anyone with money can buy a ticket. In that sense, at least, Pacha is more inclusive than some of its smaller competitors. Prices for shows this summer range from forty-five dollars, for general admission, to hundreds or thousands of dollars.

For Sunday of opening weekend, Rampa planned a kind of New York carnival: he put a skate ramp in one of the warehouse-like buildings on the property and organized an appearance by two members of the Wu-Tang Clan, Ghostface Killah and Raekwon. Around ten o’clock, the local d.j. Kitty Ca\$h played one of the weekend’s best sets, starting gently, with remixes of Aretha Franklin and Nina Simone, and then pushing into tougher,

more percussive tracks, creating an immersive music-history lesson. It began to rain, softly at first. “This rain ain’t stopping shit, y’all,” she said, as employees erected a portable tent above the d.j. booth and distributed ponchos to the audience. But the rain grew heavier and was soon accompanied by thunder and lightning, which meant that the party was officially and Clan-lessly over. (Rampa later announced that he was waiving his Sunday fee, and Pacha offered attendees a choice between a refund and free admission to a makeup party.) Some people headed for the exits, while others piled into the room with the half-pipe, where Rampa played a couple of tracks, briefly turning a rained-out club night into an old-fashioned warehouse party. At least some of the revellers seemed to enjoy this impromptu set. On TikTok, one of them offered high praise: “Felt like an og rave.” ♦

Poems

- “Campbell’s Tomato Soup”
- “requiem for a barber”

By [Campbell McGrath](#)

Begin with the flavor, which is not rank but insipid.
Made with water it tastes of a rusty farmyard pump,
made with milk it resembles melted vinyl in a pan
yet veers too closely toward its cousin, tomato sauce,
in all its richer culinary regalia, to say nothing
of that lordly fruit disguised as a vegetable,
the selfsame tomato in its myriad shapely tropes.
The larger question is existential: why soup?
As a snack it's too much, too fussy, a hassle.
As a meal it leaves you hanging, perplexed.
Certainly, I have sampled delicious soups in my day—
split pea in Les Halles in the autumn of 1985,
a tureen of Tibetan lentil-and-potato chowder
after escaping a blizzard in Durango, Colorado—
but when I summon the flavor of Campbell's soup
I am propped on pillows with a terrible cough,
home from school watching game shows on TV.
Soup is a peasant artifact steadily vanishing
from our lives, as illustrated in the financial section
by the travails of the Campbell Soup Company,
their offerings rejected by the modern marketplace
as drab, old-school, too salty, out of touch,
though for me the problem remains the name.
Despite its ubiquity *Campbell* poses a phonetic puzzle,
mysteriously pronounced, hard to spell,
derived from Scots Gaelic, meaning *crooked mouth*.
Hmm, why crooked—was it the soup?
The only compensatory value I've received
for sharing a moniker with its iconic red-and-white can
is when ordering takeout food over the phone,
at which time I invariably give my name as
“Campbell—like the soup.”
For decades a foolproof formula, a passkey,
a universal monad like Elvis or Shakira,
but now it elicits a head shake. *Cannonball? Chimbu?*
Once, I arrived to pick up spring rolls and pad Thai

to find myself mislabelled *Gumbo*—like the soup!
Better, these days, to be christened Goya, or Warhol.
But time marches onward, tastes change,
corporate empires rise and fall and each passing year
more of us live in happy ignorance
of this paradigmatic meme of twentieth-century Americana,
blossoming generations for whom tomato soup
might as well be tomato aspic,
while I continue torn between gauzy
cinematic phantoms of the past
and live-streamed promises for the future.
One last story: the other Wednesday
I stopped on the way to work
to pick up a sandwich from a vintage delicatessen
where the ancient woman at the counter,
her skin translucent as parchment paper,
stapled the receipt to a bag
and asked me,
“Campbell—like the soup?”
Yes indeed! “Are you related?” *Sadly not.*
“You know,” she said, “I grew up in Camden, New Jersey,
and the farmers used to line up day and night
all summer long
in trucks full of tomatoes
at the gates of the Campbell’s factory
with those big brick smokestacks gushing steam—
it was a long, long time ago
but I still remember the delicious smell
of tomato soup on every block in the neighborhood.”

By [Jan Wagner](#)

on monday all must rest, so monday it will stay:
shade the mirrors, slip the edges of the shears away.

who will knead his fingers now, circling, until the cloud
of shampoo gathers over us; who will conduct his crowd

of bottles on the shelf, the oils and fragrances,
with small hands? who will blast the glorious organs

of blow dryers and let them roar, let them swell?
of all the tints, take the black; darken up the colorful.

now, magnificently, slowly as a tent, there are no capes
to drape across the body; now anyone who stops

will not know where they're going or what they'll find
but that the hair keeps growing fuller, growing wild.

(Translated, from the German, by David Keplinger.)

This is drawn from "[Wisp.](#)"

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